

FINITE-USE DISCRIMINATION GEOMETRY OF ORDERED QUANTUM MEASUREMENTS

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ABSTRACT. We study ordered quantum measurements with separate acquisition and receiver resources. A common-barycenter variational principle and attained Hermitian-majorant dual describe every finite two-call reset experiment, with rank restrictions resolving the retained receiver registers. On a finite shared-basis experiment, we evaluate the entire rank-constrained optimum at every prescribed pure initial acquisition. Its value is a concave spectral function: a least rank-constrained majorant followed by one secular equation. A recorded instrument with at most d outcomes attains it without changing the initial acquisition. We determine exactly when communicating a receiver outcome to the next preparation is strictly useful, and give its exact gain and the initial-rank saturation threshold. The proof combines a rank-sensitive filtered-swap optimization with a spectral evaluation of concave envelopes. The equal-prior dimension hierarchy, complete initial equality set and quantitative projection rigidity remain complementary results on the same family. The local theory classifies fixed quadratic tangent tubes by the complete support-covariance trichotomy: a profile with $N = \sum n_j$ and $V = \sum n_j^2$ has trace orders $\min\{1, \sqrt{Ns}\}$, $\min\{1, \sqrt{Vs}\}$ or $\min\{1, Ns\}$; a hard square budget Q gives coherent order $\min\{1, \sqrt{Qs}\}$. All finite remainder terms and the unknown-remainder independence of the lower readouts are retained. The general variational characterization, solved finite family and fixed-tube local orders have distinct quantifiers.

1. INTRODUCTION

A measurement used as an unknown device returns a classical label, but its discrimination need not be classical. Correlations may be created inside an acquisition, retained in an external receiver, or transmitted coherently into a later acquisition. These are distinct resources. This article gives a local geometric classification with sharp allocation laws, a general variational principle for reset decisions, and exact finite experiments separating receiver storage from coherent acquisition. These results concern the same classical-output interface and retain the order of its labels.

1.1. Local geometry and exact experiments. An ordered measurement is a tuple $E = (E_1, \dots, E_k)$ of positive $d \times d$ matrices with $\sum_j E_j = I$. Its channel is $\mathcal{M}_E(\rho) = \sum_j \text{tr}(E_j \rho) |j\rangle\langle j|$. No post-measurement device system is available. Known preparations, ancillary systems and controls are allowed. Trace norms and diamond norms below are unhalved; success probabilities are ordinary probabilities.

The first result resolves the local geometry for a prescribed allocation. Fix E , a nonzero Hermitian zero-sum tuple H in the one-sided positive tangent cone, and a remainder allowance $\Lambda \geq 0$. For $\|R\|_\Sigma = \sum_j \|R_j\|_{\text{op}}$, consider every legal F with

$$(1.1) \quad \|F - E - sH\|_\Sigma \leq \Lambda s^2.$$

Write $P_j = \text{supp } E_j$, $Q_j = I - P_j$, $J_j = Q_j H_j Q_j$, $\Gamma_E(H) = \frac{i}{2} \sum_j [P_j, H_j]$, and $\mathcal{W}_E = \sum_j P_j \mathcal{H}_d P_j$, where $\mathcal{H}_d$ is the real Hermitian matrix space. The cone condition is $J_j \succeq 0$. For positive

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integer acquisition sizes $\mathbf{n} = (n_1, \dots, n_\ell)$, put $N = \sum_r n_r$ and $V = \sum_r n_r^2$. Theorem 13.3 proves, respectively, the orders

$$(1.2) \quad \min\{1, \sqrt{N}s\}, \quad \min\{1, \sqrt{V}s\}, \quad \min\{1, Ns\}$$

when $H \in \text{Ran } \mathcal{C}_E$, when all $J_j = 0$ and $\Gamma_E(H) \notin \mathcal{W}_E$, and when some $J_j \neq 0$. The support-kernel identity shows these cases exhaust the nonzero cone. The same orders hold for independent complete acquisition groups and for prescribed-profile reset feedback. At a reset cut the entire fresh probe, reference and within-block memory is conditionally independent of the receiver's entire old memory. The receiver may keep arbitrarily large quantum registers and act on them jointly; it cannot import them coherently into a fresh acquisition. Preparations and controls are one common rule on every formal history, including null histories. The conditional old receiver states need not coincide.

For a history-dependent policy π , hard pathwise reservations define $N_\pi = \sup_\zeta \sum_{h \in \zeta} n(h)$ and $Q_\pi = \sup_\zeta \sum_{h \in \zeta} n(h)^2$. Theorem 14.3 proves that the optimum over $N_\pi \leq N$, $Q_\pi \leq Q$, $N \leq Q \leq N^2$, has coherent order $\min\{1, \sqrt{Q}s\}$; its other two rows are unchanged. The constants and the interval $0 < s \leq s_0$ depend on the fixed E, H, Λ , but not on the profile, budget or unknown remainder. A common bounded readout gives the lower without learning that remainder. The individual-policy upper retains both $\alpha\sqrt{Q_\pi}$ and $\sqrt{rN_\pi}$; a large cost is not a lower information certificate for a policy which erases its output.

The second result concerns exact success, rather than local orders. For every $d \geq 2$, a finite family of ordered bases with simultaneous Haar second moments exists. Compare the scalar measurement $C_y = I/d$ with a once-chosen member $E_y^b(t) = (1-t)I/d + tP_y^b$, using that same device twice. The decision is scalar versus the family. Put $L = 2(d+1) - t^2$ and assign these alternatives priors $(d+1-t^2)/L$ and $(d+1)/L$. Theorem 16.2 gives the exact values

$$(1.3) \quad \begin{aligned} P_{\text{ca}} &= (d+1)/L, \\ P_{\text{reset}} &= (d+1)/L + t^2(d-1)/(2dL), \\ P_{\text{all}} &= (d+1+t^2)/L. \end{aligned}$$

Here the first class completely measures and reprepares between calls; the second allows arbitrary unit-reset receiver feedback; the third allows all adaptive controls. For $t > 0$ the inequalities are strict, including the full-rank interval $0 < t < 1$. Two independent maximally entangled pairs attain the middle value; an antisymmetric two-input state attains the last. A swap-filter spectral identity bounds every reset policy, including all intervening receiver instruments. For the scalar qubit measurement and the six ordered Pauli measurements, the exact values are $3/5$, $13/20$, and $4/5$. This finite ensemble result neither asserts a gap for every fixed pair nor identifies a coherent-memory-dimension hierarchy.

1.2. A general optimization principle and equal priors. Theorem 15.4 applies to every finite two-call ordered-measurement decision experiment, with arbitrary priors and rewards. If $g_y(\rho)$ is the best fresh-reference value after a first label without receiver feedback and c_y is its upper concave envelope on all density matrices, then

$$P_{\text{reset}} = \max_{\rho} \sum_y c_y(\rho).$$

A common first marginal is essential. At most $(\text{rank } \rho)^2$ instrument outcomes per first label suffice, and the quantum receiver has dimension at most $d_1 d_2$. Theorem 15.5 supplies exact Hermitian majorants and contact conditions. Mixed density-matrix atoms cannot be replaced silently by pure states. This is a finite-dimensional characterization of optimal values, not a polynomial-time optimization claim.

For the same finite basis devices, Theorem 17.2 removes the special prior in (1.3). With equal hypothesis priors its exact values are

$$\frac{1}{2} + \frac{t^2(d-1)}{2d(d+1)}, \quad \frac{1}{2} + \frac{t^2(d-1)}{2d^2}, \quad \frac{1}{2} + \frac{t^2}{2d}.$$

For the seven qubit devices at $t = 1$ these become $7/12 < 5/8 < 3/4$. The middle upper permits all receiver feedback, but its attaining strategy uses none. The alternative index is chosen once for both calls; independent redraws give no signal. The explicit qubit calculation and a channel/prior/law stability estimate follow the theorem.

Theorem 15.8 identifies the classical optimum by restricting the envelope atoms to pure states. Thus two explicit convexifications distinguish complete classical readout from quantum receiver storage in every finite experiment. Corollary 15.9 gives necessary and sufficient primal–dual certificates for a strict gap, and contact certificates for equality. Their majorization inequalities are universal; no sampling procedure is asserted to certify them.

These statements share one operational interface and one precise fresh-preparation cut, but have different quantifiers. The local profile law classifies shrinking fixed-tube signals at arbitrary hard resources. The variational principle treats exact two-call values for arbitrary finite experiments. The two basis theorems solve particular experiments inside that general class. Neither exact calculation supplies uniform constants for the local theorem, and the local theorem is not a premise of either exact score proof.

1.3. Geometry underlying the allocation law. The positive covariance operator $\mathcal{C}_E$ acts on zero-sum Hermitian tuples and incorporates the coupled measurement normalization. For $M = (E + F)/2$ and $H = F - E$, the global certificate is

$$(1.4) \quad D_N(E, F) \leq \min \left\{ 2, 2\sqrt{k+1} \sqrt{N \langle H, (\mathcal{C}_M + N^{-1} \text{Id})^{-1} H \rangle} \right\}.$$

Theorem 4.1 identifies its complete kernel in support coordinates. This is an upper certificate for arbitrary pairs, not an asserted global matching metric. On a nonstationary fixed unitary orbit $E_j(t) = e^{itG} E_j e^{-itG}$, the finite-angle theorem is

$$(1.5) \quad D_N(E, E(t)) \asymp_{E,G} \begin{cases} \min\{1, \sqrt{N}|t|\}, & G \in \mathcal{W}_E, \\ \min\{1, N|t|\}, & G \notin \mathcal{W}_E. \end{cases}$$

The support span is the Hermitian Kraus-product span. The channel metrological criterion and correction principle are credited to Zhou–Jiang [40]; the finite-pair estimates retain explicit quadratic channel errors. Normalized tangent realizations extend these estimates to the whole legal tube, including unknown remainders and one-sided support openings.

The width-cap consequence, for $1 \leq b \leq N$, is

$$(1.6) \quad D_N^{[b]}(E, F) \asymp_{E,H,\Lambda} \begin{cases} \min\{1, \sqrt{N}s\}, & H \in \text{Ran } \mathcal{C}_E, \\ \min\{1, \sqrt{Nb}s\}, & J_j = 0 \ (\forall j), \Gamma_E(H) \notin \mathcal{W}_E, \\ \min\{1, Ns\}, & J_j \neq 0 \ (\exists j). \end{cases}$$

The complete profile refines this cap through V , rather than through its largest group alone. The upper uses conditional fidelity, not unconditional tensorization after feedback. The lower first pays the full $m_r \kappa s^2$ block errors and then combines unequal signals. These distinctions are essential to uniformity in the unknown remainder.

The variational and equality statements also have quantitative forms. Corollary 15.6 decomposes a certified reward loss into nonnegative spectral, majorization, and leaf-decision defects. Lemma 17.4 and Corollary 17.5 control, at the relative scale ε/t^2 , the deviation of near-optimal

reset Gram matrices from scalar matrices. This concerns the selected normal form, not unique dilations or physical calibration.

1.4. A dimension-resolved reset hierarchy. The two receiver resources can be resolved separately rather than compared only at their classical and unrestricted endpoints. At the first recording cut let the retained old receiver have dimension at most k , and let the independently prepared second reference have dimension at most ℓ . The initial reference and temporary workspaces before that cut are not counted by these thresholds. Theorem 18.3 expresses the exact reward as the same common-barycenter optimization, with atom rank at most k and fresh Gram rank at most ℓ . Both primal and dual are attained. A dimension-preserving Kraus and mixture refinement, rather than a quantum dilation retained across the cut, proves the operational reduction. Thus every finite experiment has an exact monotone hierarchy whose endpoint is the full reset optimum.

For the fixed shared-basis family, this hierarchy is explicitly solved at every pair of thresholds. With $r = \min\{k, \ell\}$, Theorem 18.5 gives the equal-prior value

$$P_{k,\ell}^{\text{eq}} = \frac{1}{2} + \frac{t^2(dr - 1)}{2dr(d + 1)}.$$

The consecutive gains are $t^2/[2d(d + 1)r(r + 1)]$. They are strictly positive for $t > 0$, throughout the same full-rank deformation. Cyclic rank- r compression instruments attain them; matching upper bounds cover all mixed, randomized and history-dependent competitors in the specified class. A corresponding exact formula holds for the previous biased priors. The saturation is at the full reset value, not at the unrestricted coherent-acquisition value. This is a complete hierarchy for the two stated recording cuts and is not a classification of auxiliary workspaces at every time in a general multi-time process.

1.5. Optimal initial spectra and quantitative rigidity. The value hierarchy has a geometric equality set. On the same shared-basis family, let $r = \min\{k, \ell\}$ and fix the initial Gram matrix ρ . Theorem 19.1 shows that its rank- r optimum is attainable precisely when $\rho \preceq I/r$. Every such matrix is a convex combination of at most d commuting rank- r projections divided by r . A finite interval construction provides the recorded instrument, so the assertion is both necessary and sufficient. If the first reference is also bounded by q , the exact value depends on $\min\{q, k, \ell\}$ and is attained without feedback on a fixed subspace. This counts three specified cuts, not the maximum workspace at every time.

Away from the exceptional qubit rank-one face, the centered-swap deficit controls squared trace distance of both Gram factors to a common normalized rank- r projection. The resulting reward-loss estimate supplies an approximate projection decomposition of the initial Gram matrix and a spectral obstruction proportional to $t^2[\text{tr}(\rho - I/r)_+]^2$. The constants are explicit, and the relative scale is the score loss divided by t^2 . At the exceptional face the exact equality condition is commutation, and an exact pinching identity replaces common-projection rigidity. No uniqueness of instruments or uncalibrated device certification is inferred from these statements.

1.6. Exact values for a fixed initial acquisition. An equality set gives only a partial description of an optimization problem. Theorem 20.1 evaluates the biased basis experiment at every prescribed pure initial Gram ρ and every pair of retained dimensions. For $r = \min\{k, \ell\}$, write $q^{(r)}(\lambda(\rho))$ for its least rank- r spectral majorant, and let $\chi(u)$ be the positive root of $\sum_i u_i/(u_i + \chi) = 1$, with value zero on vectors of rank at most one. Then

$$P_{k,\ell}^{\text{b}}(\rho) = \frac{d + 1}{2(d + 1) - t^2} + \frac{t^2\chi(q^{(r)}(\lambda(\rho)))}{2[2(d + 1) - t^2]}.$$

A finite commuting orbit decomposition supplies an instrument on the actual initial Schmidt support. Lemma 20.4 evaluates the corresponding rank-restricted concave envelope for every

continuous symmetric concave spectral objective; the measurement payoff selects the particular secular function above.

The spectral formula also determines the value of a receiver message. Keep the full old receiver, bound the fresh reference by r , and forbid communication from measurements of the old receiver to the fresh source. The optimum replaces the majorant by the leading r eigenvalues. For $t > 0$, allowing such communication improves the value exactly when $2 \leq r < \text{rank } \rho$. Its gain is attained with at most d recorded outcomes and no dependence on the first device label. This fixed-initial-state distinction is not contradicted by globally optimal preparations needing no feedback: those preparations change ρ . The saturated dimension at a fixed initial Gram is exactly its rank. These statements use the biased priors explicitly; no unstated equal-prior spectral-value formula is imported.

1.7. Notation, organization and scope. For positive trace-class R, S , $f(R, S) = \|\sqrt{R}\sqrt{S}\|_1$ is root fidelity. For states, $d_B(R, S)^2 = 2(1 - f(R, S))$; the inherited symbol β for a state Bures distance denotes this same distance when used with two state arguments. Named coefficients such as β_h are bound coefficients, not an additional distance. $D(\pi; E, F)$ denotes one policy's separation; a resource subscript denotes a supremum. V/N is an effective allocation width, not an integer or memory dimension.

The proof proceeds from covariance and its support kernel to normalized curves, tangent neighborhoods, complete groups, conditional reset accumulation, and hard quadratic budgets. Section 16 then gives the exact finite hierarchy and its stability. Balanced-interior geometry, coding, learning and affine repair are retained with their full proofs in the current linked supplement. The complete research edition also retains every earlier mathematical section. The independent structural article is not a premise of either main result.

The fixed-tube constants may deteriorate at changing support floors, vanishing tangents or closing correction gaps. Neither local theorem is a uniform full-boundary metric, entropy or learning theorem. The finite hierarchy supplies a different, exact, all-dimension statement on its specified ensemble family. The historical analytic programme and independent specialist priority assessment are separate obligations. Our comparison with the memory-dimension hierarchy of Zonnios–Binder [52] and with accessible post-measurement states [53] keeps these interfaces and resources distinct.

2. A COVARIANCE PRINCIPLE ON THE CLOSED MEASUREMENT BODY

Let $\mathcal{P}_{d,k}$ denote the complete body of ordered k -outcome measurements on $\mathbb{C}^d$: its elements are Hermitian tuples $E = (E_1, \dots, E_k)$ with $E_j \succeq 0$ and $\sum_j E_j = I_d$. Here $d \geq 1$, $k \geq 2$, and $N \geq 1$ are integers. Zero effects, changing ranks, and noncommuting supports are allowed throughout this section. The operational distance D_N uses unhalved trace norm, retained references, common adaptive control, and at most N calls with bounded public stopping. The device consumes its input and returns only its ordered classical label.

The normalized tangent space and its inner product are

$$\mathcal{T}_{d,k} = \{H_j = H_j^* : \sum_j H_j = 0\}, \quad \langle H, K \rangle = \sum_j \text{tr}(H_j K_j).$$

All operator inequalities below on tuple spaces refer to this real Hilbert structure, unless stated otherwise. For any Hermitian tuple K , not necessarily of zero sum, put

$$(2.1) \quad S_E(K) = \sum_j E_j K_j,$$

$$(2.2) \quad (\mathcal{C}_E K)_j = \frac{1}{2}(E_j K_j + K_j E_j - E_j S_E(K) - S_E(K)^* E_j).$$

The matrix $S_E(K)$ need not be Hermitian. Keeping its adjoint in (2.2) is essential. The operator $\mathcal{C}_E$ is polynomial in the effects, annihilates constant tuples, and has range in $\mathcal{T}_{d,k}$. In particular its restriction to $\mathcal{T}_{d,k}$ is well defined.

For $\tau = N^{-1}$ set

$$(2.3) \quad g_{N,E}(H)^2 = N \langle H, (\mathcal{C}_E + \tau \text{Id})^{-1} H \rangle,$$

$$(2.4) \quad \mathcal{Q}_N(E, F) = g_{N,(E+F)/2}(F - E).$$

The inverse acts on $\mathcal{T}_{d,k}$. Positivity proved below makes it nonsingular even on the spectral boundary. We do not identify $\mathcal{Q}_N$ with an operational distance or assume a triangle inequality for it.

Theorem 2.1 (Closed-body adaptive bound). *For every $E, F \in \mathcal{P}_{d,k}$,*

$$(2.5) \quad D_N(E, F) \leq \min\{2, 2\sqrt{k+1} \mathcal{Q}_N(E, F)\}.$$

More generally, every continuously differentiable path $E(t)$ in $\mathcal{P}_{d,k}$ satisfies

$$(2.6) \quad D_N(E(0), E(1)) \leq \sqrt{k+1} \int_0^1 g_{N,E(t)}(E'(t)) dt.$$

These statements are uniform over all ranks and support patterns. For $k = 2$, $E = (A, I_d - A)$ and $H = (B, -B)$, the restriction is exactly

$$(2.7) \quad (\mathcal{C}_E H)_1 = A(I_d - A)B + BA(I_d - A), \quad \mathcal{Q}_N(E, F)^2 = 2Q_N(A, F_1)^2,$$

where Q_N is the binary comparison modulus in (S4.4).

Thus the binary variance operator has a normalized-tuple extension, not an independently regularized list of effects. The theorem gives a global upper certificate. The two-sided balanced-interior classification and its entropy theorem remain Theorems S22.1 and S22.2; a matching general lower bound for (2.4) is a separate question.

2.1. Positive covariance and horizontal decomposition.

Lemma 2.2 (Covariance identities). *On all Hermitian tuples, $\mathcal{C}_E$ is self-adjoint and positive semidefinite, and*

$$(2.8) \quad \begin{aligned} \langle K, \mathcal{C}_E K \rangle &= \sum_j \text{tr}(E_j K_j^2) - \|S_E(K)\|_{\text{HS}}^2 \\ &= \sum_j \|E_j^{1/2}(K_j - S_E(K))\|_{\text{HS}}^2. \end{aligned}$$

For $E_t = (1-t)E + tF$, $0 \leq t \leq 1$,

$$(2.9) \quad \langle K, [\mathcal{C}_{E_t} - (1-t)\mathcal{C}_E - t\mathcal{C}_F]K \rangle = t(1-t)\|S_F(K) - S_E(K)\|_{\text{HS}}^2.$$

The kernel consists precisely of tuples satisfying $E_j^{1/2}(K_j - S_E(K)) = 0$ for every j . Moreover $\mathcal{C}_E = 0$ on $\mathcal{T}_{d,k}$ if and only if the E_j are mutually orthogonal projections, with zero projections permitted.

Proof. Write $A_j = E_j^{1/2}$ and let A be the vertical stack of the A_j . Then $A^*A = I_d$, so $P = I - AA^*$ is an orthogonal projection. On complex matrix tuples with real Hilbert–Schmidt inner product, define

$$(T_A B)_j = A_j^* B_j + B_j^* A_j.$$

Its real adjoint is $(T_A^* K)_j = 2A_j K_j$. Direct multiplication gives

$$(2.10) \quad \mathcal{C}_E = \frac{1}{4} T_A P T_A^*.$$

This proves self-adjointness, positivity, and (2.8). It also proves the kernel claim. Constant tuples are killed and the displayed formula has zero sum, so no extra projection onto $\mathcal{T}_{d,k}$ is implicit.

In the first line of (2.8), the first term is linear in E and $S_E(K)$ is linear in E . Expanding its squared norm gives (2.9).

If the restriction of $\mathcal{C}_E$ to $\mathcal{T}_{d,k}$ vanishes, subtracting the arithmetic mean from any tuple shows that its form vanishes on all tuples. Take $K_j = I_d$ at one index and zero elsewhere. Its form is $\text{tr}(E_j - E_j^2) = 0$. Since $0 \preceq E_j \preceq I_d$, this forces $E_j^2 = E_j$. Their sum being I_d , they are mutually orthogonal: each positive term in $\sum_{i \neq j} E_j E_i E_j = 0$ must vanish. Conversely, for orthogonal projections the summands $E_j K_j$ have mutually orthogonal ranges, and $\|\sum_j E_j K_j\|_{\text{HS}}^2 = \sum_j \|E_j K_j\|_{\text{HS}}^2$. Their covariance form is zero. $\square$

Lemma 2.3 (Regularized horizontal energy). *For $H \in \mathcal{T}_{d,k}$ and $\tau > 0$,*

$$(2.11) \quad \langle H, (\mathcal{C}_E + \tau \text{Id})^{-1} H \rangle = \min_{\substack{A^* B = 0, R \in \mathcal{T}_{d,k} \\ T_A B + R = H}} \left(4\|B\|_{\text{HS}}^2 + \tau^{-1} \|R\|_{\text{HS}}^2 \right).$$

If $K = (\mathcal{C}_E + \tau \text{Id})^{-1} H$, a minimizer is

$$(2.12) \quad B_j = \tfrac{1}{2} A_j (K_j - S_E(K)), \quad R = \tau K.$$

Proof. The constraint $A^* B = 0$ means $PB = B$. Thus $B = PT_A^* K/4$ gives $T_A B = \mathcal{C}_E K$ and is feasible with $R = \tau K$. Its objective is $\langle K, \mathcal{C}_E K \rangle + \tau \|K\|^2 = \langle H, K \rangle$. For any feasible perturbation $(\Delta B, \Delta R)$, its linear objective term is

$$8\langle B, \Delta B \rangle + 2\tau^{-1} \langle R, \Delta R \rangle = 2\langle K, T_A \Delta B + \Delta R \rangle = 0.$$

The remaining terms are nonnegative. This proves minimality, including at singular E ; no inverse of an effect was used. $\square$

2.2. Adaptive differentiation and boundary passage.

Proof of Theorem 2.1. First suppose every effect at the point of differentiation is positive definite. The tangent $H_0 = T_A B$ in (2.12) has zero sum. For small real s , $E_j + s(H_0)_j$ is positive and the tuple remains normalized. If $R_j(s)$ is its positive square root, then

$$U_j = (B_j - R'_j(0)) A_j^{-1}$$

is anti-Hermitian. Indeed, multiplication of $U_j + U_j^*$ on the left and right by A_j gives $A_j B_j + B_j^* A_j - A_j R'_j(0) - R'_j(0) A_j = 0$. The factors $\exp(s U_j) R_j(s)$ therefore realize the derivative B_j while preserving the effects exactly.

Use the measurement isometry with an observed label and an inaccessible duplicate label, together with the factor environment. At the point under consideration it obeys

$$W^* \dot{W} = \sum_j A_j^* B_j = 0, \quad \|\dot{W}\|_{\text{op}} \leq \|B\|_{\text{HS}}.$$

Purify any fixed common N -slot tester. In the inner product of terms differentiated at distinct slots, cancel the common later isometries; the later differentiated slot contains $W^* \dot{W}$ or its adjoint and vanishes. Thus the observed output derivative in the direction H_0 has trace norm at most $2\sqrt{N} \|B\|_{\text{HS}}$. This holds for every reference and memory dimension. The environments used for comparison are not returned to the tester.

The output derivative of the same tester is linear in the channel tangent. For the residual tuple R , one call on a joint state ρ has reference blocks $\text{tr}_A[(R_j \otimes I)\rho]$. Duality against Hermitian

contractions bounds the trace norm of the j th block by $\|R_j\|_{\text{op}}$. Inserting the residual at each slot and using trace-norm contraction hence costs at most

$$N \sum_j \|R_j\|_{\text{op}} \leq N\sqrt{k}\|R\|_{\text{HS}}.$$

This is a differential decomposition, not a proposed physical mixture of measurements. With $\tau = N^{-1}$, Cauchy–Schwarz and (2.11) now give

$$\begin{aligned} 2\sqrt{N}\|B\|_{\text{HS}} + N\sqrt{k}\|R\|_{\text{HS}} &\leq \sqrt{k+1}\sqrt{4N\|B\|_{\text{HS}}^2 + N^2\|R\|_{\text{HS}}^2} \\ &= \sqrt{k+1} g_{N,E}(H). \end{aligned}$$

Integrate for the fixed tester, then take the supremum. Padding a publicly stopped experiment with discarded dummy calls makes its original stopped output a common postprocessing of an N -slot experiment, so the same bound applies.

For a boundary path, use $E^\epsilon(t) = (1 - \epsilon)E(t) + \epsilon U$, where $U_j = I_d/k$ and $0 < \epsilon < 1$. The covariance is polynomial and positive semidefinite, so its regularized inverse is continuous. Moreover $g_{N,E}(H) \leq N\|H\|_{\text{HS}}$, uniformly on the closed body. Dominated convergence applies to the integrals. The hybrid bound $D_N(E, F) \leq N \sum_j \|E_j - F_j\|_{\text{op}}$ gives convergence of the endpoint distances. This proves (2.6) on the entire body.

For the segment $E(t) = (1 - t)E + tF$, put $M = (E + F)/2$ and $H = F - E$. Concavity and positivity imply, for $0 < t \leq 1/2$,

$$\mathcal{C}_{E(t)} + \tau \text{Id} \succeq 2t(\mathcal{C}_M + \tau \text{Id}).$$

Inverse order bounds $g_{N,E(t)}(H)$ by $(2t)^{-1/2}g_{N,M}(H)$; use $2(1 - t)$ on the second half. The two scalar integrals sum to 2, proving (2.5). Finally, substituting $K = (B, -B)$ into (2.2) gives (2.7). The factor two in the quadratic form comes from the two equal Hilbert–Schmidt contributions. $\square$

2.3. Classical postprocessing.

Proposition 2.4 (Data processing of horizontal energy). *Let $T = (T_{aj})$ be a column-stochastic classical channel and $F_a = \sum_j T_{aj}E_j$. Write $(T^*L)_j = \sum_a T_{aj}L_a$. Then, on all Hermitian tuples,*

$$(2.13) \quad \langle L, \mathcal{C}_F L \rangle \geq \langle T^*L, \mathcal{C}_E(T^*L) \rangle.$$

Consequently the extended quadratic energy

$$\mathcal{I}_E(H) = \sup_K \{2\langle H, K \rangle - \langle K, \mathcal{C}_E K \rangle\}$$

satisfies $\mathcal{I}_F(TH) \leq \mathcal{I}_E(H)$. The value may be infinite at a boundary; when finite it is $\langle H, \mathcal{C}_E^\dagger H \rangle$.

Proof. The two S matrices coincide. The difference of the covariance forms is therefore

$$\sum_j \text{tr} \left[E_j \left(\sum_a T_{aj} L_a^2 - \left(\sum_a T_{aj} L_a \right)^2 \right) \right] \geq 0.$$

The bracket equals the positive sum $\frac{1}{2} \sum_{a,b} T_{aj} T_{bj} (L_a - L_b)^2$. Substitution in the variational definition and enlargement from $K = T^*L$ to all K prove the energy inequality. The pseudoinverse statement follows by diagonalizing the real self-adjoint covariance; a component along its kernel makes the supremum infinite. $\square$

2.4. A uniform noise crossover, sharp in horizon and noise.

Theorem 2.5 (Regularization of arbitrary support patterns). *For $0 \leq \epsilon \leq 1$ put $E^\epsilon = (1-\epsilon)E + \epsilon U$. On $\mathcal{T}_{d,k}$,*

$$(2.14) \quad \mathcal{C}_{E^\epsilon} \succeq (1-\epsilon)\mathcal{C}_E + \frac{\epsilon}{k} \text{Id}.$$

In particular, for arbitrary $E, F \in \mathcal{P}_{d,k}$,

$$(2.15) \quad D_N(E^\epsilon, F^\epsilon) \leq \min \left\{ 2, \frac{2\sqrt{k+1}(1-\epsilon)N}{\sqrt{1+N\epsilon/k}} \|F - E\|_{\text{HS}} \right\}.$$

The dependence on N and ϵ is attained, up to constants depending only on k , by a two-dimensional family. Namely, let $P = |0\rangle\langle 0|$, $P_\theta = |u_\theta\rangle\langle u_\theta|$, $u_\theta = \cos\theta|0\rangle + \sin\theta|1\rangle$, and

$$E = (P, I_2 - P, 0, \dots, 0), \quad F = (P_\theta, I_2 - P_\theta, 0, \dots, 0).$$

For $0 \leq \theta \leq \pi/2$, $0 \leq \epsilon \leq 1/2$, and $z = (1-\epsilon)N \sin\theta / \sqrt{1+N\epsilon}$,

$$(2.16) \quad \frac{\min\{1, z\}}{16384} \leq D_N^{\text{na}}(E^\epsilon, F^\epsilon) \leq D_N(E^\epsilon, F^\epsilon) \leq 4\sqrt{k(k+1)} \min\{1, z\}.$$

Proof. For $K \in \mathcal{T}_{d,k}$, $S_U(K) = 0$ and $\mathcal{C}_U K = K/k$. Concavity proves (2.14). Apply it at the midpoint and use Theorem 2.1 to obtain (2.15).

For the displayed family, keep labels one and two and send every other label to an independent fair bit. This common postprocessing produces the binary effects $B = \beta P + \epsilon I_2/2$ and $B_\theta = \beta P_\theta + \epsilon I_2/2$, where $\beta = 1 - \epsilon$. The equality holds on subnormalized reference states. It therefore preserves the nonadaptive lower witnesses as well as channel postprocessing. The midpoint eigenvalues are $(1 \pm \beta \cos\theta)/2$, and direct substitution in (S4.5) gives

$$Q_N(B, B_\theta)^2 = \frac{4N^2\beta^2 \sin^2\theta}{2 + N(1 - \beta^2 \cos^2\theta)} \geq \frac{4z^2}{2 + z^2}.$$

Here $1 - \beta^2 \leq 2\epsilon$ and $N \geq 1$ were used. It follows that $\min\{1, Q_N(B, B_\theta)\} \geq \min\{1, z\}$. The binary lower theorem in dimension two gives the first inequality of (2.16). For the upper, $\|F - E\|_{\text{HS}} = 2 \sin\theta$ and $(1 + N\epsilon)/(1 + N\epsilon/k) \leq k$. Combine (2.15) with $D_N \leq 2$. $\square$

The family passes from projections and zero effects at $\epsilon = 0$ to strictly positive effects at every $\epsilon > 0$. Thus a bound uniform over the closed body must accommodate both the N scale and the $\sqrt{N/\epsilon}$ scale. This conclusion does not assert a matching entropy law on all multi-outcome boundary strata.

2.5. Exact evaluation without effect square roots.

Proposition 2.6 (Polynomial-size rational certificate). *For Gaussian-rational $E, F \in \mathcal{P}_{d,k}$ and integer $N \geq 1$, $Q_N(E, F)^2$ is rational and can be evaluated by exact linear algebra on a positive definite real system of dimension $p = (k-1)d^2$. The bit complexity is polynomial in d, k , the total input bit length, and $\log N$. Legality is decidable by exact positive-semidefinite tests. No effect square root, ODE integration, rank decision by tolerance, or enumeration of a dictionary is required.*

Proof. Use the rational Hermitian matrix basis consisting of diagonal units and real and imaginary off-diagonal pairs. For each of the first $k-1$ slots, insert one matrix basis element there and its negative in the last slot. These tuples B_a form a rational basis of $\mathcal{T}_{d,k}$; they need not be orthonormal. Form

$$G_{ab} = \langle B_a, B_b \rangle, \quad L_{ab} = \langle B_a, \mathcal{C}_{(E+F)/2} B_b \rangle, \quad b_a = \langle B_a, F - E \rangle.$$

All entries are rational, G is positive definite, and $L \succeq 0$. Solve $(L + N^{-1}G)x = b$. Then $Q_N^2 = Nb^T x$. Omitting the Gram matrix would compute a different regularization. Fraction-free elimination performs polynomially many rational arithmetic operations; clearing input denominators and applying determinant bounds gives polynomial bit length for all minors and for the output. The same reasoning applies to exact symmetric elimination for each effect's positivity. The complexity statement concerns this certificate evaluator, not the exhaustive public code or collective-readout synthesis. $\square$

3. A FINITE PRODUCT ESTIMATE

All classical distances in this paper are unhalved ℓ^1 distances. The following elementary estimate is the Bernoulli-product input to Theorems 5.1 and S22.1. We include its proof so that the support and discrimination theorems can be checked without the binary supplement.

Lemma 3.1 (A uniform finite Bernoulli lower bound). *For $p, q \in [0, 1]$ and every integer $N \geq 1$,*

$$(3.1) \quad \|\text{Ber}(p)^{\otimes N} - \text{Ber}(q)^{\otimes N}\|_1 \geq \frac{1}{128} \min\{1, \sqrt{N}|p - q|\}.$$

In particular, if a differentiable two-sided probability curve satisfies $p'(0) \neq 0$, then, on a sufficiently small two-sided interval, its product experiment separates $p(0)$ from $p(t)$ by at least $c \min\{1, \sqrt{N}|t|\}$ for every N , with $c > 0$ depending on that curve. Necessarily $0 < p(0) < 1$.

Proof. Set $h = |p - q|$. First let $m \geq 1$ and $\sqrt{mh} \leq 1/4$. For $u = m^{-1/2}$ define $z(a) = 1 - a + ae^{iu}$, $0 \leq a \leq 1$. The argument of $z(a)$ has a continuous choice since its real part is positive. Direct differentiation and $0 < u \leq 1$ give

$$|z(a)|^2 \geq 1 - \frac{1}{4m}, \quad \frac{u}{2} \leq \frac{d}{da} \arg z(a) = \frac{\sin u}{|z(a)|^2} \leq \frac{4u}{3}.$$

Bernoulli's inequality gives $|z(a)|^m \geq \sqrt{3/4} > 3/4$. The arguments of $z(p)^m$ and $z(q)^m$ differ by an angle $\vartheta \in [\sqrt{mh}/2, 4\sqrt{mh}/3] \subset [0, 1/3]$. For two complex numbers of moduli at least $3/4$,

$$|z(p)^m - z(q)^m| \geq \frac{3}{2} \sin(\vartheta/2) \geq \frac{3}{2\pi} \vartheta \geq \frac{1}{16} \sqrt{mh}.$$

These complex numbers are the expectations of $\exp(iu \sum_{a=1}^m X_a)$ under the two Bernoulli product laws. This statistic has modulus one, so their ℓ^1 distance is at least its expectation difference.

If $h \leq 1/4$ and $\sqrt{N}h \leq 1/4$, use $m = N$. If $h \leq 1/4$ and $\sqrt{N}h > 1/4$, use $m = \lfloor (16h^2)^{-1} \rfloor \leq N$. Then $1/(4\sqrt{2}) \leq \sqrt{mh} \leq 1/4$, since $\lfloor x \rfloor \geq x/2$ for $x \geq 1$. Discarding the other observations is a common processing and gives a lower bound $1/(64\sqrt{2}) > 1/128$. For $h > 1/4$ the one-copy distance is $2h > 1/2$. The case $h = 0$ is immediate. Finally differentiability gives $|p(t) - p(0)| \geq |p'(0)||t|/2$ near zero. A two-sided probability curve cannot have a nonzero first derivative at either endpoint. $\square$

4. SUPPORT GEOMETRY OF THE COVARIANCE KERNEL

Let $\mathcal{H}_d$ be the real Hilbert space of Hermitian $d \times d$ matrices, with inner product $\text{tr}(AB)$. Throughout this section, $E = (E_1, \dots, E_k)$ is an arbitrary ordered measurement: $E_j \succeq 0$ and $\sum_j E_j = I_d$. Put

$$P_j = \text{supp } E_j, \quad Q_j = I_d - P_j, \quad r_j = \text{rank } E_j.$$

The support of a zero effect is the zero projection. The real subspaces

$$(4.1) \quad \mathcal{W}_E = \sum_{j=1}^k P_j \mathcal{H}_d P_j, \quad \mathcal{A}_E = \mathcal{W}_E^\perp = \{A \in \mathcal{H}_d : P_j A P_j = 0 \text{ for every } j\}$$

will be called the *support span* and its orthogonal complement. The sum in (4.1) is a sum of real linear subspaces, not an assertion that $\mathcal{W}_E$ is an algebra. In particular, for higher-rank effects it need not equal the span of the effects themselves. We use the zero-sum tuple space $\mathcal{T} = \{K \in \mathcal{H}_d^k : \sum_j K_j = 0\}$ with the sum Hilbert–Schmidt inner product, and the covariance of Lemma 2.2.

Theorem 4.1 (Complete support description of the kernel). *For $A \in \mathcal{A}_E$ and $Z_j \in Q_j \mathcal{H}_d Q_j$, set*

$$(4.2) \quad X_j = i[P_j, A] + Z_j, \quad K_j = X_j - \frac{1}{k} \sum_{\ell=1}^k X_\ell.$$

This is a real linear bijection from $\mathcal{A}_E \oplus \bigoplus_j Q_j \mathcal{H}_d Q_j$ onto $\ker(\mathcal{C}_E|_{\mathcal{T}})$. Consequently

$$(4.3) \quad \dim \ker(\mathcal{C}_E|_{\mathcal{T}}) = d^2 - \dim \mathcal{W}_E + \sum_j (d - r_j)^2.$$

In particular the kernel, and not just its dimension, depends only on the ordered support projections.

Proof. Write $S = \sum_j E_j K_j$. The nonnegative variance decomposition in Lemma 2.2 shows that K is in the kernel exactly when

$$(4.4) \quad P_j K_j = P_j S \quad \text{for every } j.$$

Indeed multiplication by $\sqrt{E_j}$ has precisely the same kernel as multiplication by P_j ; no inverse at a singular effect is used. Decompose $S = B + iA$, where A, B are Hermitian. Since $P_j K_j P_j$ and $P_j B P_j$ are Hermitian, compressing (4.4) gives $P_j A P_j = 0$. Furthermore $P_j(K_j - B) = iP_j A$, with the adjoint identity on the other side. These determine the two off-diagonal support blocks and the support block of $K_j - B$; its remaining block is arbitrary. Thus

$$K_j = B + i[P_j, A] + Z_j, \quad Z_j = Q_j Z_j Q_j = Z_j^*.$$

The condition $\sum_j K_j = 0$ forces $B = -k^{-1} \sum_j (i[P_j, A] + Z_j)$, proving necessity.

Conversely, let A, Z_j be as stated. Because $P_j A P_j = 0$, $E_j A P_j = 0$, and hence

$$\sum_j E_j X_j = i \sum_j E_j A = iA.$$

For the mean-subtracted tuple, therefore, $S = iA - \bar{X}$, where $\bar{X} = k^{-1} \sum_j X_j$ is Hermitian. We have $P_j X_j = iP_j A$, which proves (4.4). To check injectivity, suppose that all K_j vanish. Then all X_j equal a common Hermitian matrix D , so the last display gives $D = iA$. A matrix that is both Hermitian and anti-Hermitian is zero. Therefore $A = D = 0$ and all $Z_j = 0$. Finally $\dim(Q_j \mathcal{H}_d Q_j) = (d - r_j)^2$; rank-nullity in (4.1) proves (4.3). $\square$

Proposition 4.2 (Unitary directions and the support span). *Let $G \in \mathcal{H}_d$ and $H_j = i[G, E_j]$. Then*

$$(4.5) \quad H \in \text{Ran}(\mathcal{C}_E|_{\mathcal{T}}) \iff G \in \mathcal{W}_E.$$

More precisely, the pairing of H with the kernel tuple (4.2) is $-2 \text{tr}(GA)$.

Proof. We have $Q_j H_j Q_j = 0$, so the Z_j terms have zero pairing with H ; its zero tuple sum also removes $\bar{X}$. Trace cyclicity gives

$$\sum_j \text{tr}(H_j i[P_j, A]) = \text{tr} \left(A \sum_j i[H_j, P_j] \right).$$

Expanding in the displayed order yields

$$\sum_j i[H_j, P_j] = -2G + \sum_j (E_j G P_j + P_j G E_j).$$

Every summand in the last sum belongs to $P_j \mathcal{H}_d P_j$; it is therefore orthogonal to A . The pairing is $-2 \operatorname{tr}(GA)$ as asserted. A self-adjoint linear operator in finite dimension has range equal to the orthogonal complement of its kernel. Theorem 4.1 now proves (4.5). $\square$

Corollary 4.3 (Regularized tangent scales). *Let Π_0 be the orthogonal projection onto $\ker(\mathcal{C}_E|_{\mathcal{T}})$, and let u_ℓ be an orthonormal basis of its complement with $\mathcal{C}_E u_\ell = \lambda_\ell u_\ell$, $\lambda_\ell > 0$. For every $H \in \mathcal{T}$,*

$$(4.6) \quad g_{N,E}(H)^2 = N^2 \|\Pi_0 H\|_{\text{HS}}^2 + N \sum_{\ell} \frac{|\langle H, u_\ell \rangle|^2}{\lambda_\ell + N^{-1}}.$$

Thus a nonzero horizontal direction has order $\sqrt{N}$, whereas a direction with nonzero kernel component has order N , at a fixed measurement. For $H = i[G, E]$ the first alternative is exactly $G \in \mathcal{W}_E$ and $H \neq 0$.

Proof. Apply the finite-dimensional spectral theorem to $N(\mathcal{C}_E + N^{-1} \text{Id})^{-1}$ and use Proposition 4.2. $\square$

The corollary concerns a comparison form, not yet an operational lower bound. The distinction matters at the boundary; the next section proves the corresponding finite-pair statement on entire unitary orbits.

Proposition 4.4 (Exact represented-input decision). *On Gaussian-rational effects and a Gaussian-rational Hermitian generator, membership in $\mathcal{W}_E$, stationarity of the orbit, and the kernel dimension (4.3) are decidable by exact rational linear algebra in polynomial bit complexity. The same calculation gives the orthogonal projection of G onto $\mathcal{W}_E$.*

Proof. First use exact positive-semidefinite and normalization tests to validate the input. Select independent columns V_j of E_j by rational elimination. For a nonzero effect its support is

$$P_j = V_j(V_j^* V_j)^{-1} V_j^*;$$

for a zero effect use $P_j = 0$. Compress a rational Hermitian basis by all P_j and select a real-linearly independent subfamily $W_1, \dots, W_q$. The positive Gram system $\sum_b \operatorname{tr}(W_a W_b) c_b = \operatorname{tr}(W_a G)$ gives $\Pi_{\mathcal{W}_E} G = \sum_b c_b W_b$. This tests membership exactly. Commutators test stationarity. Formula (4.3) then gives the dimension. There are at most kd^2 candidate matrices and d^2 independent matrices. Clearing denominators and applying fraction-free elimination gives polynomial bit bounds by the determinant bounds used in Proposition 2.6. This is a decision about supports and a specified tangent; no optimum over testers or quantum-control synthesis is computed. $\square$

5. FINITE-ANGLE DISTINGUISHABILITY ON UNITARY ORBITS

Fix an ordered measurement E and a Hermitian generator G , and write

$$(5.1) \quad E_j(t) = e^{itG} E_j e^{-itG}, \quad H_j = i[G, E_j].$$

Both E and G are known in this pair-comparison problem. A tester must use the same controls under the two hypotheses E and $E(t)$; its design may depend on this known pair. It has access to the classical device labels and its own reference and memory, never to an unobserved measurement environment.

The Hamiltonian-not-in-Kraus-span criterion and its quantum-error-correction interpretation are established in quantum channel metrology [40, Theorems 1–3]. The support criterion below is its measurement-channel specialization. We prove a finite-angle, finite-use trace-distance statement directly, including an explicit accumulated error estimate. Thus no lower bound is inferred solely from an infinitesimal Fisher-information scaling.

Theorem 5.1 (Finite-use trichotomy on every fixed unitary orbit). *Let $d, N \geq 1$, $k \geq 2$, and let E and G be as in (5.1), with no rank or strict-positivity assumption. The following alternatives are exhaustive.*

- (1) *If $[G, E_j] = 0$ for every j , then $D_N(E, E(t)) = 0$ for all N and all t .*
- (2) *If $H \neq 0$ and $G \in \mathcal{W}_E$, there are constants $c, C, t_0 > 0$, depending on E, G , such that*

$$(5.2) \quad c \min\{1, \sqrt{N}|t|\} \leq D_N(E, E(t)) \leq C \min\{1, \sqrt{N}|t|\}, \quad |t| \leq t_0.$$

The lower bound is attained by a product-input, nonadaptive experiment.

- (3) *If $G \notin \mathcal{W}_E$, there are constants $c, C, t_0 > 0$, depending on E, G , such that*

$$(5.3) \quad c \min\{1, N|t|\} \leq D_N(E, E(t)) \leq C \min\{1, N|t|\}, \quad |t| \leq t_0.$$

The lower bound admits a common adaptive experiment carrying a logical qubit between calls, with a reference of dimension at most $2d$ at an individual call and ideal trusted controls.

The constants are not uniform as the measurement or generator changes. All upper bounds allow arbitrary retained reference and adaptive memory, and bounded public stopping. This theorem classifies a fixed unitary orbit; it does not assert two-sided equivalence of the midpoint covariance modulus on arbitrary pairs in the complete measurement body.

We first record the error-correction statement in the form needed for finite angles. Its exact-correction premise is the usual Knill–Laflamme condition [41]; a proof is included to specify which systems are available to the recovery.

Lemma 5.2 (A corrected logical channel and its quadratic error). *Let $\mathcal{M}$ be a channel on an input space $\mathbf{A}$, and let $J : \mathbb{C}^2 \rightarrow \mathbf{A} \otimes \mathbf{R}$ be an isometry. Suppose a channel $\mathcal{R}$ from the actual output of $\mathcal{M} \otimes \text{Id}_{\mathbf{R}}$ to $\mathbb{C}^2$ satisfies*

$$\mathcal{R} \circ (\mathcal{M} \otimes \text{Id}_{\mathbf{R}}) \circ \mathcal{J} = \text{Id}_2, \quad \mathcal{J}(X) = JXJ^*.$$

For a Hermitian G on $\mathbf{A}$, put $G_L = J^(G \otimes I)J$, $\mathcal{U}_t(X) = e^{-itG_L}Xe^{itG_L}$, and*

$$\Phi_t = \mathcal{R} \circ (\mathcal{M} \otimes \text{Id}_{\mathbf{R}}) \circ \text{Ad}(e^{-it(G \otimes I)}) \circ \mathcal{J}.$$

In unhalved diamond norm,

$$(5.4) \quad \|\Phi_t - \mathcal{U}_t\|_{\diamond} \leq 4t^2\|G\|_{\text{op}}^2, \quad \|\Phi_t^{\text{om}} - \mathcal{U}_t^{\text{om}}\|_{\diamond} \leq 4mt^2\|G\|_{\text{op}}^2.$$

Proof. Let $\mathcal{T} = \mathcal{R} \circ (\mathcal{M} \otimes \text{Id})$ and choose Kraus operators V_ℓ for $\mathcal{T}$. Since $\mathcal{T} \circ \mathcal{J}$ is the identity channel, whose Choi operator has rank one, $V_\ell J = c_\ell I_2$ with $\sum_\ell |c_\ell|^2 = 1$. Trace preservation on the entire input space gives

$$\sum_\ell c_\ell^* V_\ell = \sum_\ell (V_\ell J)^* V_\ell = J^* \sum_\ell V_\ell^* V_\ell = J^*.$$

It follows by differentiating that $\Phi_0 = \text{Id}_2$ and $\Phi'_0 = -i[G_L, \cdot]$. For any Hermitian B , the generator of $\text{Ad}(e^{-itB})$ has diamond norm at most $2\|B\|_{\text{op}}$. Its second derivative has diamond norm at most $4\|B\|_{\text{op}}^2$, since unitary channels have diamond norm one. Taylor's integral remainder and contractivity under the surrounding channels bound the remainder for Φ_t by $2t^2\|G\|_{\text{op}}^2$. The remainder for $\mathcal{U}_t$ has the same bound because $\|G_L\|_{\text{op}} \leq \|G\|_{\text{op}}$. Their zeroth and first derivatives agree, proving the first inequality. Telescoping the two m -fold compositions of channels proves the second. $\square$

Lemma 5.3 (A reference-preserving code from the support complement). *Suppose $G \notin \mathcal{W}_E$ and put*

$$(5.5) \quad A = G - \Pi_{\mathcal{W}_E} G, \quad s = \text{tr } A_+ = \text{tr } A_- > 0, \quad \rho_{\pm} = A_{\pm}/s.$$

There is a code $J : \mathbb{C}^2 \rightarrow \mathbb{C}^d \otimes \mathbb{R}$ with $\dim \mathbb{R} \leq 2d$, corrected exactly by a channel acting only on the classical label and $\mathbb{R}$ after a call to E . The logical generator $G_L = J^*(G \otimes I)J$ is diagonal with gap

$$(5.6) \quad \Delta = \text{tr}(G(\rho_+ - \rho_-)) = \|A\|_{\text{HS}}^2/s > 0.$$

Proof. The identity belongs to $\mathcal{W}_E$ because $\sum_j E_j = I$; thus A is traceless. Its nonzero positive and negative parts have equal trace. Also

$$P_j \rho_+ P_j = P_j \rho_- P_j \quad \text{for every } j.$$

Purify ρ_+ and ρ_- in $\mathbb{C}^d \otimes \mathbb{C}^d$ and attach orthogonal flags to the second factor. Map the two logical basis vectors to these flagged purifications. They are orthonormal, and the flags make all off-diagonal code matrix elements of $B \otimes I$ vanish. The two diagonal elements are equal for every $B \in P_j M_d(\mathbb{C}) P_j$ by the compression identity above.

Choose measurement Kraus operators $L_{j,a} = |j\rangle\langle a| \sqrt{E_j}$, $1 \leq a \leq d$. Products with different labels are zero; for one label their span is $P_j M_d(\mathbb{C}) P_j$. Hence, with $\mu = (j, a)$,

$$(5.7) \quad J^*(L_\mu^* L_\nu \otimes I)J = \lambda_{\mu\nu} I_2$$

for a positive matrix λ of trace one. Diagonalize λ by a unitary change of Kraus representation. If its positive eigenvalues are λ_b , the corresponding operators $(\tilde{L}_b \otimes I)J/\sqrt{\lambda_b}$ are isometries with mutually orthogonal ranges. A channel projects onto those ranges and applies their inverse isometries; on the orthogonal complement it prepares a fixed logical state. This is completely positive and trace preserving on the entire output space, and it corrects the code exactly. Its input space is just the actual classical-label register tensor $\mathbb{R}$. It does not contain the lost input or a dilation environment. Equivalently, since the actual state is label diagonal, the same channel may first read the label and then act on the retained reference.

The logical generator is diagonal by the flags, with entries $\text{tr}(G\rho_\pm)$. Orthogonality of A to $\Pi_{\mathcal{W}_E} G$ gives $\text{tr}(GA) = \text{tr}(A^2)$, proving (5.6). $\square$

Proof of Theorem 5.1. The first assertion follows directly from (5.1). For the second, Proposition 4.2 gives

$$h^2 = \langle H, \mathcal{C}_E^\dagger H \rangle < \infty.$$

Conjugating all tuple components by e^{itG} intertwines the covariances and their tangent directions, so this value is constant along the orbit. Inverse spectral comparison gives $g_{N,E(t)}(E'(t)) \leq \sqrt{N}h$. The complete-body path theorem therefore yields

$$(5.8) \quad D_N(E, E(t)) \leq \min\{2, \sqrt{k+1} h \sqrt{N}|t|\}.$$

There is an index j with $H_j \neq 0$. Choose a unit vector v with $\langle v, H_j v \rangle \neq 0$. The two Bernoulli parameters obtained by input v and the event “label j ” have gap at least $c_0|t|$ on a sufficiently small two-sided interval. The finite Bernoulli-product bound used in Theorem S22.1 gives $(1/128) \min\{1, c_0 \sqrt{N}|t|\}$. This proves (5.2), including all finite N .

For the third alternative, use Lemma 5.3. The equality $\mathcal{M}_{E(t)} = \mathcal{M}_E \circ \text{Ad}(e^{-itG})$ holds also with a retained reference. Encode, make one device call, recover, and repeat $m \leq N$ times. Under E the logical channel is the identity. Under $E(t)$ it is Φ_t^m from Lemma 5.2. Start in the equal superposition of the two logical basis states. The ideal logical unitary produces unhalved trace distance $2|\sin(mt\Delta/2)|$ from that state. Consequently, for every real t ,

$$(5.9) \quad D_N(E, E(t)) \geq \max_{1 \leq m \leq N} \left(2|\sin(mt\Delta/2)| - 4mt^2 \|G\|_{\text{op}}^2 \right)_+.$$

This bound concerns one specified common tester for each m , not an inaccessible choice of a different recovery under the two hypotheses.

For $t \neq 0$ choose

$$|t| \leq t_0 := \min \left\{ \frac{1}{2\Delta}, \frac{\Delta}{4\pi\|G\|_{\text{op}}^2} \right\}, \quad m = \min \left\{ N, \left\lfloor \frac{1}{|t|\Delta} \right\rfloor \right\}.$$

Then $x = m|t|\Delta \leq 1$ and $x \geq \frac{1}{2} \min\{1, N|t|\Delta\}$. Since $2\sin(x/2) \geq 2x/\pi$ and the error in (5.9) is at most x/π , we obtain

$$(5.10) \quad D_N(E, E(t)) \geq \frac{1}{2\pi} \min\{1, N|t|\Delta\}.$$

The elementary hybrid estimate for the input unitaries gives $D_N(E, E(t)) \leq \min\{2, 2N|t|\|G\|_{\text{op}}\}$, which proves (5.3). The zero-angle case is immediate. The protocol uses deterministic $m \leq N$ calls, hence belongs to the class allowing bounded public stopping. The upper estimates hold for that entire class. Finally if the first alternative held with $G \notin \mathcal{W}_E$, then $H = 0$ would contradict Proposition 4.2; the alternatives are disjoint. $\square$

Corollary 5.4 (Full-rank, rank-one and projective measurements). *If one effect is positive definite, every nonconstant unitary orbit has the square-root law (5.2). If all nonzero effects have rank one, every nonconstant orbit has that law precisely when the measurement is informationally complete, meaning that its effects span $\mathcal{H}_d$. If such a rank-one measurement is not informationally complete, some generator has the linear law (5.3). For a projection-valued measurement every orbit is either constant or has the linear law.*

Proof. A full-rank support contributes $\mathcal{H}_d$ to $\mathcal{W}_E$. For rank-one supports, $P_j \mathcal{H}_d P_j = \mathbb{R}E_j$, so that $\mathcal{W}_E$ is exactly the effect span. If this span is proper, a nonzero element of its orthogonal complement is a generator outside $\mathcal{W}_E$ and cannot commute with all effects by Proposition 4.2. For a projection-valued measurement $\mathcal{W}_E$ is the block-diagonal Hermitian space of the orthogonal supports, which is exactly the space of generators commuting with every effect. Apply Theorem 5.1. $\square$

Proposition 5.5 (Support spans under output processing). *Let $T_{aj} \geq 0$ satisfy $\sum_a T_{aj} = 1$, and put $F_a = \sum_j T_{aj} E_j$. Then $\mathcal{W}_E \subseteq \mathcal{W}_F$. Thus classical processing of the ordered output cannot turn a nonconstant square-root orbit into a linear orbit. A reversible splitting of labels preserves the support span and the classification.*

Proof. For positive matrices, the kernel of a positive weighted sum is the intersection of the kernels of its positively weighted summands. Thus the support of F_a contains P_j whenever $T_{aj} > 0$. Every column has such a row, giving $P_j \mathcal{H}_d P_j \subseteq \mathcal{W}_F$. This proves the inclusion. The orbit commutes with classical processing, so Theorem 5.1 applies with the same generator. A reverse stochastic map gives the opposite inclusion for a reversible splitting. $\square$

Remark 5.6 (What the lower construction does not assert). The code and recovery depend on the known pair through E, G . They do not constitute a common estimator of an unknown measurement. Their existence is proved in finite-dimensional quantum mechanics; polynomial gate synthesis, physical execution, and parameter-uniform bounds for approximate controls are not inferred. Formula (5.9) quantifies the finite-angle channel remainder under ideal controls. It is not a claim of perfect finite-use discrimination. The full-body entropy and growing- k learning questions are separate from this orbit classification.

6. FINITE-USE CLASSIFICATION OF SMOOTH BOUNDARY CURVES

The support criterion extends beyond unitary conjugation. Here $E(t) = (E_1(t), \dots, E_k(t))$, $|t| < a$, is a fixed two-sided C^2 curve of ordered measurements, with $E = E(0)$ and $H = E'(0) \neq 0$. The curve may change ranks at zero and need not have a differentiable exact Kraus representation there. All constants below belong to this fixed curve, not uniformly to the measurement body.

Fix $0 < a_0 < a$. Whenever a remainder below uses L , it means

$$(6.1) \quad L = \sup_{|u| \leq a_0} \sum_{j=1}^k \|E_j''(u)\|_{\text{op}} < \infty.$$

This is the supremum of the sum, on the stated fixed interval.

Lemma 6.1 (The support generator of a two-sided tangent). *Let $P_j = \text{supp } E_j$ and $Q_j = I - P_j$. Then $Q_j H_j Q_j = 0$. Conversely every Hermitian zero-sum tuple with these vanishing blocks is the derivative of a real-analytic two-sided measurement curve. Define the Hermitian matrix*

$$(6.2) \quad \Gamma_E(H) = \frac{i}{2} \sum_j [P_j, H_j].$$

On the real zero-sum tuple space,

$$(6.3) \quad H \in \text{Ran } \mathcal{C}_E \iff \Gamma_E(H) \in \mathcal{W}_E.$$

For a unitary tangent $H_j = i[G, E_j]$, $\Gamma_E(H) - G \in \mathcal{W}_E$. In particular every stationary unitary generator belongs to $\mathcal{W}_E$.

Proof. For $v \in \text{Ran } Q_j$, the nonnegative function $\langle v, E_j(t)v \rangle$ has a two-sided minimum at zero; its derivative vanishes. Polarization on that subspace gives $Q_j H_j Q_j = 0$. The sum of the H_j vanishes by normalization. Pairing with the complete kernel tuple in Theorem 4.1, both the mean and the missing-support blocks disappear, leaving

$$\langle H, K(A, Z) \rangle = \text{tr} \left(A \sum_j i[H_j, P_j] \right) = -2 \text{tr}(A \Gamma_E(H)).$$

Finite-dimensional self-adjointness proves (6.3). For a unitary tangent, expansion gives

$$\Gamma_E(H) = G - \frac{1}{2} \sum_j (E_j G P_j + P_j G E_j),$$

and the last sum belongs to $\mathcal{W}_E$. If the tangent is zero, this identity also proves the final assertion. For the converse realizability statement, the normalized-factor construction in the proof of Lemma 6.4 is real analytic near zero and has precisely the prescribed value and derivative. $\square$

Theorem 6.2 (Two-sided smooth-curve classification). *For the fixed C^2 curve above there are $c, C, t_0 > 0$ such that, for all integers $N \geq 1$ and $|t| \leq t_0$,*

$$(6.4) \quad c \min\{1, N^\alpha |t|\} \leq D_N(E, E(t)) \leq C \min\{1, N^\alpha |t|\}, \quad \alpha = \begin{cases} 1/2, & \Gamma_E(H) \in \mathcal{W}_E, \\ 1, & \Gamma_E(H) \notin \mathcal{W}_E. \end{cases}$$

The first lower bound uses repeated product inputs. The second has a known-pair adaptive realization with ideal encoding and recovery, a logical qubit, and a per-call external reference of dimension at most $2d$. The same controls are used under both hypotheses. Constants may degenerate with the tangent, nonzero support eigenvalues, curvature, or support-complement gap.

The exponents and the error-correction mechanism have the established channel-metrology antecedents [40, 41]. The content here is a direct finite-pair theorem from C^2 effect coordinates: the support formula decides its branch even at second-order rank openings, without assuming a smooth exact Kraus family or inferring a finite-angle lower from Fisher information. It is a theorem for each fixed nonzero first-order curve, not a uniform arbitrary-pair midpoint equivalence.

Lemma 6.3 (A horizontal curve with prescribed first derivative). *If $H \in \text{Ran } \mathcal{C}_E$, there are factors $A_j = \sqrt{E_j}$ and B_j satisfying*

$$A^*A = I, \quad A^*B = 0, \quad A_j^*B_j + B_j^*A_j = H_j,$$

where A, B denote the vertically stacked matrices. Put $b = \|B\|_{\text{op}}$ and $S = B^*B$. The factors

$$(6.5) \quad A(t) = (A + tB)(I + t^2S)^{-1/2}$$

define a normalized measurement $\tilde{E}_j(t) = A_j(t)^*A_j(t)$ with the same zeroth and first derivatives as $E(t)$, and

$$A(t)^*A'(t) = 0, \quad \|A'(t)\|_{\text{op}} \leq b, \quad \|A''(t)\|_{\text{op}} \leq 7b^2 \quad (|t|b \leq 1).$$

If $L = \sup_{|s| \leq a_0} \sum_j \|E_j''(s)\|_{\text{op}}$ on a closed subinterval of the given curve, then, for $|t| \leq a_0$ and $|t|b \leq 1$,

$$(6.6) \quad D_N(E, E(t)) \leq \min \left\{ 2, 2b\sqrt{N}|t| + \frac{1}{2}(L + 16b^2)Nt^2 \right\}.$$

Proof. The factorization $\mathcal{C}_E = T_A(I - AA^*)T_A^*/4$ in Lemma 2.2 gives a horizontal solution: for $K = \mathcal{C}_E^\dagger H$ take $B = (I - AA^*)T_A^*K/4$. Thus $T_AB = H$ and $A^*B = 0$. In fact $4\|B\|_{\text{HS}}^2 = \langle H, \mathcal{C}_E^\dagger H \rangle$. Formula (6.5) is normalized because $(A + tB)^*(A + tB) = I + t^2S$. Its derivative is

$$A'(t) = (B - tAS)(I + t^2S)^{-3/2}, \quad A'(t)^*A'(t) = S(I + t^2S)^{-2}.$$

These identities prove horizontality and the first derivative bound. Differentiation once more gives, in the displayed order,

$$A''(t) = -AS(I + t^2S)^{-3/2} - 3t(B - tAS)S(I + t^2S)^{-5/2}.$$

Since $\|A\| = 1$, $\|S\| = b^2$, and $|t|b \leq 1$, its norm is at most $b^2 + 3|t|(b + |t|b^2)b^2 \leq 7b^2$.

Duplicate each output label into an inaccessible environment to form a Stinespring isometry $W(t)$ from $A(t)$. It has exactly the same operator derivative bounds and $W(t)^*W'(t) = 0$. For one fixed purified adaptive tester, differentiating its final vector gives a sum over slots. Distinct terms are orthogonal: after cancelling later common isometries, the later differentiated slot contains W^*W' or its adjoint. Each diagonal term has squared norm at most b^2 . The final vector derivative is at most $\sqrt{N}b$, and the density derivative at most $2\sqrt{N}b$. Integration followed by common partial trace proves $D_N(E, \tilde{E}(t)) \leq 2b\sqrt{N}|t|$. Padding stopped branches by discarded fixed-input calls includes bounded public stopping; the original stopped output is a common postprocessing of the padded experiment.

In the following application $\sum_j R_j = 0$. Thus the map is a Hermiticity-preserving, trace-annihilating channel difference or derivative, not a positive channel. Lemma 8.1 also proves its bound on arbitrary amplified trace-class inputs by adjoint duality, without a state-optimization assumption.

For a Hermitian tuple R , the measurement derivative has diamond norm at most $\sum_j \|R_j\|_{\text{op}}$. Indeed each reference output block is the partial trace against R_j ; its trace norm on a state is at most $\|R_j\|_{\text{op}}$ by duality. The Hermiticity-preserving diamond norm may be optimized over states with a reference. The Stinespring second derivative bound similarly gives $\|\mathcal{M}_{E(t)}''\|_{\diamond} \leq 2\|W''(t)\| + 2\|W'(t)\|^2 \leq 16b^2$. Taylor's integral formula, with common value and derivative at zero, therefore bounds the one-call channel difference by $(L + 16b^2)t^2/2$. Hybrid telescoping over N calls proves (6.6). $\square$

Lemma 6.4 (A logical derivative for a general measurement curve). *Suppose $\Gamma = \Gamma_E(H) \notin \mathcal{W}_E$. Use the support-complement code of Lemma 5.3 with Γ in place of G , and let $\mathcal{R}$ be its fixed CPTP recovery. Put*

$$\Phi_t = \mathcal{R} \circ (\mathcal{M}_{E(t)} \otimes \text{Id}) \circ \mathcal{J}, \quad \Gamma_L = J^*(\Gamma \otimes I)J.$$

Here L is exactly the local curvature quantity $\sup_{|u| \leq a_0} \sum_j \|E_j''(u)\|_{\text{op}}$ from (6.1); the remainder does not infer it from E, H . Then $\Phi_0 = \text{Id}_2$, $\Phi'_0 = -i[\Gamma_L, \cdot]$, and, for $|t| \leq a_0$,

$$(6.7) \quad \|\Phi_t - \text{Ad}(e^{-it\Gamma_L})\|_{\diamond} \leq \kappa t^2, \quad \kappa = \frac{1}{2}L + 2\|\Gamma\|_{\text{op}}^2.$$

Its logical gap is $\Delta = \|Z\|_{\text{HS}}^2 / \text{tr } Z_+ > 0$, where $Z = \Gamma - \Pi_{\mathcal{W}_E} \Gamma$.

Proof. To compute the derivative, not to factor the unknown curve itself, put

$$A_j = \sqrt{E_j}, \quad B_j = (\sqrt{E_j})^+ H_j (I - P_j/2).$$

The inverse here is only on the fixed support; it is zero on its complement. Since $Q_j H_j Q_j = 0$, $A_j^* B_j + B_j^* A_j = H_j$ and

$$\sum_j A_j^* B_j = \frac{1}{2} \sum_j [P_j, H_j] = -i\Gamma.$$

In particular this sum is anti-Hermitian. The normalized factors $(A + tB)(I + t^2 B^* B)^{-1/2}$ have derivative B and induce the same channel value and derivative as $E(t)$, irrespective of its second-order rank changes.

Write $L_\mu = |j\rangle\langle a| A_j$ and $\dot{L}_\mu = |j\rangle\langle a| B_j$ for $\mu = (j, a)$, with identity on the reference understood. If R_ℓ are Kraus operators of the completed recovery, exact correction gives $R_\ell L_\mu J = c_{\ell\mu} I_2$. The left first-order term of the logical channel is therefore

$$\sum_{\ell, \mu} \bar{c}_{\ell\mu} R_\ell \dot{L}_\mu J = J^* \sum_{\mu} L_\mu^* \dot{L}_\mu J = -i\Gamma_L,$$

where completeness of the recovery is used in the first equality. The adjoint term proves the commutator derivative. The actual logical second derivative is at most L in diamond norm, and that of the logical unitary is at most $4\|\Gamma\|_{\text{op}}^2$. Taylor's integral remainder proves (6.7). The support-code lemma gives the gap and the available systems. $\square$

Proof of Theorem 6.2. If H is in the covariance range, let $x = \sqrt{N}|t|$. For $x \leq 1$, (6.6) is at most $[2b + (L + 16b^2)/2]x$; for $x \geq 1$ use the bound two. For the lower, choose j, v with $\langle v, H_j v \rangle \neq 0$. The fixed input v and the event “label j ” have a first-order probability gap, and Lemma 3.1 proves the result.

If H is outside the range, apply Lemma 6.4. Starting from the equal logical superposition and telescoping m corrected calls gives, for $|t| \leq a_0$,

$$(6.8) \quad D_N(E, E(t)) \geq \max_{1 \leq m \leq N} (2|\sin(mt\Delta/2)| - m\kappa t^2)_+.$$

Choose

$$t_0 \leq \min\{a_0, (2\Delta)^{-1}, \Delta/(2\pi\kappa)\}, \quad m = \min\{N, \lfloor (|t|\Delta)^{-1} \rfloor\}.$$

For $0 < |t| \leq t_0$ the floor is at least two, and $x = m|t|\Delta$ satisfies $\frac{1}{2} \min\{1, N|t|\Delta\} \leq x \leq 1$. The sine term is at least $2x/\pi$ and the error at most $x/(2\pi)$. This gives a positive constant times $\min\{1, N|t|\}$. For the upper, the finite constant $L_1 = \sup_{|s| \leq a_0} \sum_j \|E_j'(s)\|_{\text{op}}$ and hybrid telescoping give $D_N \leq \min\{2, NL_1|t|\}$. The point $t = 0$ is immediate. $\square$

Corollary 6.5 (Rank opening without a unitary orbit). *Let $P = |0\rangle\langle 0|$ on $\mathbb{C}^2$, $G = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $U_t = e^{itG}$, and for $|t| < 1/2$ set*

$$E_1(t) = (1 - t^2)U_t P U_t^* + \frac{1}{2}t^2 I_2, \quad E_2(t) = I_2 - E_1(t).$$

Both effects are full rank for $t \neq 0$ and projective at zero. Nevertheless $D_N(E(0), E(t)) \asymp \min\{1, N|t|\}$ on a fixed small interval, uniformly over $N \geq 1$.

Proof. At zero the derivative is $i[G, (P, I - P)] \neq 0$ and the support span is diagonal, while G is not. Lemma 6.1 and Theorem 6.2 apply. The eigenvalues of the first effect are $t^2/2$ and $1 - t^2/2$, proving the rank assertion. $\square$

A vanishing first derivative is intentionally excluded from the classification: it does not imply stationarity of a general curve. For example the scalar curve $(t^2, 1 - t^2)$ has distance $2[1 - (1 - t^2)^N]$ from $(0, 1)$, of order $\min\{1, Nt^2\}$. Likewise a one-sided rank opening can have a nonzero missing-support first-order block and is not covered by the two-sided tangent lemma.

7. CERTIFIED ERRORS IN THE DISCRIMINATION CONTROLS

The preceding lower bounds first concern ideal known-pair controls. Their stability has a direct finite formulation. An error bound on a control channel is a bound in unhalved diamond norm, valid on every input including its memory reference. It is not a bound measured only on one calibration state.

Theorem 7.1 (A finite control-error ledger). *Use the code and recovery from Lemma 6.4, with gap Δ and remainder constant κ . In this statement $L = \sup_{|u| \leq a_0} \sum_j \|E_j''(u)\|_{\text{op}}$ and $\kappa = L/2 + 2\|\Gamma_E(H)\|_{\text{op}}^2$, as in (6.1) and (6.7). Suppose an implemented initial logical state differs from the ideal one in trace norm by at most e_0 . At cycle r , suppose its implemented logical channel differs from the corresponding ideal corrected channel by at most ν_r under each hypothesis. Suppose the implemented final readout differs from the ideal Helstrom readout by at most e_f in diamond norm. For every integer $m \leq N$ and $|t| \leq a_0$, the actual classical output laws obey*

$$(7.1) \quad \|\tilde{p}_{0,m} - \tilde{p}_{t,m}\|_1 \geq \left(2|\sin(mt\Delta/2)| - m\kappa t^2 - 2e_0 - 2e_f - 2 \sum_{r=1}^m \nu_r \right)_+.$$

Here one may choose the Helstrom readout for the ideal unrotated and logical-unitary states; it depends on the known pair and on m . The cycle error may be bounded by the sum of the encoding and recovery errors, because the unknown measurement itself is a channel.

In particular, put $z = \min\{1, N|t|\Delta\}$ and choose m, t_0 as in the proof of Theorem 6.2. For $0 < |t| \leq t_0$, the sufficient conditions

$$(7.2) \quad e_0 + e_f \leq z/(16\pi), \quad \nu_r \leq |t|\Delta/(8\pi) \quad (1 \leq r \leq m)$$

imply $\|\tilde{p}_{0,m} - \tilde{p}_{t,m}\|_1 \geq z/(2\pi)$. All errors are deterministic uniform channel bounds; no independence of control errors is required.

Proof. Choose the ideal two-outcome Helstrom readout for the equal logical superposition and its image under $e^{-imt\Gamma_L}$. Its classical ℓ^1 separation is $2|\sin(mt\Delta/2)|$. Replacing the logical unitary by the actual ideal corrected channel loses at most $m\kappa t^2$, by channel telescoping and Lemma 6.4. Under either hypothesis, telescoping the implemented preparation, cycles and readout changes its output law in ℓ^1 by at most $e_0 + e_f + \sum_r \nu_r$. The triangle inequality proves (7.1). Replacing encoding and recovery around one unchanged unknown channel costs no more than the sum of their diamond errors, which proves the stated decomposition of ν_r .

For the indicated choice set $x = m|t|\Delta$, so $z/2 \leq x \leq 1$. The ideal sine term minus curvature is at least $3x/(2\pi)$. Conditions (7.2) give $e_0 + e_f \leq x/(8\pi)$ and $\sum_r \nu_r \leq x/(8\pi)$. The total loss is therefore at most $x/(2\pi)$, leaving $x/\pi \geq z/(2\pi)$. $\square$

For a unitary orbit, Lemma 5.2 gives the same result with $\kappa = 4\|G\|_{\text{op}}^2$ and the original support-code gap. An error allowance fixed independently of the angle need not meet (7.2) as $t \rightarrow 0$. The theorem does not certify a particular piece of hardware, synthesize a gate circuit, or allow target-dependent advice in common learning. It specifies exactly which certified approximation errors suffice to preserve the finite-use lower bound.

Proposition 7.2 (Exact first-order decisions). *For Gaussian-rational E and a Gaussian-rational Hermitian zero-sum H , two-sided first-order realizability and the branch of Theorem 6.2 for $H \neq 0$ are decidable in polynomial represented-input bit complexity. The decision gives $\Gamma_E(H)$, its projection on $\mathcal{W}_E$, and its orthogonal residual. A zero tangent is reported as a higher-order question, not as stationarity of an unspecified curve.*

Proof. The exact support projectors in Proposition 4.4 are rational. Test $Q_j H_j Q_j = 0$ for every j and compute (6.2). Necessity and sufficiency of the test follow from Lemma 6.1 and the normalized-factor construction in Lemma 6.4: for any such tangent it gives a real-analytic measurement curve with that derivative. Real coordinate rank selection and the Hilbert–Schmidt Gram system then project $\Gamma_E(H)$ exactly. Rational matrix arithmetic and fraction-free elimination have the bit bounds already proved in Proposition 4.4. Only these finite operations are performed. A numerical curvature bound for an unspecified curve, an optimal tester, and a compiled recovery are not outputs of this decision. $\square$

8. ONE-SIDED TANGENT NEIGHBORHOODS

The preceding curve results admit a finite-pair formulation. The base measurement and its direction are fixed, but the second-order perturbation may vary arbitrarily within a prescribed bound. This also permits first-order opening of a missing support. Write $\|R\|_\Sigma = \sum_j \|R_j\|_{\text{op}}$ for a Hermitian tuple. All trace and diamond norms in this section are unhalved.

Lemma 8.1 (A reference-uniform channel bound). *For a Hermitian tuple R , let $\mathcal{D}_R(X) = \sum_j \text{tr}(R_j X) |j\rangle\langle j|$. Then*

$$(8.1) \quad \|\mathcal{D}_R\|_\diamond \leq \|R\|_\Sigma.$$

The map is Hermiticity preserving, and is trace annihilating when $\sum_j R_j = 0$. In particular this applies to measurement-channel differences, derivatives, and Taylor remainders.

Proof. At every reference dimension the output is block diagonal. In trace norm duality it therefore suffices to use a block-diagonal test $Y = \bigoplus_j Y_j$, with $\|Y_j\|_{\text{op}} \leq 1$. Its adjoint image is $\sum_j R_j \otimes Y_j$, of norm at most $\sum_j \|R_j\|_{\text{op}}$. Duality bounds the amplified map on *every* trace-class input X , not only on states. Taking the supremum over reference dimensions proves (8.1). The remaining assertions follow by taking adjoints and traces. $\square$

Lemma 8.2 (The one-sided cone and its lineality). *Fix a measurement E , put $P_j = \text{supp } E_j$ and $Q_j = I - P_j$, and let H be a Hermitian zero-sum tuple. It is the right derivative at zero of a one-sided real-analytic measurement curve if and only if*

$$(8.2) \quad J_j := Q_j H_j Q_j \succeq 0 \quad \text{for every } j.$$

The largest vector subspace of this cone is given by $J_j = 0$ for all j . For every Hermitian zero-sum tuple, not necessarily in the cone,

$$(8.3) \quad H \in \text{Ran } \mathcal{C}_E \iff J_j = 0 \ (1 \leq j \leq k), \quad \Gamma_E(H) \in \mathcal{W}_E.$$

Here $\Gamma_E(H) = \frac{i}{2} \sum_j [P_j, H_j]$ and $\mathcal{W}_E = \sum_j P_j \mathcal{H}_d P_j$ are real Hermitian objects.

Proof. Right differentiation of $\langle v, E_j(t)v \rangle \geq 0$ for $v \in \text{Ran } Q_j$ proves necessity. For sufficiency, for each nonzero support let $\lambda_j > 0$ be the smallest eigenvalue of E_j on that support. Choose

$$c \geq 1 + \max_{j: P_j \neq 0} 2\|H_j\|_{\text{op}}^2 / \lambda_j.$$

For all sufficiently small $s \geq 0$, the P_j block of $E_j + sH_j + cs^2 I$ is at least $\lambda_j P_j / 2$. Its Schur complement on Q_j is bounded below by

$$sJ_j + s^2 \left(c - 2\|H_j\|_{\text{op}}^2 / \lambda_j \right) Q_j \succeq 0.$$

If $P_j = 0$, condition (8.2) is $H_j \succeq 0$ and positivity is immediate. Thus the rational matrix curve

$$(8.4) \quad E_j^c(s) = \frac{E_j + sH_j + cs^2I}{1 + kcs^2}$$

is positive, sums exactly to I , and has right derivative H_j . This proves sufficiency, including singular J_j with nonzero cross blocks. The cone and its negative intersect precisely at $J_j = 0$.

Pair H with the complete kernel in Theorem 4.1. Taking only its independent missing-support parameters Z_j shows that orthogonality to the kernel forces $Q_j H_j Q_j = 0$; the tuple-mean term vanishes because $\sum_j H_j = 0$. Once those blocks vanish, the remaining pairing is $-2 \operatorname{tr}(A \Gamma_E(H))$ for every $A \in \mathcal{W}_E^\perp$. Self-adjointness of $\mathcal{C}_E$ proves (8.3). $\square$

Lemma 8.3 (A finite normalized-factor remainder). *Let $A_j^* A_j = E_j$, $A^* A = I$, and $A_j^* B_j + B_j^* A_j = H_j$, with $\sum_j H_j = 0$. Set $b = \|B\|_{\text{op}}$, $h = \max_j \|H_j\|_{\text{op}}$ and*

$$R_s = (I + s^2 B^* B)^{-1/2}, \quad G_j(s) = R_s (A_j + s B_j)^* (A_j + s B_j) R_s.$$

This defines an analytic measurement near zero, and, for $0 \leq s \leq 1$,

$$(8.5) \quad \|G(s) - E - sH\|_\Sigma \leq K_b s^2, \quad K_b = kb^2(2 + h).$$

When $sb \leq 1$, its stacked factors differ from A in operator norm by at most $3bs/2$. If $A^ B = 0$, then also*

$$(8.6) \quad D_N(E, G(s)) \leq \min\{2, 2b\sqrt{N}s\}.$$

Proof. The zero tuple sum gives $A^* B + B^* A = 0$, so the stacked normalization is $I + s^2 B^* B$. This matrix is positive definite for all real s ; its inverse square root is analytic near zero. Functional calculus gives $\|R_s\| \leq 1$ and $\|R_s - I\| \leq s^2 b^2/2$. Expanding the effect as $R_s(E_j + sH_j + s^2 B_j^* B_j)R_s$ bounds its remainder by $s^2 b^2(1 + sh) + s^2 \|B_j\|^2$. Sum over j , use $\sum_j \|B_j\|^2 \leq kb^2$, and obtain (8.5). The factor difference is at most $sb + s^2 b^2/2$. For horizontal B , the ordered identities in Lemma 6.3 give $A(s)^* A'(s) = 0$ and $\|A'(s)\| \leq b$. Copying the classical label into a discarded environment is an isometric embedding of the stacked factors, so it preserves these derivative norms. The orthogonal-slot argument of that lemma gives (8.6), including bounded public stopping. The environment is not supplied to the tester. $\square$

Definition 8.4 (A quadratic tangent neighborhood). Fix E , a Hermitian zero-sum $H \neq 0$ satisfying (8.2), and a number $\Lambda \geq 0$. For $s > 0$, the neighborhood consists of all legal measurements F with

$$(8.7) \quad \|F - E - sH\|_\Sigma \leq \Lambda s^2.$$

The number Λ is a bound on a *finite remainder*, not an inferred curvature bound. No curve, or differentiability of F as a function of s , is part of this definition.

Theorem 8.5 (Finite-pair classification in a fixed tangent neighborhood). *For each fixed E, H, Λ in Definition 8.4 there are $c, C, s_0 > 0$ such that, for every integer $N \geq 1$, every $0 < s \leq s_0$, and every legal F satisfying (8.7),*

$$(8.8) \quad c \min\{1, N^\alpha s\} \leq D_N(E, F) \leq C \min\{1, N^\alpha s\}, \quad \alpha = \begin{cases} 1/2, & H \in \operatorname{Ran} \mathcal{C}_E, \\ 1, & H \notin \operatorname{Ran} \mathcal{C}_E. \end{cases}$$

The constants are uniform over the stated remainder neighborhood, but not over E, H or Λ . If some $J_j \neq 0$, the linear lower is attained by repeated product inputs and a classical event. If all $J_j = 0$ but $\Gamma_E(H) \notin \mathcal{W}_E$, it is attained by the known-direction correction code on the actual label and a retained reference, with ideal controls. Neither construction is a common unknown-measurement learner.

There are two different reasons for a linear scale. First-order support opening creates an event impossible at the base measurement. Within the lineality space of the cone, coherent correction produces the established Kraus-span metrological mechanism [40, 41]. The result above makes this distinction in support coordinates and controls finite remainders uniformly at a fixed base and direction. It is not a uniform arbitrary-pair equivalence for the midpoint covariance certificate.

Proof. Put $h = \max_j \|H_j\|_{\text{op}} > 0$ and $h_1 = \|H\|_{\Sigma}$. Choose a unit eigenvector of one H_j with expectation of magnitude h . On that fixed input the event “label j ” has probability gap at least $hs - \Lambda s^2$. For s sufficiently small this is at least $hs/2$. Lemma 3.1 therefore gives a uniform product lower of order $\min\{1, \sqrt{N}s\}$, for every F in the neighborhood. All pairs also satisfy the elementary upper

$$(8.9) \quad D_N(E, F) \leq \min\{2, N(h_1 s + \Lambda s^2)\},$$

by Lemma 8.1 and channel telescoping.

Suppose first that $H \in \text{Ran } \mathcal{C}_E$. The horizontal solution of Lemma 6.3 provides factors B with $A^*B = 0$ and $T_A B = H$. Lemma 8.3 and (8.7) give

$$(8.10) \quad D_N(E, F) \leq \min\{2, 2b\sqrt{N}s + (\Lambda + K_b)Ns^2\}.$$

For $x = \sqrt{N}s \leq 1$, the right side is at most $(2b + \Lambda + K_b)x$; for $x \geq 1$ it is at most two. This proves the upper in the first branch without discarding its finite Ns^2 remainder before comparison.

Next suppose some $J_j \neq 0$. It is positive semidefinite; choose a unit vector $v \in \text{Ran } Q_j$ with $\lambda = \langle v, J_j v \rangle > 0$. The base probability of label j is zero, while that under F is at least $\lambda s - \Lambda s^2 \geq \lambda s/2$ after decreasing s_0 . Repeating v gives the exact impossible-event bound

$$(8.11) \quad D_N(E, F) \geq 2 \left[1 - (1 - \lambda s/2)^N \right] \geq 2(1 - e^{-1}) \min\{1, N\lambda s/2\}.$$

Here s_0 is chosen also so that $\lambda s/2 \leq 1$. Together with (8.9) this proves the linear law. The vector and the decision event are the same for every allowed F .

It remains to consider $J_j = 0$ for all j and $\Gamma := \Gamma_E(H) \notin \mathcal{W}_E$. Use the completed recovery and logical code of Lemma 5.3, applied to Γ . The normalized first-order factors in Lemma 6.4 show algebraically that the corrected derivative $\mathcal{D}_H$ is $-i[\Gamma_L, \cdot]$. This calculation needs only E, H , not a smooth factorization of F . For $\Phi_F = \mathcal{R} \circ (\mathcal{M}_F \otimes \text{Id}) \circ \mathcal{J}$, (8.7) gives

$$\|\Phi_F - \text{Id}_2 + is[\Gamma_L, \cdot]\|_{\diamond} \leq \Lambda s^2.$$

The integral Taylor remainder for the logical unitary is at most $2\|\Gamma\|_{\text{op}}^2 s^2$. Consequently

$$(8.12) \quad \|\Phi_F - \text{Ad}(e^{-is\Gamma_L})\|_{\diamond} \leq \kappa_{\text{tube}} s^2, \quad \kappa_{\text{tube}} = \Lambda + 2\|\Gamma\|_{\text{op}}^2.$$

The gap is $\Delta = \|Z\|_{\text{HS}}^2 / \text{tr } Z_+ > 0$, where $Z = \Gamma - \Pi_{\mathcal{W}_E} \Gamma$. For every $m \leq N$, equal logical amplitudes and channel telescoping give a lower bound $(2|\sin(ms\Delta/2)| - m\kappa_{\text{tube}}s^2)_+$. Choose

$$s_0 \leq \min\{1, (2\Delta)^{-1}, \Delta/(2\pi\kappa_{\text{tube}})\}, \quad m = \min\{N, \lfloor (s\Delta)^{-1} \rfloor\}.$$

The floor is at least two, and $\frac{1}{2} \min\{1, Ns\Delta\} \leq x = ms\Delta \leq 1$. The sine term is at least $2x/\pi$ and the loss at most $x/(2\pi)$. In particular the separation is at least $\min\{1, Ns\Delta\}/(2\pi)$. The readout may be fixed to the ideal logical pair, so this lower is uniform over all the permitted F as well. Its code uses an external reference of dimension at most $2d$ per call. Finally (8.9) completes the proof. $\square$

Corollary 8.6 (All nonzero one-sided first jets). *Let $E(t)$, $0 \leq t < a$, be a fixed C^2 measurement curve, including its right derivatives at zero, with $H = E'(0) \neq 0$. Then (8.8) holds with $s = t$ and curve-dependent constants on a fixed right interval. The branch is decided by (8.3). Thus first-order support openings, as well as tangential second-order rank changes, are included.*

Proof. One-sided positivity gives (8.2). On $[0, a_0]$ set $L_+ = \sup_{0 \leq u \leq a_0} \sum_j \|E_j''(u)\|_{\text{op}}$. Taylor's formula gives (8.7) with $\Lambda = L_+/2$. Apply Theorem 8.5. This does not extend the two-sided tangent lemma by dropping one of its hypotheses; it uses the larger cone of Lemma 8.2. $\square$

Corollary 8.7 (Certified controls on the whole neighborhood). *In the tangential linear branch, Theorem 7.1 holds uniformly over (8.7) with $t = s$ and $\kappa = \kappa_{\text{tube}}$ from (8.12). In particular the same initial, cycle and final error allowances with $z = \min\{1, Ns\Delta\}$ preserve separation at least $z/(2\pi)$.*

Proof. Equation (8.12) supplies the only curvature input used in that theorem's telescoping proof. Pay it once, and pay the certified implementation errors under both hypotheses exactly as in (7.1). The ideal readout depends on E, H, s, m , not on the unknown remainder within the neighborhood. The precision allowances remain angle dependent, not fixed-noise robustness or a control-synthesis guarantee. $\square$

9. INDEPENDENT PROBES AND HIGHER-ORDER APPROACHES

A linear finite-use scale need not signal coherent accumulation. The support-opening case can already be distinguished by independent classical observations. We make the acquisition restriction precise. Let $D_N^{\text{prod}}(E, F)$ be the supremum of the final trace separation when the input is a product of at most N probe-reference states, one pair per call. The pairs may differ and may depend on the known hypotheses. All completed outputs and references may undergo an arbitrary joint final readout. There is no quantum correlation between different input pairs, and no intervening feedback into later inputs. Public randomization is allowed. This is narrower than nonadaptive acquisition with an entangled many-call input.

Theorem 9.1 (Three local acquisition regimes). *Fix E, H, Λ as in Definition 8.4. Uniformly in N , all sufficiently small $s > 0$, and every F in (8.7), the two distances have the following orders:*

Direction	$D_N^{\text{prod}}(E, F)$	$D_N(E, F)$
$J_j = 0$ for all j , $\Gamma_E(H) \in \mathcal{W}_E$	$\min\{1, \sqrt{N}s\}$	$\min\{1, \sqrt{N}s\}$
$J_j = 0$ for all j , $\Gamma_E(H) \notin \mathcal{W}_E$	$\min\{1, \sqrt{N}s\}$	$\min\{1, Ns\}$
Some $J_j \neq 0$	$\min\{1, Ns\}$	$\min\{1, Ns\}$

Every comparison constant and the common small interval depend only on the fixed E, H, Λ . In particular, collective processing of independently acquired outputs does not recover the linear scale of the second row. The assertion does not bound arbitrary entangled parallel inputs or all classical-feedback acquisition models.

Proof. Theorem 8.5 gives the adaptive column. Its repeated-input lower gives the product lower in the first two rows; (8.11) and (8.9) give both product bounds in the last row. It remains to prove a reference-uniform product upper when all $J_j = 0$.

Use the canonical first-order factors $A_j = \sqrt{E_j}$ and $B_j = (\sqrt{E_j})^+ H_j (I - P_j/2)$ from Lemma 6.4. They need not be horizontal. Put $b = \|B\|$, let $G(s)$ be their normalized measurement in Lemma 8.3, and set

$$a = \frac{3}{2}b + \sqrt{\Lambda + K_b}.$$

For every input state with any reference, denote the corresponding output states under $E, G(s), F$ by ρ_0, ρ_g, ρ_f . Their Bures distance $\beta(\rho, \sigma) = \sqrt{2 - 2f(\rho, \sigma)}$, where $f(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ is root fidelity, satisfies

$$\beta(\rho_0, \rho_g) \leq \frac{3}{2}bs, \quad \beta(\rho_g, \rho_f)^2 \leq \|\rho_g - \rho_f\|_1 \leq (\Lambda + K_b)s^2.$$

The first inequality uses purifications from the two stacked factors and their operator-norm difference; it holds uniformly over the input. The second uses (8.1), (8.5), and the fidelity-trace

inequalities. The triangle inequality for β follows by choosing common purifications at the middle state and applying the Hilbert-space triangle inequality. Thus $1 - f(\rho_0, \rho_f) \leq a^2 s^2 / 2$. These standard fidelity properties and their unhalved trace convention are as in [1, Proposition 3.16 and Theorems 3.22, 3.33].

For different independent probe-reference pairs the output states are still products. Root fidelity is multiplicative, and $1 - \prod_r f_r \leq \sum_r (1 - f_r)$. Therefore

$$\left\| \bigotimes_r \rho_{0,r} - \bigotimes_r \rho_{f,r} \right\|_1 \leq 2 \sqrt{1 - \prod_r f_r^2} \leq 2a\sqrt{N}s.$$

A joint final readout contracts trace norm. Convexity gives the same bound for public randomization, including a retained record of that randomization. Capping the result at two proves the required upper. $\square$

Corollary 9.2 (Higher-order approaches with a quadratic remainder). *Let $q \geq 1$ be an integer, and let a legal measurement family $F(t)$, $0 < t < a$, satisfy for fixed $E, H \neq 0, \Lambda$*

$$(9.1) \quad \|F(t) - E - t^q H\|_\Sigma \leq \Lambda t^{2q}.$$

Then H satisfies (8.2), and the conclusions of Theorems 8.5 and 9.1 hold with $s = t^q$. This includes zero first derivatives when $q > 1$.

Proof. Compress (9.1) to each missing support, divide by t^q , and let $t \downarrow 0$; positivity gives $J_j \geq 0$. Substitute $s = t^q$ in the two finite-pair theorems. No inverse reparametrization is differentiated. In particular, the conclusion holds for $F(t) = E_*(t^q)$ whenever E_* is a fixed one-sided C^2 curve with nonzero right derivative. It also holds for a C^{2q} curve whose first nonzero derivative has order q and whose intermediate derivatives of orders $q+1, \dots, 2q-1$ vanish, by Taylor's remainder after the leading nonzero coefficient. $\square$

Condition (9.1) is substantive. A first nonzero coefficient alone does not imply it. The following exact family displays what an intervening support-opening term can contribute.

Proposition 9.3 (Mixed higher-order rates). *Let $1 \leq a < b < 2a$ be integers. For scalar input dimension and three outcomes put*

$$E = (\tfrac{1}{2}, \tfrac{1}{2}, 0), \quad F(t) = ((1-t^b)(\tfrac{1}{2} + t^a), (1-t^b)(\tfrac{1}{2} - t^a), t^b), \quad 0 < t \leq \tfrac{1}{4}.$$

There are absolute positive constants c, C such that for all $N \geq 1$

$$(9.2) \quad c \min\{1, \sqrt{N}t^a + Nt^b\} \leq D_N(E, F(t)) = D_N^{\text{prod}}(E, F(t)) \leq C \min\{1, \sqrt{N}t^a + Nt^b\}.$$

The leading coefficient $(1, -1, 0)$ is in the covariance range. Its order- a contribution and the order- b opening must both be retained to obtain the finite-use law.

Proof. A scalar-input device only produces independent classical draws. Every tester output, including one with bounded stopping, is a common postprocessing of the full N -draw record. Thus its optimal distance is exactly the ℓ^1 distance of the two product laws. The event that label three occurs gives the lower $2[1 - (1-t^b)^N] \geq 2(1-e^{-1}) \min\{1, Nt^b\}$. Send labels one and two to their corresponding bits and label three to a fair bit. This gives Bernoulli parameters $1/2$ and $1/2 + (1-t^b)t^a$; their gap is at least $t^a/2$. Lemma 3.1 gives a second lower of order $\min\{1, \sqrt{N}t^a\}$. The larger of these lower bounds controls one half of the truncated sum.

Under $F(t)$ the probability of no third label is $(1-t^b)^N$; conditioned on that event the record is Bernoulli with parameter $1/2 + t^a$. Splitting off the complementary event gives the upper

$$2[1 - (1-t^b)^N] + \|\text{Ber}(1/2)^{\otimes N} - \text{Ber}(1/2 + t^a)^{\otimes N}\|_1.$$

For $u = t^a$, the one-copy root fidelity obeys $f^2 = (1 + \sqrt{1 - 4u^2})/2$, so $1 - f^2 \leq 2u^2$. Multiplicativity and the trace upper give a bound $2\sqrt{2N}u$ for the Bernoulli term. Use $1 - (1-t^b)^N \leq Nt^b$ and

cap at two. For the range assertion, the scalar support generator is zero and the missing-support component of $(1, -1, 0)$ vanishes; apply (8.3). Since $b < 2a$, the opening remainder is not $O(t^{2a})$, so this example does not satisfy (9.1) with that leading coefficient. $\square$

For example, $a = 2, b = 3$ gives $\min\{1, \sqrt{N}t^2 + Nt^3\}$ despite a vanishing first derivative. The scalar family $(t^q, 1 - t^q)$, in contrast, has the exact distance $2[1 - (1 - t^q)^N]$ and satisfies (9.1) in the opening branch. A zero tangent supplied without higher-order data is still an undetermined higher-order problem, not a statement of stationarity.

Proposition 9.4 (Exact cone and finite-neighborhood certificates). *For Gaussian-rational E, H , one-sided first-order realizability, lineality, covariance-range membership and the three mechanisms of Theorem 9.1 are decidable in polynomial represented-input bit complexity. For rational $s > 0$, $\Lambda \geq 0$ and $\lambda_j \geq 0$ with $\sum_j \lambda_j \leq \Lambda$, legality of F and the sufficient finite-neighborhood inequalities*

$$(9.3) \quad -\lambda_j s^2 I \preceq F_j - E_j - sH_j \preceq \lambda_j s^2 I \quad (1 \leq j \leq k)$$

are also exactly decidable with that complexity. A nonzero opening supplies a rationally represented witness and its exact output probability at a supplied pair. No local interval, curvature bound, recovery circuit or exact adaptive distance is inferred from a first jet alone.

Proof. The rational support projectors of Proposition 4.4 give every J_j . Exact positive-semidefinite tests establish (8.2); zero tests establish lineality. When the J_j vanish, the same real-rank and Hilbert–Schmidt Gram projection decides whether $\Gamma_E(H)$ belongs to $\mathcal{W}_E$. Lemma 8.2 gives necessity and sufficiency, not only a formal positivity prefilter.

If $J_j \succeq 0$ is nonzero, some diagonal element $(J_j)_{rr}$ is positive. Put $v = Q_j e_r$. Then $v \neq 0$, $v^* H_j v = (J_j)_{rr} > 0$, and $\|v\|^2 = (Q_j)_{rr} > 0$. Both the rate $v^* H_j v / \|v\|^2$ and, for a supplied F , the probability $v^* F_j v / \|v\|^2$ are rational. The unit vector itself need not have rational coordinates: its rational vector and squared normalization specify the state exactly. Finally (9.3) is a finite list of rational PSD tests. Rational elimination, support projection and these tests have the polynomial bit bounds of the retained exact kernels. Verifying one supplied pair does not verify a neighborhood for an unspecified curve, or that a supplied s lies in the theorem’s small interval. The exact impossible-event probability, when provided, has its finite N -draw interpretation without that caveat. $\square$

10. NONEMPTY TANGENT NEIGHBORHOODS

A prescribed remainder allowance need not admit any legal perturbation. The following quantitative realization makes the nonvacuous range explicit. It is a sufficient allowance, not a minimal one. The underlying positive-semidefinite tangent condition is standard semidefinite variational geometry; see [45]. What is needed here is one common normalization and a finite remainder bound for the whole measurement tuple.

Proposition 10.1 (A sufficient allowance). *Let E be an ordered k -outcome measurement and let H be a Hermitian zero-sum tuple with $Q_j H_j Q_j \succeq 0$ for every j . Choose $a > 0$ and $\lambda > 0$ such that*

$$\|H_j\|_{\text{op}} \leq a, \quad E_j \succeq \lambda P_j \quad (1 \leq j \leq k), \quad P_j = \text{supp } E_j.$$

Set

$$(10.1) \quad c = 1 + 2a^2/\lambda, \quad s_* = \min\{1, \lambda/(2a), 1/a\}, \quad \Lambda_* = ck(1 + 2k).$$

Then, for every $0 \leq s \leq s_$,*

$$(10.2) \quad F_j^c(s) = \frac{E_j + sH_j + cs^2 I_d}{1 + kcs^2}$$

is a legal measurement and satisfies

$$(10.3) \quad \|F^c(s) - E - sH\|_\Sigma \leq \Lambda_* s^2.$$

Thus every allowance $\Lambda \geq \Lambda_*$ gives a nonempty neighborhood at every $s \in (0, s_*]$. For arbitrary real input, (10.2) is a rational function of s with matrix coefficients. For Gaussian-rational E, H one may choose rational $a, \lambda, c, s_*, \Lambda_*$, giving Gaussian-rational coefficients.

Proof. For $P_j \neq 0$, the support block of the numerator in (10.2) is at least $\lambda P_j/2$. Its cross block is $sP_j H_j Q_j$. The Schur complement on the missing support is at least

$$sQ_j H_j Q_j + s^2(c - 2a^2/\lambda)Q_j \succeq 0.$$

If $P_j = 0$, the hypothesis says $H_j \succeq 0$, so the numerator is positive directly. This also covers zero outcomes without assigning them a smallest positive eigenvalue. The common denominator is positive for all real s , and summing the numerators gives exactly $(1 + kcs^2)I_d$.

Direct subtraction, with no series expansion, gives

$$F_j^c(s) - E_j - sH_j = \frac{cs^2(I_d - kE_j - ksH_j)}{1 + kcs^2}.$$

Every effect has norm at most one, and $sa \leq 1$ on the given interval. The component remainder is therefore at most $c(1 + 2k)s^2$. Summing proves (10.3). There is a positive common support floor since there are finitely many effects and at least one is nonzero. Rational minorants and majorants give the last assertion. $\square$

Proposition 10.2 (Exact nonemptiness certificates). *On Gaussian-rational represented input, a sufficient certificate (10.1) can be constructed and verified in polynomial bit complexity. It certifies the whole explicit curve (10.2), not only sampled values. Its interval s_* is an interval of legality and remainder control; it is not a computed small interval for the discrimination laws.*

Proof. Use the rational support projectors of Proposition 4.4. The maximum absolute row sum, with $|\Re z| + |\Im z|$ in place of $|z|$, gives a rational upper bound $a \geq 1$ for every Hermitian H_j . Starting at $\lambda = 1$, halve until the exact positive-semidefinite tests $E_j - \lambda P_j \succeq 0$ all hold. Nonzero eigenvalues of a rational positive matrix have a lower bound $2^{-\text{poly}(d, L)}$ in terms of its represented bit length L : clear denominators and use the nonzero coefficient of lowest degree in the characteristic polynomial, together with the row-sum bound on the other eigenvalues. Thus only polynomially many halvings are required. All remaining operations in (10.1) are rational arithmetic. The displayed PSD predicates and algebraic proof above verify the certificate on the continuum. A resource-limited implementation must reject an exhausted cap, rather than return a partial certificate. No claim that $\Lambda < \Lambda_*$ is infeasible follows from this sufficient construction. $\square$

11. ENTANGLEMENT WIDTH AND PARALLEL DISCRIMINATION

We now allow correlations between inputs to several calls, but keep independent groups of bounded size. The bound is imposed on the complete probe-reference groups; a reference shared between groups would be an additional correlation resource.

Definition 11.1 (Independent entangled blocks). For integers $1 \leq b \leq N$, a block- b tester chooses a partition $n_1 + \dots + n_\ell \leq N$, $1 \leq n_r \leq b$, and an input state

$$\sigma = \bigotimes_{r=1}^{\ell} \sigma_r, \quad \sigma_r \in \mathcal{S}((\mathbb{C}^d)^{\otimes n_r} \otimes R_r).$$

The reference spaces are arbitrary. All inputs are prepared before any call; there is no feedback into an input. All classical outputs and retained references may be processed jointly after the

calls. Public randomization may choose both the partition and the states, with its classical record retained. Write $D_N^{[b]}$ for the supremum of the unhalved final trace separation in this class.

Thus $D_N^{[1]} = D_N^{\text{prod}}$. At $b = N$ there is no restriction on the joint input and reference, and

$$(11.1) \quad D_N^{[N]}(E, F) = D_N^{\text{par}}(E, F) = \|\mathcal{M}_E^{\otimes N} - \mathcal{M}_F^{\otimes N}\|_{\diamond}.$$

Unused calls can be padded and discarded. Always $D_N^{[1]} \leq D_N^{[b]} \leq D_N^{\text{par}} \leq D_N$. Separable mixtures over complete probe–reference pairs are included by public randomization. Separability only on the probes, while an arbitrary shared reference remains, is not the same restriction.

Lemma 11.2 (A finite block upper). *Fix E, H, Λ as in Definition 8.4, with $Q_j H_j Q_j = 0$ for every j . Let B be the canonical first-order factor solution in the proof of Theorem 9.1, and put*

$$\beta_0 = \|B\|_{\text{op}}, \quad K = k\beta_0^2(2 + \max_j \|H_j\|_{\text{op}}), \quad r = \Lambda + K.$$

For $0 < s \leq 1$ and $s\beta_0 \leq 1$, every legal F in the quadratic neighborhood satisfies, for a tester with block sizes n_r ,

$$(11.2) \quad \|\rho_E - \rho_F\|_1 \leq \min \left\{ 2, 2s \left[\sum_r \left(\frac{3}{2}\beta_0 n_r + \sqrt{r n_r} \right)^2 \right]^{1/2} \right\}$$

$$(11.3) \quad \leq \min \{ 2, 2(\frac{3}{2}\beta_0 + \sqrt{r})s\sqrt{Nb} \}.$$

Both inequalities are uniform over entangled states within a block, all reference dimensions, and final collective processing.

Proof. Let $G(s)$ be the normalized surrogate from Lemma 8.3, and let W_s, W_0 be its and the base measurement's common Stinespring factors, with the duplicated label in a discarded environment. That lemma gives $\|W_s - W_0\| \leq 3\beta_0 s/2$ and $\|\mathcal{M}_F - \mathcal{M}_{G(s)}\|_{\diamond} \leq r s^2$. For one block of n calls, telescoping tensor products of isometries gives $\|W_s^{\otimes n} - W_0^{\otimes n}\| \leq 3\beta_0 n s/2$. Purify any block input, including its reference. The corresponding output states have Bures distance at most $3\beta_0 n s/2$. Channel telescoping also gives

$$\|\mathcal{M}_F^{\otimes n} - \mathcal{M}_{G(s)}^{\otimes n}\|_{\diamond} \leq n r s^2.$$

The Bures distance from the surrogate output to the F output is therefore at most $s\sqrt{r n}$. Notice the retained $n r s^2$ channel remainder: it is not set to zero when the block grows. The Bures triangle inequality bounds the block distance by $s(3\beta_0 n/2 + \sqrt{r n})$.

Before final processing the outputs of different blocks are products. Writing f_r for their root fidelities, their squared Bures distance is $2(1 - \prod_r f_r) \leq \sum_r 2(1 - f_r)$. The unhalved trace norm is at most twice the Bures distance. This proves (11.2). Since $n_r \geq 1$ and $\sum_r n_r^2 \leq b \sum_r n_r \leq N b$, it also proves (11.3). A final channel contracts trace norm. For a retained randomization record, the states are direct sums with the same weights under both hypotheses; their trace norm difference is the weighted sum of the conditional differences. Hence the same uniform bound holds for randomized preparations and partitions. The fidelity facts used here are those stated in the proof of Theorem 9.1, not an assumption of fidelity multiplicativity inside an entangled block. $\square$

Lemma 11.3 (Parallel corrected phase accumulation). *Suppose $Q_j H_j Q_j = 0$ for all j and $\Gamma = \Gamma_E(H) \notin \mathcal{W}_E$. There are a fixed logical encoding $\mathcal{J}$ and a fixed recovery $\mathcal{R}$ such that*

$$\Phi_E = \text{Id}_2, \quad \|\Phi_F - \mathcal{U}_s\|_{\diamond} \leq \kappa s^2, \quad \kappa = \Lambda + 2\|\Gamma\|_{\text{op}}^2,$$

where $\Phi_F = \mathcal{R}(\mathcal{M}_F \otimes \text{Id})\mathcal{J}$ and $\mathcal{U}_s = \text{Ad}(e^{-is\Gamma_L})$. The two eigenvalues of Γ_L have a gap $\Delta > 0$. For every integer $m \geq 1$, a parallel block of m calls has separation at least

$$(11.4) \quad (2|\sin(ms\Delta/2)| - m\kappa s^2)_+.$$

It also has a fixed binary readout whose base probability is $1/2$ and whose alternative probability differs from $1/2 + \sin(ms\Delta)/2$ by at most $m\kappa s^2/2$. The encoding, recovery and readout are independent of the unknown remainder $F - E - sH$.

Proof. The one-call assertion, including complete trace-preserving recovery on the actual label and reference, is exactly the corrected-channel calculation in Theorem 8.5. We change the acquisition, not that calculation. In the logical eigenbasis prepare

$$|\text{GHZ}_m\rangle = 2^{-1/2}(|0\rangle^{\otimes m} + |1\rangle^{\otimes m}),$$

and apply the tensor encoding $\mathcal{J}^{\otimes m}$ before any call. All m device inputs are now fixed. After all calls apply $\mathcal{R}^{\otimes m}$ to the actual labels and retained references. The output is $\Phi_F^{\otimes m}$ applied to this GHZ state, and channel tensor telescoping gives

$$\|\Phi_F^{\otimes m} - \mathcal{U}_s^{\otimes m}\|_{\diamond} \leq m\kappa s^2.$$

Under the base channel it is the original GHZ state. The ideal alternative multiplies its two components by phases whose difference is $ms\Delta$, giving the pure-state trace separation $2|\sin(ms\Delta/2)|$ and (11.4).

On their two-dimensional span choose the sign of the Pauli Y observable so that its ideal expectation is $\sin(ms\Delta)$; extend it by zero on the orthogonal complement. The effect $(I + Y)/2$ is defined on the entire output, and has base probability $1/2$. The probability of one event changes by at most half the trace norm between normalized states. This proves the stated binary readout bound. The reference dimension in the encoding is at most $2d$ per call, by Lemma 5.3. No recovered logical output is sent into a later device call; all recoveries are final operations. In particular this construction is parallel, not a deferred-description of a sequential input choice. $\square$

Theorem 11.4 (Entanglement-width law). *Fix $E, H \neq 0, \Lambda$ as in Definition 8.4. There exist $c, C, s_0 > 0$, depending only on these fixed data, such that for every pair of integers $1 \leq b \leq N$, every $0 < s \leq s_0$, and every legal F with $\|F - E - sH\|_{\Sigma} \leq \Lambda s^2$,*

$$(11.5) \quad c\omega_{N,b}(s) \leq D_N^{[b]}(E, F) \leq C\omega_{N,b}(s),$$

where

$$\omega_{N,b}(s) = \begin{cases} \min\{1, \sqrt{N}s\}, & H \in \text{Ran } \mathcal{C}_E, \\ \min\{1, \sqrt{Nb}s\}, & Q_j H_j Q_j = 0 \text{ for all } j, \quad \Gamma_E(H) \notin \mathcal{W}_E, \\ \min\{1, Ns\}, & \text{some } Q_j H_j Q_j \neq 0. \end{cases}$$

The same c, C, s_0 work for all b, N and all permitted remainders. The lower testers may be chosen without knowing that remainder. Nonemptiness is guaranteed, for instance, by Proposition 10.1 when $\Lambda \geq \Lambda_*$; for a smaller prescribed allowance the theorem quantifies only over its legal members. The controls in the coherent branch are ideal known-direction controls, not a common learner or a gate-synthesis claim.

Proof. The first row follows from the product lower and adaptive upper of Theorem 8.5, by the inclusions of strategy classes. Its finite upper remains $2\beta_h\sqrt{N}s + (\Lambda + K_h)Ns^2$, capped at two, for the *horizontal* factor solution. The third row similarly follows from its impossible-event product lower and hybrid adaptive upper. Neither row requires entanglement.

For the second row, Lemma 11.2 proves the upper using the canonical, generally nonhorizontal, solution. Use Δ, κ from Lemma 11.3, decrease s_0 so that

$$s_0 \leq \frac{1}{4\Delta}, \quad s_0 \leq \frac{\Delta}{\pi\kappa},$$

and set

$$(11.6) \quad m = \min\{b, \lfloor (2\Delta s)^{-1} \rfloor\}, \quad \ell = \lfloor N/m \rfloor, \quad x = m\Delta s.$$

The floor before taking the minimum is at least two; hence $1 \leq m \leq b$ and $\ell \geq N/(2m)$. Prepare ℓ independent GHZ blocks of size m , discard unused calls, and use the binary readout from the preceding lemma on each block. The outputs are independent Bernoulli variables with common base parameter $1/2$. Since $0 < x \leq 1/2$, their alternative parameter q_F satisfies

$$q_F - \frac{1}{2} \geq \frac{1}{2} \sin x - \frac{1}{2} m \kappa s^2 \geq \frac{x}{\pi} - \frac{\kappa s}{2\Delta} x \geq \frac{x}{2\pi}.$$

This holds for every allowed F and pays its complete finite remainder. Lemma 3.1 gives the lower

$$(11.7) \quad D_N^{[b]}(E, F) \geq \frac{1}{128} \min\{1, \sqrt{\ell} x / (2\pi)\}.$$

If $m = b$, then $\sqrt{\ell} x \geq \Delta s \sqrt{Nb/2}$, as required. If $m < b$, the floor bound implies $x \geq 1/4$, so (11.7) is at least $1/(1024\pi)$, which also bounds a fixed positive multiple of the truncated target rate. For example the coherent lower constant may be taken as $128^{-1} \min\{1, \Delta/(2\sqrt{2}\pi), 1/(8\pi)\}$. This completes both cases without losing uniformity when b grows. $\square$

Corollary 11.5 (Parallel and adaptive local orders). *In every fixed quadratic tangent neighborhood with $H \neq 0$,*

$$D_N^{\text{par}}(E, F) \asymp_{E, H, \Lambda} D_N(E, F)$$

uniformly for all $N \geq 1$ and sufficiently small $s > 0$. Both have square-root order in the covariance range and linear order outside it. In the coherent tangential branch the gain over independent acquisitions is realized by pre-existing inter-call entanglement; feedback is not necessary for that local exponent.

Proof. Set $b = N$ in Theorem 11.4 and compare with Theorem 8.5. $\square$

This compares orders, not exact distances, perfect distinguishability, or asymptotic error exponents at a fixed nonlocal pair. It therefore makes no general assertion that adaptation is useless in channel discrimination. The product converse in Theorem 9.1 remains restricted to its stated class; the parallel lower is supplied by a different proof here.

A complete qubit probability law. Let $E = (|0\rangle\langle 0|, |1\rangle\langle 1|)$, $H = (\sigma_y, -\sigma_y)$, and

$$F_0(s) = \frac{1}{1+s^2} \begin{pmatrix} 1 & -is \\ is & s^2 \end{pmatrix}, \quad F_1(s) = I - F_0(s).$$

These are legal for every real s , and $\|F(s) - E - sH\|_{\Sigma} = 2s^2/\sqrt{1+s^2} \leq 2s^2$. On m parallel inputs prepare $2^{-1/2}(|+\rangle^{\otimes m} - i|-\rangle^{\otimes m})$, where $|\pm\rangle$ are the σ_x eigenvectors. Direct multiplication of the rank-one effects gives, for every ordered binary string z ,

$$(11.8) \quad p_s(z) = 2^{-m} (1 + (-1)^{|z|} \text{Im}(w_s^m)), \quad w_s = \frac{(1+is)^2}{1+s^2}.$$

The two diagonal GHZ terms each give 2^{-m} before their mixture weights; their cross term gives the displayed signed imaginary part. The base law is uniform, and summing over all strings yields $\|p_s - p_0\|_1 = |\text{Im}(w_s^m)| = |\sin(2m \arctan s)|$. Parity is a sufficient readout for this pair of laws. This is a concrete instance of standard GHZ phase accumulation, without references or feedback, rather than a claim of a new metrological mechanism. Here the actual allowance $\Lambda = 2$ suffices even though the general construction with $a = \lambda = 1$ gives the coarse sufficient value $\Lambda_* = 30$.

Corollary 11.6 (Controlled power approaches). *If a legal family satisfies $\|F(t) - E - t^q H\|_{\Sigma} \leq \Lambda t^{2q}$ with $H \neq 0$ and an integer $q \geq 1$, then Theorem 11.4 and Corollary 11.5 hold with $s = t^q$. In the coherent branch the block rate is $\min\{1, \sqrt{Nb} t^q\}$. No such conclusion is asserted from a leading coefficient without the stated remainder bound.*

Proof. The cone condition follows by missing-support compression and t^{-q} rescaling, as in Corollary 9.2. Substitute $s = t^q$; no inverse parametrization is differentiated. $\square$

12. CLASSICAL FEEDBACK AND COHERENT RESET WIDTH

We use normal quantum instruments on trace-class operators. Countably many recorded outcomes form an ℓ^1 direct sum. All identities below follow from finite truncations retaining a remainder outcome and passage to the trace-norm limit; positivity gives monotone convergence of trace sums. The hard finite depth makes the conditional induction applicable. The independent blocks of Definition 11.1 do not exhaust the strategies with bounded coherent acquisition. We now allow both classical feedback between blocks and adaptive quantum control within a block. The receiver may retain and process quantum memory throughout. The restriction is on importing that memory into the next acquisition, not on its dimension or final readout.

Definition 12.1 (Reset protocols). Fix integers $1 \leq b \leq N$. At a classical history h , a protocol may stop, or choose a positive integer $n(h) \leq b$ and prepare a new complete probe-reference system in a state σ_h . Conditional on h , this state is the same under both hypotheses and is in a tensor product with the receiver's entire existing memory. A common $n(h)$ -call adaptive experiment acts on the new system alone; its internal inputs, quantum memory and controls are unrestricted. After acquisition, its whole accessible output is delivered to the receiver. A common quantum instrument may act jointly on this output and all previous receiver memory, retaining quantum memory and producing the next classical history. No quantum subsystem of that memory enters a subsequent acquisition. On every history,

$$\sum_r n(h_r) \leq N.$$

The integers are reserved call budgets: unused internal slots can be padded and are still charged. Public stopping at block boundaries is allowed. All classical histories are finite strings with finite or countable outcome sets; conditional state preparations and instruments are defined also at histories null under one hypothesis. Let $D_N^{\text{reset},b}(E, F)$ be the supremum of the unhalved trace separation of the final accessible states, including the classical record and arbitrary final joint processing.

The complete new system, including every reference and purification used in acquisition, must be independent of the receiver conditional on h . Separability of the probe marginal alone is not sufficient. The receiver memory need not be classical, and can be entangled with previous block outputs. Measuring that memory to choose a fresh input is allowed; transmitting it coherently into that input is not. In particular,

$$(12.1) \quad D_N^{[b]} \leq D_N^{\text{reset},b} \leq D_N, \quad D_N^{\text{reset},N} = D_N.$$

The first inclusion uses no feedback and postpones receiver processing; the equality uses one unrestricted adaptive block. At $b = 1$ the receiver can still adapt future preparations through classical messages. Equality with the independent-product distance is not asserted.

For positive, possibly subnormalized operators, write $f(R, S) = \|\sqrt{R}\sqrt{S}\|_1$. For normalized states put $d_B(R, S)^2 = 2(1 - f(R, S))$. We use

$$(12.2) \quad \begin{aligned} f(R \otimes \rho, S \otimes \sigma) &= f(R, S)f(\rho, \sigma), \\ f\left(\bigoplus_z R_z, \bigoplus_z S_z\right) &= \sum_z f(R_z, S_z), \\ f(\mathcal{A}(R), \mathcal{A}(S)) &\geq f(R, S) \end{aligned}$$

for a channel $\mathcal{A}$. The last assertion for unnormalized operators follows by separate normalization and homogeneity. For normalized states,

$$(12.3) \quad d_B(R, S)^2 \leq \|R - S\|_1 \leq 2\sqrt{1 - f(R, S)^2} \leq 2d_B(R, S).$$

These are standard fidelity identities and inequalities [1, Chapter 3].

Lemma 12.2 (Conditional fidelity accumulation). *Consider a reset protocol in which the fresh output states at node h , before receiver processing, satisfy*

$$f(\omega_{0,h}, \omega_{1,h}) \geq (1 - a_h)_+, \quad a_h \geq 0.$$

Suppose every root-to-terminal history has $\sum_h a_h \leq A$. For any initial receiver operators $R_0, R_1 \succeq 0$, the final classical-quantum operators satisfy

$$(12.4) \quad f(R_{0,\text{out}}, R_{1,\text{out}}) \geq (1 - A)_+ f(R_0, R_1).$$

Thus the receiver's quantum processing and classical feedback need not preserve a product state across blocks.

Proof. The assertion is immediate for $A \geq 1$. For $A < 1$, prove the stronger continuation statement at each node, for arbitrary incoming positive operators and a bound $A_h < 1$ on the remaining path sum. At a terminal node a final common channel only increases fidelity. At a nonterminal node let $a = a_h$ and bound every child's remaining sum by $A_h - a$. Appending the fresh output multiplies the incoming fidelity by at least $1 - a$. Apply the receiver instrument, retaining its classical outcome. If its two subnormalized outputs at outcome z are $R_{0,z}, R_{1,z}$, channel monotonicity and direct-sum additivity give

$$\sum_z f(R_{0,z}, R_{1,z}) \geq (1 - a) f(R_0, R_1).$$

The induction hypothesis applies separately at every z , even though the two conditional receiver states differ. The continued output fidelity is at least

$$[1 - (A_h - a)](1 - a) f(R_0, R_1) \geq (1 - A_h) f(R_0, R_1).$$

Here $0 \leq a \leq A_h < 1$, so the product estimate has the stated sign. Each acquisition costs at least one call, hence the depth is at most N and backwards induction terminates. Countably many outcomes are handled by the same nonnegative sum; no normalization at a zero-probability history is required. Conditional public randomization is another retained instrument outcome. This proves the statement, including stopping and final joint processing. $\square$

Lemma 12.3 (A fresh adaptive block). *Fix E, H, Λ in Definition 8.4, and assume $Q_j H_j Q_j = 0$ for every j . Let $A_j = \sqrt{E_j}$ and choose the canonical factors B_j of Lemma 6.4 with $A_j^* B_j + B_j^* A_j = H_j$. Set*

$$\beta_0 = \|B\|_{\text{op}}, \quad K = k\beta_0^2(2 + \max_j \|H_j\|_{\text{op}}), \quad r = \Lambda + K, \quad \alpha = 3\beta_0/2.$$

If $0 < s \leq 1$ and $s\beta_0 \leq 1$, the outputs of every common fresh n -call adaptive experiment obey

$$(12.5) \quad d_B(\omega_E, \omega_F) \leq s(\alpha n + \sqrt{rn}), \quad 1 - f(\omega_E, \omega_F) \leq \frac{1}{2}(\alpha + \sqrt{r})^2 n^2 s^2,$$

for every legal F with $\|F - E - sH\|_{\Sigma} \leq \Lambda s^2$. The constants are independent of the input, reference, internal controls and permitted remainder.

Proof. The normalized surrogate in Lemma 8.3 has isometries W_0, W_s with $\|W_s - W_0\| \leq \alpha s$ and

$$\|\mathcal{M}_F - \mathcal{M}_{G(s)}\|_{\diamond} \leq r s^2.$$

Purify the fresh input and all internal common controls. Inserting the difference $W_s - W_0$ successively in the n slots bounds the final purified vector difference by $n\alpha s$: every other factor is an isometry. This argument is valid for adaptive controls, not only tensor-power acquisition. The discarded dilation environments are used in the proof but are unavailable to the experiment. Purification and contraction give the same Bures upper for the accessible E and surrogate outputs. Replacing the surrogate by F costs at most nrs^2 in unhalved trace norm by adaptive channel telescoping. The first inequality of (12.3) and the Bures triangle inequality give the first bound

in (12.5). The second follows from $1 - f = d_B^2/2$ and $\sqrt{n} \leq n$. Internal early stopping is padded within the already reserved n slots and then discarded. This retains the full nrs^2 error before taking its square root. $\square$

The canonical factor above is generally not horizontal. Its linear-in- n purified displacement is appropriate for a coherent block. The separate horizontal construction remains essential to the stronger regular-direction bound for unrestricted adaptive strategies.

Theorem 12.4 (Reset-width interpolation). *Fix a measurement E , a nonzero Hermitian zero-sum tuple H with $J_j = Q_j H_j Q_j \succeq 0$, and $\Lambda \geq 0$. There are $c, C, s_0 > 0$, depending only on E, H, Λ , such that for every $N \geq 1$, $1 \leq b \leq N$, $0 < s \leq s_0$, and every legal measurement F satisfying*

$$\|F - E - sH\|_\Sigma := \sum_j \|F_j - E_j - sH_j\|_{\text{op}} \leq \Lambda s^2,$$

one has $c\tau \leq D_N^{\text{reset},b}(E, F) \leq C\tau$, where

$$(12.6) \quad \tau = \begin{cases} \min\{1, \sqrt{N}s\}, & H \in \text{Ran } \mathcal{C}_E, \\ \min\{1, \sqrt{Nb}s\}, & J_j = 0 \ (\forall j), \quad \Gamma_E(H) \notin \mathcal{W}_E, \\ \min\{1, Ns\}, & J_j \neq 0 \ (\exists j). \end{cases}$$

Here $\Gamma_E(H) = \frac{i}{2} \sum_j [P_j, H_j]$ and $\mathcal{W}_E = \sum_j P_j \mathcal{H}_a P_j$. All constants are uniform in N, b, F , not in E, H, Λ . For each N, b, s , the lower experiment can be chosen independently of the unknown remainder. It can in fact be chosen without inter-block feedback and with parallel acquisition inside each block.

Proof. In the tangential cases put $\gamma = (\alpha + \sqrt{r})^2/2$ as in Lemma 12.3. For a particular public policy let

$$Q = \sup_{\text{terminal histories}} \sum_r n(h_r)^2 \leq bN.$$

At a node, the fresh experiment has normalized outputs independent of the incoming receiver memory, conditional on that history. Lemma 12.3 bounds its fidelity deficit by $a_h = \gamma n(h)^2 s^2$. Lemma 12.2, starting from the same normalized initial memory, therefore gives

$$f(\rho_{E,\text{out}}, \rho_{F,\text{out}}) \geq (1 - \gamma Q s^2)_+.$$

If $\gamma Q s^2 \leq 1$, use (12.3) and $1 - (1 - x)^2 \leq 2x$; otherwise the trace bound two suffices. In both cases we obtain the finite policy-sensitive estimate

$$(12.7) \quad D(\text{this protocol}; E, F) \leq \min\{2, 2(\alpha + \sqrt{r})s\sqrt{Q}\} \leq \min\{2, 2(\alpha + \sqrt{r})s\sqrt{bN}\}.$$

The uniform per-block bound permits every history-dependent choice of inputs and internal controls, and arbitrary common receiver instruments. The two output states are not assumed to be products across feedback branches.

For the coherent row, the lower construction and the two integer allocation cases of Theorem 11.4 belong to the reset class by (12.1). Its fixed binary event pays $m\kappa s^2$ before the finite Bernoulli amplification. Thus the same constants and remainder-independent lower apply here. For the regular row, the product lower and the fully adaptive horizontal bound (8.10) sandwich this class. That upper retains $(\Lambda + K_h)Ns^2$, which is controlled by $\sqrt{N}s$ when $\sqrt{N}s \leq 1$ and by the trace cap otherwise. For the opening row, the fixed impossible event (8.11) and the adaptive hybrid bound (8.9) give the result already without entanglement. Take the minimum of the finitely many fixed scale restrictions. The full range criterion (8.3) makes the cases disjoint and exhaustive. $\square$

Corollary 12.5 (Classical feedback at unit reset width). *In a fixed coherent tangent neighborhood,*

$$D_N^{\text{reset},1}(E, F) \asymp \min\{1, \sqrt{N}s\}, \quad D_N^{\text{par}}(E, F) \asymp D_N(E, F) \asymp \min\{1, Ns\}.$$

The first class permits arbitrary receiver quantum memory and classical feedback, but no coherent import of that memory into a fresh probe–reference preparation. These are local comparison orders, not equality of exact distances or fixed-pair error exponents.

Proof. Apply Theorem 12.4 at $b = 1$ and combine the parallel endpoint in Corollary 11.5 with the adaptive tube theorem. All intervals and constants refer to the same fixed E, H, Λ . $\square$

The role of feedback in channel discrimination has extensive prior study [47, Theorem 9 and Remark 10]. Our assertion is a finite, remainder-uniform consequence of a conditional fidelity estimate, with arbitrary receiver memory and variable reserved block sizes. It does not introduce the classical-feedback resource model or the standard-versus-coherent metrological mechanism. The isometric-extension estimates likewise have direct antecedents in channel Bures bounds [49, Theorems 17 and 19].

12.1. Finite policy certificates. For a finite public decision tree, its pathwise call maximum and quadratic acquisition cost are computed recursively. At a terminal node set $N_h = Q_h = 0$; at an acquisition node set

$$N_h = n(h) + \max_z N_{hz}, \quad Q_h = n(h)^2 + \max_z Q_{hz}.$$

These are maxima over complete histories, not expected costs under either hypothesis. Exact integer arithmetic verifies $N_h \leq N$ and $n(h) \leq b$ on the whole policy. Given rational s and a separately established rational majorant $\bar{\gamma} \geq \gamma$,

$$f_{\text{out}} \geq (1 - \bar{\gamma}Qs^2)_+, \quad \|\rho_{E,\text{out}} - \rho_{F,\text{out}}\|_1^2 \leq \min\{4, 8\bar{\gamma}Qs^2\}.$$

The same induction also gives the stronger recursion $F_h = (1 - \bar{\gamma}n(h)^2s^2)_+ \min_z F_{hz}$ with terminal value one. A policy certificate checks these budget consequences. It does not verify the physical reset condition or establish the supplied one-block fidelity majorant. Those remain hypotheses. In particular a numerical or declared reference dimension is not a certificate of conditional independence.

The explicit nonempty interval s_* of Proposition 10.1 continues to certify legality and the sufficient allowance only. It is not the discrimination interval s_0 above; that interval also depends on the lower witness and logical correction gap. The sufficient Λ_* is neither minimal nor canonical. Zero tangents, intervening higher-order terms, arbitrary-pair boundary equivalence and growing-output learning retain their separate hypotheses and questions.

13. ARBITRARY ALLOCATION PROFILES

The largest permitted block does not determine how that resource is actually used. We retain the entire list of block sizes. Throughout this section the groups are complete probe–reference systems, not merely probe marginals. We first specify independent acquisition, then allow the reset-constrained feedback of Definition 12.1 while keeping the complete reservation profile fixed.

Definition 13.1 (A prescribed allocation). Let $\mathbf{n} = (n_1, \dots, n_\ell)$ be a nonempty finite list of positive integers. Write

$$N_{\mathbf{n}} = \sum_{r=1}^{\ell} n_r, \quad V_{\mathbf{n}} = \sum_{r=1}^{\ell} n_r^2.$$

An allocation tester has an input $\bigotimes_r \sigma_r$, where σ_r is an arbitrary state on $(\mathbb{C}^d)^{\otimes n_r} \otimes R_r$. Reference dimensions and final joint processing are unrestricted. Public mixtures, with the mixture label retained, are allowed for this fixed allocation. A group may use fewer than its assigned slots by

padding with fixed inputs and discarding their outputs. Denote the supremum of the unhalved final trace separation by $D^n(E, F)$. For the same list, a prescribed-profile reset tester reserves n_r calls in its r th fresh block, independent of the classical history. The preparation and all internal adaptive controls may depend on that history, but the complete fresh system is conditionally independent of all old receiver memory. This is one common conditional rule on every formal history, not an equality of the two conditional receiver states. Early termination can be padded by unused reserved blocks. Our convention records a deterministic terminal symbol at each remaining round, executes a fixed fresh preparation and discards its output. All positive reservations in the prescribed profile are still charged. Zero-length terminal symbols are not additional profile components. Write $D^{\text{reset}, n}$ for this larger supremum. Both N_n and V_n are hard budgets, not expectations. Then

$$D^n \leq D^{\text{reset}, n} \leq D_{N_n}.$$

The first moment N_n governs independent statistical accumulation; the second moment V_n governs coherent accumulation within these groups. The second-moment principle has established metrological antecedents [33, 34]. The point here is a finite trace-distance statement uniform over a whole measurement neighborhood, including its unknown quadratic remainder. We first record the finite classical estimate that permits unequal blocks without grouping them into dyadic size classes.

Lemma 13.2 (A common readout for unequal signals). *Let $0 < \alpha \leq 1$, $0 \leq x_r \leq 1/2$, and $\alpha x_r \leq u_r \leq x_r$. Let P_0 be the law of independent fair signs $X_r \in \{-1, 1\}$, and let P_u be the independent law with $P_u\{X_r = 1\} = 1/2 + u_r$. There is a binary randomized readout depending only on (x_r) , and not on (u_r) , for which the unhalved separation is at least*

$$(13.1) \quad \frac{\alpha}{64} \min \left\{ 1, \left(\sum_r x_r^2 \right)^{1/2} \right\}.$$

Consequently the full product laws have at least this separation. In the application, (x_r) is fixed by E, H, Λ, s and the public profile. The readout has precisely this public dependence and no dependence on the unknown tuple $F - E - sH$ or actual biases (u_r) .

Proof. Put $S = \sum_r x_r^2$. For $S = 0$ the assertion is immediate. Otherwise set

$$A = \max\{\sqrt{S}, S\}, \quad t_r = \frac{x_r}{16A}, \quad T(X) = \sin \left(\sum_r t_r X_r \right).$$

Then $0 \leq t_r \leq 1/16$ and $\sum_r t_r^2 \leq 1/256$. Under P_0 the expectation of T is zero. Independence under P_u gives

$$\mathbb{E}_u \exp \left(i \sum_r t_r X_r \right) = \prod_r (\cos t_r + 2iu_r \sin t_r) = Re^{i\phi}, \quad \phi = \sum_r \arctan(2u_r \tan t_r).$$

Each factor has positive real part. Also

$$R \geq \prod_r \cos t_r \geq 1 - \frac{1}{2} \sum_r t_r^2 \geq \frac{1}{2}.$$

For $0 \leq t \leq 1/16$, $t \leq \tan t \leq 2t$. The arguments of $\arctan$ above lie in $[0, 1]$, where $v/2 \leq \arctan v \leq v$. It follows that

$$\frac{\alpha S}{16A} \leq \sum_r u_r t_r \leq \phi \leq 4 \sum_r u_r t_r \leq \frac{S}{4A} \leq \frac{1}{4}.$$

Since $\sin \phi \geq \phi/2$ on this interval, $\mathbb{E}_u T = R \sin \phi \geq \alpha S/(64A)$. Accept with probability $(1 + T(X))/2$. The resulting binary laws have unhalved separation $|\mathbb{E}_u T - \mathbb{E}_0 T|$. Finally $S/A = \min\{1, \sqrt{S}\}$. This proves the assertion with one readout for every permitted vector of biases. $\square$

Theorem 13.3 (Allocation-profile law). *Fix an ordered measurement E , a nonzero Hermitian tuple H with $\sum_j H_j = 0$ and $Q_j H_j Q_j \succeq 0$, and a number $\Lambda \geq 0$. Here $P_j = \text{supp } E_j$, $Q_j = I - P_j$, and $\|R\|_\Sigma = \sum_j \|R_j\|_{\text{op}}$. There are $c, C, s_0 > 0$, depending on E, H, Λ alone, such that for every allocation $\mathbf{n}$, every $0 < s \leq s_0$, and every legal measurement F satisfying*

$$(13.2) \quad \|F - E - sH\|_\Sigma \leq \Lambda s^2,$$

one has

$$(13.3) \quad c\omega_{\mathbf{n}}(s) \leq D^n(E, F) \leq D^{\text{reset}, \mathbf{n}}(E, F) \leq C\omega_{\mathbf{n}}(s),$$

where

$$\omega_{\mathbf{n}}(s) = \begin{cases} \min\{1, \sqrt{N_{\mathbf{n}}}s\}, & H \in \text{Ran } \mathcal{C}_E, \\ \min\{1, \sqrt{V_{\mathbf{n}}}s\}, & Q_j H_j Q_j = 0 \ (1 \leq j \leq k), \quad \Gamma_E(H) \notin \mathcal{W}_E, \\ \min\{1, N_{\mathbf{n}}s\}, & \text{some } Q_j H_j Q_j \neq 0. \end{cases}$$

These constants are uniform in the number and sizes of groups, and in the entire permitted remainder. In every row a single lower tester can be chosen from $E, H, \Lambda, s, \mathbf{n}$, independently of that remainder. The coherent construction uses ideal known-direction controls. It is neither a common learner nor a statement about separable probe marginals with a shared reference. Nonemptiness and a sufficient allowance are as in Proposition 10.1; its legality interval s_* does not replace the discrimination interval s_0 .

Proof. Write $N = N_{\mathbf{n}}$ and $V = V_{\mathbf{n}}$. Product-reference inputs are admissible in every allocation, and every prescribed-profile reset tester is an adaptive tester. The regular and opening rows therefore follow from Theorem 8.5. In the regular row its finite upper is

$$2\beta_h \sqrt{N}s + (\Lambda + K_h)Ns^2,$$

capped at two, for the horizontal factor solution. This is of the asserted order since $x^2 \leq x$ for $x = \sqrt{N}s \leq 1$. The opening lower uses the fixed impossible-output event, not a coherent code.

First consider the independent coherent upper. Use the canonical, generally nonhorizontal, factor and the constants β_0, r_0 of Lemma 11.2, writing r_0 for its remainder constant r . With root fidelity $f(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ and Bures distance $\beta^2 = 2(1 - f)$, the normalization is

$$\beta(\rho, \sigma)^2 \leq \|\rho - \sigma\|_1 \leq 2\beta(\rho, \sigma).$$

The actual channel remainder in a group of n_r calls is $n_r r_0 s^2$; its Bures contribution is $s\sqrt{r_0 n_r}$. The block estimate consequently gives the finite upper

$$(13.4) \quad D^n(E, F) \leq \min \left\{ 2, 2s \left[\sum_r \left(\frac{3}{2}\beta_0 n_r + \sqrt{r_0 n_r} \right)^2 \right]^{1/2} \right\}$$

$$(13.5) \quad \leq \min\{2, 2s(\frac{3}{2}\beta_0 \sqrt{V} + \sqrt{r_0 N})\}.$$

The last line is the Euclidean triangle inequality. Since $N \leq V$, it implies the required upper. Fidelity multiplies only between independent group outputs. A retained public mixture gives a direct sum with common weights, whose trace separation is the weighted sum of the conditional separations; the same bound follows. Final joint processing is unrestricted and contracts trace norm.

The same finite upper holds for prescribed-profile reset testers, for a different reason. Lemma 12.3 gives the normalized fresh output deficit

$$a_r = \frac{1}{2}s^2(\alpha n_r + \sqrt{r_0 n_r})^2, \quad \alpha = \frac{3}{2}\beta_0,$$

uniformly at every history in round r . These deficits have the same deterministic sum along every full reservation path. Apply Lemma 12.2 to positive subnormalized receiver operators. If

$A = \sum_r a_r \leq 1$, then $\|\rho_E - \rho_F\|_1 \leq 2\sqrt{2A}$; if $A > 1$, use the trace cap two. This proves (13.4) and (13.5) with $D^{\text{reset},n}$ in place of D^n . It does not multiply unconditional output fidelities after feedback. Countable histories are interpreted as trace-class direct sums; all fidelity summands are nonnegative.

For the lower let $\Delta, \kappa > 0$ be as in Lemma 11.3, decreasing s_0 if necessary so that

$$s_0 \leq \frac{1}{4\Delta}, \quad s_0 \leq \frac{\Delta}{\pi\kappa}.$$

In group r use

$$m_r = \min\{n_r, \lfloor (2\Delta s)^{-1} \rfloor\}, \quad x_r = m_r \Delta s,$$

logical GHZ inputs, with the remaining slots padded and discarded. All encodings precede acquisition and all recoveries follow it. The reference dimension is at most $2d$ per used call; the reference of a whole group may grow with its size. The fixed binary event of that lemma has base probability $1/2$ and alternative probability $1/2 + u_r$, with

$$\left| u_r - \frac{1}{2} \sin x_r \right| \leq \frac{1}{2} m_r \kappa s^2.$$

This is the full finite tensor-channel error, paid before statistical amplification. Since $0 < x_r \leq 1/2$, the displayed choice of s_0 implies

$$\frac{x_r}{2\pi} \leq u_r \leq \frac{x_r}{2} + \frac{\kappa s}{2\Delta} x_r \leq x_r.$$

Independent group outputs need not have equal biases. Lemma 13.2, with $\alpha = 1/(2\pi)$, gives

$$(13.6) \quad D^n(E, F) \geq \frac{1}{128\pi} \min \left\{ 1, \Delta s \left(\sum_r m_r^2 \right)^{1/2} \right\}.$$

Both the individual events and the final randomized classical readout are independent of $F - E - sH$.

If every $m_r = n_r$, the last minimum is $\min\{1, \Delta s \sqrt{V}\}$, at least $\min\{1, \Delta\} \min\{1, s \sqrt{V}\}$. If some $m_r < n_r$, then $(2\Delta s)^{-1} \geq 2$ and the floor bound gives $x_r \geq 1/4$. The last minimum is consequently at least $1/4$. The target $\min\{1, s \sqrt{V}\}$ is at most one, so this also controls it by an absolute positive factor. In particular the coherent lower constant may be taken as $\min\{\Delta, 1/4\}/(128\pi)$. This proves the uniform lower for both classes, including profiles with very unequal or very large groups. No output is used to prepare a later input. $\square$

Corollary 13.4 (Width budgets and effective width). *For $1 \leq b \leq N$, put $q = \lfloor N/b \rfloor$ and $r = N - qb$. Among allocations with $\sum n_j \leq N$ and $n_j \leq b$,*

$$(13.7) \quad \max V_n = qb^2 + r^2, \quad \frac{1}{2}Nb \leq qb^2 + r^2 \leq Nb.$$

Thus the coherent width law can equivalently be written with $\sqrt{qb^2 + r^2} s$, with constants independent of N, b . For a prescribed allocation the effective width is $b_{\text{eff}} = V_n/N_n$, not necessarily its largest part. Coarsening an allocation increases its coherent scale and leaves the other two scales unchanged.

Proof. Adding unused slots increases the objective. If two parts are below b , transfer slots from the smaller to the larger until one is zero or the other is b . Their sum of squares does not decrease. Iteration gives q parts equal to b and, if $r > 0$, one part r . The upper bound follows from $n_j^2 \leq bn_j$. Since $r < b \leq qb$, one has $qb \geq N/2$, which proves the lower. Merging parts u, v adds $2uv$ to their squared sum. For independent allocation testers, the operational inclusion also holds because product inputs within a merged group remain admissible. No analogous set inclusion is asserted for arbitrary reset-feedback testers when a receiver-interaction boundary is removed; their local orders are the preceding theorem's statement. $\square$

For example, with $N = L^2$ calls and largest block L , the allocation $(L, 1, \dots, 1)$ has $V = 2L^2 - L$, whereas L groups each of size L have $V = L^3$. Before saturation their coherent scales differ by a factor of order $\sqrt{L}$, despite equal call counts and equal largest blocks. At the single-group endpoint, the preceding theorem recovers matching parallel and adaptive *local orders*; it makes no claim of equality of their exact distances or global error exponents.

The comparison concerns the best tester for a prescribed reservation profile. It does not state that each feedback policy using that profile is informative. For example, a common final channel that forgets every output has zero separation. Likewise, an arbitrary policy's worst-path cost is not itself a matching lower certificate: a costly branch can be uninformative or rarely visited.

14. QUADRATIC ACQUISITION BUDGETS AND RECEIVER MEMORY

A width bound can conceal unused coherent acquisition. The natural policy quantity is the worst-path sum of squared reservations. We first retain the separate finite remainder, then identify the sharp local order under this resource bound. None of these quantities measures the dimension of a quantum memory.

Proposition 14.1 (Finite policy profile). *Fix a legal quadratic tube as in Definition 8.4, and suppose $Q_j H_j Q_j = 0$ for every j . Use the constants α, r and scale restrictions of Lemma 12.3. For a reset policy π , define the hard path bounds*

$$N_\pi = \sup_{\zeta} \sum_{h \in \zeta} n(h), \quad Q_\pi = \sup_{\zeta} \sum_{h \in \zeta} n(h)^2,$$

where ζ ranges over formal terminal histories, and put

$$B_\pi = \sup_{\zeta} \sum_{h \in \zeta} (\alpha n(h) + \sqrt{r n(h)})^2.$$

For every permitted remainder, the final separation of this policy satisfies

$$(14.1) \quad \begin{aligned} D(\pi; E, F) &\leq \min\{2, 2s\sqrt{B_\pi}\} \\ &\leq \min\{2, 2s(\alpha\sqrt{Q_\pi} + \sqrt{rN_\pi})\}. \end{aligned}$$

The receiver dimension and all receiver instruments are unrestricted. The complete fresh acquisition remains conditionally independent of old receiver memory, including all within-block references and memory.

Proof. At history h the fresh adaptive-block lemma gives a deficit at most $a_h = s^2(\alpha n(h) + \sqrt{r n(h)})^2/2$. The term $n(h)rs^2$ is the full tube-to-surrogate channel error before Bures conversion. Along every terminal history their sum is at most $s^2 B_\pi/2$. Conditional fidelity accumulation and $\|\rho - \sigma\|_1 \leq 2\sqrt{1 - f(\rho, \sigma)^2}$ prove the first bound, with the universal trace cap covering large deficits. On each path the Euclidean triangle inequality gives

$$\left[\sum_h (\alpha n(h) + \sqrt{r n(h)})^2 \right]^{1/2} \leq \alpha \left(\sum_h n(h)^2 \right)^{1/2} + \sqrt{r \sum_h n(h)}.$$

Taking suprema proves the second bound. No expectation of a path cost or probability of a history has been substituted for a supremum. The induction uses positive subnormalized operators and hence applies also at null histories and to trace-class countable direct sums. $\square$

Definition 14.2 (A quadratic reservation budget). For integers $N \geq 1$ and $N \leq Q \leq N^2$, let $D_{N,Q}^{\text{reset}}(E, F)$ be the supremum over reset protocols with $N_\pi \leq N$ and $Q_\pi \leq Q$. The budgets include every reserved slot on every formal terminal history, whether or not it is used internally. There is no additional width constraint. The conditional reset and common-control conventions are those of Definition 12.1.

Theorem 14.3 (Sharp quadratic budget law). *Fix E , a nonzero Hermitian zero-sum tuple H in the one-sided cone, and $\Lambda \geq 0$. There exist $c, C, s_0 > 0$, depending only on E, H, Λ , such that for all integers $N \geq 1$, $N \leq Q \leq N^2$, $0 < s \leq s_0$, and every legal F with*

$$\|F - E - sH\|_\Sigma := \sum_j \|F_j - E_j - sH_j\|_{\text{op}} \leq \Lambda s^2,$$

one has $c\omega \leq D_{N,Q}^{\text{reset}}(E, F) \leq C\omega$, where

$$(14.2) \quad \omega = \begin{cases} \min\{1, \sqrt{N}s\}, & H \in \text{Ran } \mathcal{C}_E, \\ \min\{1, \sqrt{Q}s\}, & Q_j H_j Q_j = 0 \ (\forall j), \quad \Gamma_E(H) \notin \mathcal{W}_E, \\ \min\{1, Ns\}, & Q_j H_j Q_j \neq 0 \ (\exists j). \end{cases}$$

The lower tester is independent of the unknown remainder and can be chosen with a deterministic profile, no inter-block feedback, and parallel acquisition within its blocks. Constants are uniform in N, Q, F , not in the fixed tube. The theorem asserts local orders, not exact-distance identities or a classification of memory dimensions.

Proof. For coherent tangents, Proposition 14.1 gives

$$D(\pi; E, F) \leq \min\{2, 2s(\alpha\sqrt{Q} + \sqrt{rN})\} \leq \min\{2, 2(\alpha + \sqrt{r})s\sqrt{Q}\}.$$

For the lower put $b = \lfloor Q/N \rfloor$. Then $1 \leq b \leq N$, $Nb \leq Q$, and $Nb \geq Q/2$. Take as many blocks of size b as possible, followed by the positive remainder when necessary. Their sum is N and, by Corollary 13.4,

$$Q/4 \leq Nb/2 \leq V_n \leq Nb \leq Q.$$

The lower construction of Theorem 13.3 belongs to the prescribed budget class and gives the coherent lower. In particular, the possible quadratic-budget integrality gaps cost only an absolute factor; the theorem does not claim to attain every integer Q exactly.

For regular tangents use the unrestricted adaptive horizontal upper and the independent-product lower. The all-singleton profile has square cost $N \leq Q$. The finite upper still contains $2\beta_h\sqrt{N}s + (\Lambda + K_h)Ns^2$, capped at two; putting $x = \sqrt{N}s$ gives the claimed order. For support openings the same singleton profile uses the fixed impossible-label event, and the ordinary adaptive hybrid bound gives the upper. This lower is a classical leakage experiment rather than a coherent phase code. All lower controls are chosen from the fixed tube and public budgets; the complete finite $m_r\kappa s^2$ errors were paid in Theorem 13.3 before statistical amplification. $\square$

Proposition 14.4 (Certified deviations from reset). *Fix a finite or countably branching reset policy with a hard finite call bound. At each acquisition history h , replace the ideal preparation channel $R \mapsto R \otimes \sigma_h$ by a channel $\tilde{\mathcal{P}}_h$ with the same input and output systems. Assume, including auxiliary references,*

$$\|\tilde{\mathcal{P}}_h - (\text{Id} \otimes \sigma_h)\|_\diamond \leq \delta_h, \quad \sup_{\zeta} \sum_{h \in \zeta} \delta_h \leq \varepsilon.$$

All within-block testers, receiver instruments, and reservation rules are kept fixed and common. The implemented final separation obeys

$$(14.3) \quad \tilde{D}(\pi; E, F) \leq \min\{2, D(\pi; E, F) + 2\varepsilon\}.$$

For tangential directions one may substitute the right side of (14.1) for $D(\pi; E, F)$. For the same pair of ideal and implemented channels the proof gives the two-sided estimate $|\tilde{D}(\pi; E, F) - D(\pi; E, F)| \leq 2\varepsilon$. In particular a lower $c\omega$ remains at least $c\omega/2$ if $\varepsilon \leq c\omega/4$; the condition $\varepsilon = o(\omega)$ suffices along a shrinking-signal sequence.

Proof. For either fixed device, let ε_h bound the remaining path sum from a history h . Induct backwards on remaining depth with arbitrary positive incoming operator R . Replacing the preparation changes the continued output in trace norm by at most $\delta_h \operatorname{tr} R$, since every continuation is a channel. For the ideal preparation followed by the common block and receiver instrument, the positive child operators have traces summing to $\operatorname{tr} R$. The induction bounds the remaining difference by $(\varepsilon_h - \delta_h) \operatorname{tr} R$. Their sum is $\varepsilon_h \operatorname{tr} R$. At terminals it is zero; countable sums converge monotonically in trace. Thus each hypothesis's normalized output differs by at most ε . Applying the triangle inequality to the two hypotheses proves the result. $\square$

The hypothesis is a uniform channel certificate for a specified common ideal reset preparation. Marginal separability, finite policy arithmetic, or an average calibration error does not supply it. Fixed nonzero ε can dominate a shrinking local signal; no fixed-error asymptotic robustness is inferred. The same argument applies to the particular lower construction when its preparations satisfy the displayed hypothesis, giving the lower separation minus 2ε .

14.1. A strict tester-level comparison with classical adaptation. We now give an explicit comparison with the classically adaptive model of Ohst–Zhang–Nguyen–Plávala–Quintino [51, Definition 23 and (64)]. For two calls their tester operators have the form

$$(14.4) \quad T_i = \sum_z R_z \otimes S_{i|z},$$

where R and each $S_{\cdot|z}$ are single-call testers. This is an entire first-call measurement followed by conditional new preparation, not retention of an unmeasured first-call quantum register for a later joint readout.

Proposition 14.5 (Retained receiver memory). *Every two-call tester of the form (14.4) is a unit-width reset tester. The inclusion is strict at the level of tester operators, already for two-dimensional inputs and two classical output labels. This is not an assertion of a strict optimal-score gap for every ensemble or for every pair of identical measurement channels.*

Proof. Realize R on the first call, retain only its classical outcome z , then prepare and realize $S_{\cdot|z}$ freshly. This obeys the unit-width reset condition and proves inclusion.

For strictness, write I_1, I_2 for the two qubit inputs and Y_1, Y_2 for their classical output registers. Prepare independent maximally entangled states on $I_r R_r$, one per call. Keep the first reference and label in the receiver, append the second complete pair freshly, and make a final joint measurement. One effect of that measurement is $\Pi_{R_1 R_2} \otimes |00\rangle\langle 00|_{Y_1 Y_2}$, where $\Pi = |\Phi^+\rangle\langle\Phi^+|$ and $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. For precision, order factors as I_1, Y_1, I_2, Y_2 and take the fully transposed unnormalized Choi representative $J^\sharp(E) = \sum_y E_y \otimes |y\rangle\langle y|$, obtained from $(\operatorname{Id} \otimes \mathcal{M}_E)(|\Omega\rangle\langle\Omega|)$ by full transpose, with $|\Omega\rangle = |00\rangle + |11\rangle$. Thus $\operatorname{tr}_Y J^\sharp = I$. The two normalized pairs produce the receiver operator $(J^\sharp(E) \otimes J^\sharp(F))^\top/4$. For any receiver effect M , the Born probability is $\operatorname{tr}[M^\top(J^\sharp(E) \otimes J^\sharp(F))]/4$. This proves the transpose and factor $1/4$ for arbitrary complex effects, not merely for the real Bell projector. Grouping the input factors below means the canonical permutation of the stated factor order. In the Choi convention with input marginal I , direct contraction of the two normalized maximally entangled preparations gives the tester effect

$$(14.5) \quad T_0 = \frac{1}{4} \Pi_{I_1 I_2}^\top \otimes |00\rangle\langle 00|_{Y_1 Y_2}, \quad T_1 = \frac{1}{4} I_{I_1 I_2} \otimes I_{Y_1 Y_2} - T_0.$$

Both are positive and their sum is the deterministic parallel tester with input marginal $I_{I_1 I_2}/4$. They are block diagonal in the classical outputs. The described realization has reset width one: no old reference enters the second acquisition. Indeed $0 \preceq \Pi^\top \otimes |00\rangle\langle 00| \preceq I$, so the complementary effect is positive. The deterministic sum is $\Xi \otimes I_{Y_2}$ with $\Xi = I_{I_1} \otimes I_{Y_1} \otimes I_{I_2}/4$ and $\operatorname{tr}_{I_2} \Xi = (I_{I_1}/2) \otimes I_{Y_1}$; the initial marginal has trace one. This verifies both causal tester trace constraints. The cone of positive separable operators across the complete-call cut $I_1 Y_1 \mid I_2 Y_2$ is

closed: its trace-one slice is the compact convex hull of the compact set of product states, and trace controls the scaling in its cone. Hence the limiting sums mentioned below lie in this same closed cone. Partial transpose is used only as a necessary condition for separability.

Partial transpose on the complete first-call factor $I_1 Y_1$ leaves the output projector unchanged. Since Π is real and its partial transpose has eigenvalues $1/2, 1/2, 1/2, -1/2$, the partial transpose of T_0 has an eigenvalue $-1/8$. Every positive sum (14.4) has positive partial transpose across $I_1 Y_1 | I_2 Y_2$, including limits of such sums. Hence this effect cannot occur in a tester of that form. This proves strictness. $\square$

Consequently the constrained-separability condition on complete first- and second-call tester factors in [51, (65)–(68)] cannot be imposed on the whole unit-reset class. Reset independence is a factorization at a specified conditional preparation cut; it does not imply separability of the eventual tester across call factors. The witness is a calibration of model inclusions, not a new memory-discrimination mechanism or a physical experiment.

15. A VARIATIONAL PRINCIPLE FOR TWO-CALL RESET EXPERIMENTS

The fresh-preparation cut admits an exact finite-dimensional description of optimal decision values. No symmetry, second-moment identity, or special choice of prior is needed in this section. The description is by a concave envelope over density matrices, not over pure states alone. Its finite realization bounds the receiver needed to attain a value; it does not identify reset width with a prescribed memory dimension.

15.1. Experiments and strategy classes. Let $I_\ell = \mathbb{C}^{d_\ell}$ and let Y_ℓ be a finite classical alphabet, $\ell = 1, 2$. A finite experiment consists of a probability law (p_θ) , two ordered measurements $E^{\theta, \ell}$ on I_ℓ , and rewards $r_{j\theta} \in [0, 1]$ for decisions $1 \leq j \leq m$. The index θ is sampled once. Conditional on it the two slots are the memoryless measurement channels $\mathcal{M}_{E^{\theta, \ell}}$. Repeating one unknown measurement is the case $E^{\theta, 1} = E^{\theta, 2}$. Identification and binary decisions on subfamilies are obtained by taking indicator rewards.

Definition 15.1 (Two-call classes). An unrestricted strategy prepares a state on $I_1 \otimes R$, observes the first label y , applies a common trace-preserving map from R to $I_2 \otimes R'$, and, after observing z , measures R' to decide j . A unit-reset strategy instead applies a normal instrument $(\mathcal{K}_{yh})_h$ to R , retains its old quantum output O_{yh} , and prepares a state σ_{yh} on the complete fresh system $I_2 \otimes R_{2,yh}$. Conditional on (y, h) the preparation is exactly the tensor product of the old output with σ_{yh} . No old quantum register enters I_2 or its fresh reference. After the second label, the two receivers may be measured jointly. A classically adaptive strategy is a unit-reset strategy that measures the entire old receiver and retains only a classical record before fresh preparation.

All rules are independent of θ and defined on every formal history, including histories null under a hypothesis. Receiver spaces have no dimension bound. Instruments may have countably many outcomes; all classical records are retained. Finite or countable public randomization and the finite-dimensional trace-norm closure of the resulting output-dephased Choi tester tuples are included in each class. General public seed laws are included through this closure: their finitely many tester coordinates are limits of finite convex combinations. Early stopping is implemented by a fixed, discarded fresh call and a decision independent of its label; it is not a zero-length component of a prescribed profile. Write $P_{\text{all}}, P_{\text{reset}}, P_{\text{ca}}$ for the suprema of expected reward in these classes.

This reset cut is the two-call instance of Definition 12.1. The first class in the ordering $P_{\text{ca}} \leq P_{\text{reset}} \leq P_{\text{all}}$ agrees with complete measure-and-reprepare adaptation, not with a constraint of separability on all final reset effects. In the order I_1, Y_1, I_2, Y_2 , a classically adaptive effect is a sum of positive products across $I_1 Y_1 | I_2 Y_2$ [51, Definition 23]. The positive separable cone is

closed in these finite dimensions: its trace-one slice is the compact convex hull of product states, and trace controls passage to the cone. Only this necessary condition, not its sufficiency for classical adaptive normalization, is used below.

Lemma 15.2 (Countable records). *Let (X_h) be positive trace-class operators with $\sum_h \text{tr } X_h < \infty$. Their partial sums converge in trace norm, and their recorded direct sum lies in the trace-class ℓ^1 sum. For effects $0 \preceq F_h \preceq I$, the corresponding probability series converges absolutely. Replacing all but finitely many branches of a normal instrument by one remainder branch preserves normalization; changing the continuation on that branch changes any reward in $[0, 1]$ by at most its probability.*

Proof. The norm of a positive tail equals its trace. This proves the first assertions and bounds the probability tail by $\sum_{h>n} \text{tr } X_h$. For an instrument, the sum of the omitted completely positive maps is the remainder map; its trace is precisely the omitted outcome probability. Apply the bound to each member of the finite experiment and then average. The maximum of its finitely many tail probabilities tends to zero. No conditional state is divided by a history probability. $\square$

15.2. A complete branch normal form. Fix bases on I_1, I_2 . A coefficient map $C : I_1 \rightarrow R$ represents $|C\rangle = \sum_i |i\rangle \otimes C|i\rangle$; its squared norm is $\text{tr } C^*C$. Measuring its input gives the unnormalized receiver state $CE_y^\top C^*$.

Lemma 15.3 (Unit-reset normal form and converse). *Every implemented unit-reset strategy has a purification, still respecting the reset cut, from which its original decision statistics are recoverable by common receiver channels. For this purification there are rectangular maps*

$$C_0 : I_1 \longrightarrow R_0, \quad C_{yh} : I_1 \longrightarrow O_{yh}, \quad D_{yh} : I_2 \longrightarrow R_{2,yh}$$

with, on the input copies,

$$(15.1) \quad \rho = C_0^* C_0, \quad \text{tr } \rho = 1, \quad A_{yh} = C_{yh}^* C_{yh}, \quad \sum_h A_{yh} = \rho, \quad b_{yh} = D_{yh}^* D_{yh}, \quad \text{tr } b_{yh} = 1.$$

The receiver block on the formal record (y, h, z) is

$$(15.2) \quad (C_{yh} E_y^{\theta,1 \top} C_{yh}^*) \otimes (D_{yh} E_z^{\theta,2 \top} D_{yh}^*).$$

The sum in (15.1) converges in trace norm. Conversely, every finite collection satisfying (15.1) is realizable, up to common receiver isometries, by a unit-reset strategy with old receiver dimension at most d_1 and fresh reference dimension at most d_2 . Zero-probability histories require no extra assumption. The tester closure in Definition 15.1 is used by continuity of decision values; no physical dilation is assumed for an abstract limit before its value has been realized by the finite optimization.

Proof. Purify the first preparation, retaining the purifying system in R_0 . The Schmidt support has dimension at most d_1 . For each y , dilate each branch map of the intervening instrument to a single map $V_{yh} : R_0 \rightarrow O_{yh}$, retaining its environment as old receiver memory. On the relevant support, $\sum_h V_{yh}^* V_{yh} = I$ in the strong sense. Thus $C_{yh} = V_{yh} C_0$ has $\sum_h C_{yh}^* C_{yh} = C_0^* C_0$. The positive tails converge in trace norm by Lemma 15.2. This argument uses only the finite Schmidt support even when the original receiver or a dilation environment is infinite-dimensional. Tracing those environments recovers the original instrument.

Purify the entire fresh preparation separately. Its coefficient map is D_{yh} with squared Hilbert–Schmidt norm one. The tensor-product preparation rule, not a statement about a probe marginal, gives (15.2). Common pre-query controls are absorbed into these preparations and common post-query controls into the final receiver measurement. Fresh mixing may be purified inside the fresh reference. A retained public seed may be put in the classical record; randomization before the first call is a mixture of such strategies and cannot increase the supremum of a linear reward. More

explicitly, for seed law q_s and first coefficient maps $C_{0,s}$, take $C_0 = \bigoplus_s \sqrt{q_s} C_{0,s}$ and measure the seed register as part of the first receiver instrument. The branch maps are $C_{y,(s,h)} = \sqrt{q_s} V_{ysh} C_{0,s}$, and their Gram sum is $\sum_s q_s C_{0,s}^* C_{0,s} = C_0^* C_0$ for every y . The fresh rule retains its dependence on (s, y, h) , entirely on the fresh side of the cut. Thus the original seeded decision statistics are recovered. Countable records and tester limits follow from the preceding lemma and continuity.

For the converse let $r = \text{rank } \rho$ and choose $C_0 : I_1 \rightarrow \mathbb{C}^r$ with $C_0^* C_0 = \rho$. Its restriction to $\text{supp } \rho$ is invertible onto $\mathbb{C}^r$. Let C_0^+ be the inverse on this support, and set

$$C_{yh} = \sqrt{A_{yh}}, \quad V_{yh} = \sqrt{A_{yh}} C_0^+, \quad D_{yh} = \sqrt{b_{yh}}.$$

Positivity and $\sum_h A_{yh} = \rho$ imply $\text{supp } A_{yh} \subseteq \text{supp } \rho$. Hence $V_{yh} C_0 = \sqrt{A_{yh}}$ and $\sum_h V_{yh}^* V_{yh} = I_r$. These are Kraus operators of a trace-preserving recorded instrument with outputs in $\mathbb{C}^{d_1}$. The fresh vector $|\sqrt{b_{yh}}\rangle$ is normalized and independent of that output. Polar decomposition identifies any other rectangular maps of the same Gram matrices with these maps on their supports; receiver isometries may be absorbed into the final effects. Zero Gram matrices give zero branch maps. All other null histories are handled by the same unnormalized formulas, including projective devices. $\square$

15.3. The compact primal problem. Let $\mathcal{D}_d$ denote the density matrices on $\mathbb{C}^d$, and put

$$(15.3) \quad G_{j,yz} = \sum_{\theta} p_{\theta} r_{j\theta} (E_y^{\theta,1})^{\top} \otimes (E_z^{\theta,2})^{\top}.$$

These are fixed positive matrices on the two input copies. For $A \succeq 0$ and $b \in \mathcal{D}_{d_2}$ define

$$(15.4) \quad \Phi_y(A, b) = \sum_z \max_{\substack{F_j \succeq 0 \\ \sum_j F_j = I}} \sum_j \text{tr}[F_j(\sqrt{A} \otimes \sqrt{b}) G_{j,yz}(\sqrt{A} \otimes \sqrt{b})].$$

The POVM is chosen separately at every recorded z . It acts on $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$. For $a \in \mathcal{D}_{d_1}$ let $g_y(a) = \max_{b \in \mathcal{D}_{d_2}} \Phi_y(a, b)$ and let

$$(15.5) \quad c_y(\rho) = \sup \left\{ \sum_h \lambda_h g_y(a_h) : \lambda_h \geq 0, \sum_h \lambda_h = 1, \sum_h \lambda_h a_h = \rho \right\}.$$

The envelope is taken over *all* density matrices a_h . Replacing them by rank-one states would change the resource problem. The construction of concave envelopes is standard [56]; the operational content here is the reset realization and the common barycenter constraint across all first labels.

Theorem 15.4 (Exact reset variational principle). *For every finite experiment of Definition 15.1,*

$$(15.6) \quad P_{\text{reset}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y(\rho) = \max_{\substack{A_{yh} \succeq 0, \sum_h A_{yh} = \rho \\ \rho \in \mathcal{D}_{d_1}, b_{yh} \in \mathcal{D}_{d_2}}} \sum_{y,h} \Phi_y(A_{yh}, b_{yh}).$$

All maxima are attained. For a fixed ρ of rank r , at most r^2 nonzero instrument outcomes are required after each first label. An optimal strategy exists with old receiver dimension at most d_1 , fresh reference dimension at most d_2 , and a final quantum receiver of dimension at most $d_1 d_2$, separately from the finite classical record. The dimension bounds concern the retained registers at the recording cuts, not temporary workspaces for implementing an instrument. These assertions include arbitrary priors and arbitrary fixed pairs as particular decision problems; they assert an exact optimum, not a strict advantage for every such problem.

Proof. For fixed y , Φ_y is continuous, and is positively homogeneous in A . Continuity follows from continuity of matrix square roots and maximization of a continuous function over a compact POVM set. Maximizing also over compact $\mathcal{D}_{d_2}$ makes g_y continuous. The graph $\{(a, g_y(a)) : a \in \mathcal{D}_{d_1}\}$ is compact in a real space of affine dimension at most d_1^2 . Its convex hull is compact. The largest

height above ρ is therefore attained and is exactly $c_y(\rho)$, also allowing countable decompositions. It is an upper semicontinuous concave function of ρ . Consequently $\sum_y c_y$ attains a maximum.

Carathéodory first supplies at most $r^2 + 1$ atoms at a maximizing height over a rank- r matrix: each atom is supported on $\text{supp } \rho$, since positive matrices cannot cancel on its orthogonal complement. If more than r^2 weights are nonzero, the Hermitian atoms are linearly dependent. Choose nonzero real v_h with $\sum_h v_h a_h = 0$; tracing gives $\sum_h v_h = 0$. Small perturbations $\lambda_h \pm \epsilon v_h$ preserve the barycenter and positivity. Optimality forces $\sum_h v_h g_y(a_h) = 0$, since otherwise one sign raises the height. Move in this direction until a weight vanishes. The value is unchanged. Iteration gives at most r^2 atoms. For each atom the maximum over b and the leaf POVMs is attained.

By Lemma 15.3, the value of any purified competitor is bounded by the right side of (15.6), with $A_{yh} = \lambda_{yh} a_{yh}$. Countable records follow by the positive tail bound; public mixtures and norm limits preserve the upper. Conversely, the lemma realizes every finite maximizing collection, and its optimal leaf POVMs realize every maximum in (15.4). This proves equality and the dimension bounds. The final quantum dimension excludes the separately recorded labels. $\square$

For a binary equal-prior fixed pair, its optimal unhalved final trace separation is $4P_{\text{reset}} - 2$. Thus the theorem is also a general variational formula for that operational distance. The primal has finitely many bounded matrix variables, but is not asserted to be a convex semidefinite program or an efficient synthesis algorithm.

15.4. Dual certificates and the role of receiver feedback.

Theorem 15.5 (Duality and contact conditions). *For the same experiment,*

$$(15.7) \quad P_{\text{reset}} = \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{D}_{d_1}),$$

where H_y are Hermitian $d_1 \times d_1$ matrices. The minimum is attained. It is equivalent to minimizing λ subject to the same majorizations and $\sum_y H_y = \lambda I$. For any optimal primal and dual, every atom of positive weight is a contact point, $\text{tr}(H_y a_{yh}) = g_y(a_{yh})$, and ρ is supported on the maximal eigenspace of $\sum_y H_y$. Conversely these conditions, with maximizing fresh states and leaf POVMs, certify optimality.

Proof. Every majorization of g_y majorizes its concave envelope. Therefore $\sum_y c_y(\rho) \leq \text{tr}(\rho \sum_y H_y) \leq \lambda_{\max}(\sum_y H_y)$, which proves weak duality. For completeness, the strong-duality step is finite-dimensional. Extend c_y homogeneously to positive matrices by $\hat{c}_y(A) = \text{tr } A c_y(A / \text{tr } A)$ and $\hat{c}_y(0) = 0$. Its hypograph

$$\mathcal{K}_y = \{(A, u) : A \succeq 0, u \leq \hat{c}_y(A)\}$$

is a closed convex cone. Closedness at $A = 0$ follows from boundedness of g_y ; elsewhere it follows from the compact-graph construction. The conic problem maximizing $\sum_y u_y$, with $(A_y, u_y) \in \mathcal{K}_y$, $A_y = \rho$ and $\text{tr } \rho = 1$, is precisely the primal problem. The point $A_y = \rho = I/d_1$, $u_y = -1$ is strictly interior to all hypographs, since rewards are nonnegative and A_y is positive definite. Its value is finite, in fact at most one by the operational realization. The finite-dimensional separating-hyperplane form of conic Slater duality thus gives an attained dual with no gap [57, Sections 5.2.3 and 5.9.1].

Its multipliers can be read directly: the supremum of $u - \text{tr}(H_y A)$ on $\mathcal{K}_y$ is zero exactly when $\text{tr}(H_y a) \geq g_y(a)$ for every density matrix. Taking ρ as a free Hermitian variable under its trace constraint imposes $\sum_y H_y = \lambda I$. This gives the stated dual. Conversely, for majorants with sum S , add the positive matrix $\lambda_{\max}(S)I - S$ to one H_y ; all majorizations remain valid and the sum becomes scalar. The two forms have the same minimum. Finally, the weak-duality difference is

a sum of nonnegative atom majorization defects and a nonnegative spectral defect. It vanishes exactly under the stated contact and support conditions. $\square$

The majorizations in (15.7) are universal inequalities on density matrices. A finite sampling of them is not a dual certificate.

Corollary 15.6 (Quantitative contact certificates). *Let (H_y) be any feasible all-density majorants in Theorem 15.5, put $S_H = \sum_y H_y$ and $U = \lambda_{\max}(S_H)$, and consider a finite or countable reset normal form with achieved reward v . For every nonzero branch write $w_{yh} = \text{tr } A_{yh}$, $a_{yh} = A_{yh}/w_{yh}$, and let v_{yh} be its contribution to the reward, summed over the second labels and decisions. Then*

$$(15.8) \quad \begin{aligned} U - v &= \text{tr}[\rho(UI - S_H)] \\ &+ \sum_{y,h} w_{yh} [\text{tr}(H_y a_{yh}) - g_y(a_{yh})] \\ &+ \sum_{y,h} [w_{yh} g_y(a_{yh}) - v_{yh}]. \end{aligned}$$

Every term is nonnegative. Consequently, if $U - v \leq \varepsilon$, the total Gram weight of branches with majorization defect at least $u > 0$ is at most ε/u . If the maximal eigenspace projection P of S_H has a spectral gap $\gamma > 0$ to its complement, then $\text{tr}[(I - P)\rho] \leq \varepsilon/\gamma$.

Proof. The common-barycenter constraints give $\sum_{y,h} w_{yh} \text{tr}(H_y a_{yh}) = \text{tr}(\rho S_H)$. Add and subtract this quantity and $\sum_{y,h} w_{yh} g_y(a_{yh})$. The leaf optimization bounds v_{yh} by $w_{yh} g_y(a_{yh})$. Feasible majorization and $S_H \preceq UI$ prove nonnegativity. All series converge absolutely: the first-label alphabet is finite, $\sum_h w_{yh} = 1$ for every y , and the functions and matrices are bounded. Restricting a nonnegative sum proves the first estimate; $UI - S_H \succeq \gamma(I - P)$ proves the second. $\square$

The weights in this certificate are input Gram weights, not the hypothesis-dependent probabilities of recorded histories. With an attained optimal dual, $U - v$ is precisely the loss in optimal reward. The formula requires a genuine universal majorant, not sampled tests of its inequality.

Corollary 15.7 (Label feedback versus receiver feedback). *If no receiver-instrument outcome is used to choose the second preparation, but the first device label may be used, the optimum is*

$$P_{\text{label}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y g_y(\rho).$$

Allowing receiver feedback replaces g_y by c_y in this formula. In particular, concavity of every g_y is sufficient for $P_{\text{label}} = P_{\text{reset}}$. More generally, a label-only strategy with a common density ρ is optimal for the full reset class if it attains the contact and spectral conditions of Theorem 15.5.

Proof. With one preparation choice after each y , the normal form has $A_y = \rho$; a trace-preserving receiver operation cannot improve a subsequent unrestricted joint measurement and can be postponed to it. Conversely these choices are admissible. The assertion follows from the primal formula and the definition of concave envelope. $\square$

15.5. A geometric criterion for retained receiver advantage. Let $\mathcal{P}_{d_1}$ be the rank-one density matrices. In parallel with (15.5), define

$$(15.9) \quad c_y^{\text{cl}}(\rho) = \sup \left\{ \sum_h \lambda_h g_y(a_h) : a_h \in \mathcal{P}_{d_1}, \quad \lambda_h \geq 0, \quad \sum_h \lambda_h a_h = \rho \right\}.$$

The weights sum to one by taking traces. Every density matrix admits such a decomposition, but the value of this pure-atom envelope need not equal the all-density envelope c_y .

Theorem 15.8 (Classical and quantum receiver envelopes). *For every finite experiment of Definition 15.1,*

$$(15.10) \quad P_{\text{ca}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y^{\text{cl}}(\rho) = \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{P}_{d_1}).$$

Both optima are attained. At most $(\text{rank } \rho)^2$ nonzero first-receiver outcomes per first device label suffice at an optimal ρ . The passage from complete classical readout to retained quantum receiver memory replaces the pure-atom envelopes c_y^{cl} by the all-density envelopes c_y . In particular, the invariant

$$(15.11) \quad \Gamma = \max_{\rho} \sum_y c_y(\rho) - \max_{\rho} \sum_y c_y^{\text{cl}}(\rho)$$

is exactly the optimal retained-receiver advantage in that experiment.

Proof. After complete first readout, a classical branch has a positive input Gram effect A and retains no quantum system depending on θ . Its probability under a first effect is $\text{tr}(AE_y^{\text{T}})$. Refine a spectral decomposition $A = \sum_i \lambda_i a_i$ into the classical record and keep the former second preparation and decision on each refined branch. This does not change the score and gives a rank-one decomposition with the same barycenter. Optimizing the second action separately after the refined record can only increase it.

Conversely, for a rank-one atom $a = |u\rangle\langle u|$, the old receiver in the canonical normal form has

$$\sqrt{a} E_y^{\text{T}} \sqrt{a} = \langle u, E_y^{\text{T}} u \rangle a.$$

Its normalized state is independent of the unknown device. A joint POVM with this known factor is equivalent to a POVM on the fresh reference alone, by taking its expectation against a . Thus $g_y(a)$ is attainable with only a classical first record. The converse part of Lemma 15.3, followed by discarding this known old state, realizes every finite pure-atom decomposition. These two observations give the first equality in (15.10).

The graph of g_y on the compact pure-state set is compact. The compact-convex-hull and rank-face atom-elimination arguments of Theorem 15.4 therefore apply verbatim, giving attainment and the stated branch count. Its homogeneous hypograph is again a closed convex cone, with a strictly feasible point at a positive definite common barycenter and negative heights. The proof of Theorem 15.5 applies with majorization required only on pure atoms and gives the attained dual. Majorization on pure atoms is equivalent to majorization of c_y^{cl} on every density matrix, not to majorization of g_y on every density matrix. The last assertion now follows from the two exact primal formulas. $\square$

Corollary 15.9 (Complete separation certificates). *There is a strict retained-receiver advantage if and only if there exist a finite feasible reset collection from (15.6), with achieved reward v , and Hermitian matrices H_y such that*

$$\text{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{P}_{d_1}), \quad v > \lambda_{\max} \left(\sum_y H_y \right).$$

There is no advantage if and only if an attaining classical strategy has contact and spectral equality with an all-density dual majorant of Theorem 15.5.

Proof. The displayed inequalities compare a physically attained reset lower with a valid classical upper. If the gap is positive, take an optimal finite reset collection and an optimal classical dual, whose existence was proved above. This proves both directions of the first assertion. If the gap is zero, an optimal classical strategy also attains the reset value; complementarity with any attained reset dual gives the second condition. Conversely, that condition certifies a common upper and lower for the two classes, so their values agree. $\square$

These are certificates with finitely many matrix variables and universal majorization requirements, not finite-sampling tests. The criterion applies to any fixed-pair problem as well as to ensemble decisions, but does not require every experiment to have positive Γ . For example, the equal-prior qubit calculation below gives pure-atom majorants $H_0 = H_1 = (7/24)I$, whereas a reset strategy attains $5/8$. The gap is $1/24$. Its universal pure-atom majorization follows from the classical upper in Theorem 17.2, not from the finite arithmetic regression.

15.6. The unrestricted causal normalization.

Proposition 15.10 (Physical two-call tester blocks). *In the unnormalized convention $J^\sharp(E) = \sum_y E_y \otimes |y\rangle\langle y|$, every unrestricted strategy has positive blocks $T_{j,yz}$ and positive matrices Ξ_y with*

$$(15.12) \quad \sum_j T_{j,yz} = \Xi_y, \quad \text{tr}_{I_2} \Xi_y = \rho_1, \quad \text{tr} \rho_1 = 1.$$

Its probability of decision j on the output block (y, z) is $\text{tr}[T_{j,yz}(E_y^{\theta,1} \otimes E_z^{\theta,2})]$. In particular $0 \preceq T_{j,yz} \preceq \Xi_y$ and $\text{tr} \Xi_y = 1$.

Proof. Purify the first preparation to $|\psi\rangle \in I_1 \otimes R$. For its common intervening channel $\mathcal{A}_y : R \rightarrow I_2 \otimes R'$ put

$$X_y = (\text{Id} \otimes \mathcal{A}_y)(|\psi\rangle\langle\psi|), \quad \Xi_y = \text{tr}_{R'} X_y, \quad \rho_1 = \text{tr}_R |\psi\rangle\langle\psi|.$$

For the final POVM $(F_{j,yz})_j$ on R' define

$$T_{j,yz} = \text{tr}_{R'}[(I \otimes \sqrt{F_{j,yz}})X_y(I \otimes \sqrt{F_{j,yz}})].$$

Positivity is immediate. Cyclicity in the traced factor and $\sum_j F_{j,yz} = I$ give the sum; trace preservation of $\mathcal{A}_y$ gives the marginal. Expanding the Born rule gives the contraction. Here ρ_1 is the physical input marginal, whereas the coefficient Gram matrix ρ in (15.1) is its transpose. These two conventions must not be interchanged. Every complementary effect is positive by the same construction. No factor $d_1 d_2$ occurs, because $J^\sharp$ uses the unnormalized maximally entangled vector. Countable records and limits follow by continuity. $\square$

16. AN EXACT OPERATIONAL SEPARATION OF ACQUISITION AND RECEIVER MEMORY

The local profile law does not identify exact operational distances. We now separate three strategy classes on one finite ensemble of ordered measurement channels. The separation holds in every input dimension, throughout a depolarizing family, with exact optimal success probabilities. The unknown interface still has only classical output. The experiment has a binary hypothesis decision, not a request to identify every member of the ensemble.

16.1. A finite family and its second moments. Let $d \geq 2$, let S interchange the two factors of $\mathbb{C}^d \otimes \mathbb{C}^d$, and put $\Pi_\pm = (I \pm S)/2$. A weighted family of ordered orthonormal bases consists of projections P_y^b , $1 \leq y \leq d$, and positive weights w_b summing to one. We require The weights are real; rational or algebraic coordinates are not required by the theorem. We call this a weighted exact second-moment family, not necessarily an unweighted projective design.

$$(16.1) \quad \sum_b w_b P_y^b \otimes P_z^b = \begin{cases} (I + S)/(d(d+1)), & y = z, \\ (dI - S)/(d(d^2 - 1)), & y \neq z. \end{cases}$$

Lemma 16.1 (Finite simultaneous second moments). *In every dimension $d \geq 2$ there is such a family with at most $d^6 + 1$ bases. For $d = 2$, the six ordered eigenbases of the three Pauli matrices, with both orders and equal weights, satisfy (16.1).*

Proof. Average the ordered coordinate basis over Haar measure on $U(d)$. The second moment of a single column is supported on the symmetric subspace, invariant under $U \otimes U$, and has trace one. It is therefore $2\Pi_+[d(d+1)]$. For distinct columns the invariant moment has the form $aI + bS$; its trace is one and its contraction with S is zero, since the columns are orthogonal. These two equations give the second line of (16.1). Equivalently, this line follows by summing over the other $d-1$ columns and using the first moment I/d . The vector of all d^2 Hermitian second moments lies in a real space of dimension at most d^6 . Its Haar average belongs to the convex hull of the compact set of these vectors. Carathéodory's theorem gives a convex combination of at most $d^6 + 1$ vectors; discard zero weights. For the stated qubit family expand $P = (I \pm \sigma_a)/2$ and use $S = (I \otimes I + \sum_{a=1}^3 \sigma_a \otimes \sigma_a)/2$. $\square$

This is a finite existence argument, not an efficient design construction. Second-moment designs themselves are established objects [54].

Fix $0 \leq t \leq 1$, and consider the ordered measurements

$$(16.2) \quad C_y = I/d, \quad E_y^b(t) = (1-t)I/d + tP_y^b.$$

Set

$$(16.3) \quad L = 2(d+1) - t^2, \quad p_C = \frac{d+1-t^2}{L}, \quad p_E = \frac{d+1}{L}.$$

With probability p_C the device is C ; otherwise a basis b is chosen with law w_b and the device is $E^b(t)$. This choice is made once. Both calls use the same chosen, memoryless, classical-output device. The task is to decide C versus the alternative family. In particular, the second moments cannot be replaced by two independent first moments. Every effect is strictly positive for $t < 1$. Taking the partial trace in (16.1) gives $\sum_b w_b E_y^b(t) = C_y$. Thus the two hypotheses have identical one-call channels, including on entangled inputs, and the one-call optimum is p_E . The two-call signal concerns a device chosen once; it is absent if the alternative is independently redrawn at each call.

We compare: (i) complete first-call measurement followed by measure-and-reprepare classical adaptation, as in [51, Definition 23]; (ii) unit-reset protocols of Definition 12.1, with profile (1, 1) and arbitrary receiver quantum memory and classical feedback; and (iii) all two-call adaptive protocols. Let their optimal Bayes success probabilities be P_{ca} , P_{reset} and P_{all} , respectively. Early stopping is allowed and can be implemented by a discarded fresh call. The three closed optimization classes are precisely those of Definition 15.1. Lemma 15.3 gives the full branch normal form, and Proposition 15.10 gives the physical causal normalization independently of this example.

Theorem 16.2 (Exact two-call hierarchy). *The alternative basis index in this experiment is drawn once and reused at both calls. Independently redrawing it instead gives the scalar product channel and no improvement over the larger prior. For every finite family satisfying (16.1), every $d \geq 2$, and every $0 \leq t \leq 1$, the game (16.2)–(16.3) has the exact optimal values*

$$(16.4) \quad \begin{aligned} P_{\text{ca}} &= \frac{d+1}{L}, \\ P_{\text{reset}} &= \frac{d+1}{L} + \frac{t^2(d-1)}{2dL}, \\ P_{\text{all}} &= \frac{d+1+t^2}{L}. \end{aligned}$$

For $t > 0$ both inequalities are strict. At $t = 0$ all three values are $1/2$. The priors sum to one and $p_E - p_C = t^2/L$, so p_E is the constant-decision baseline. The middle value is attained by independent maximally entangled input–reference pairs and a final joint reference readout, without feedback. Its upper bound allows all unit-reset feedback and unrestricted receiver dimension. The

last value is attained by a parallel antisymmetric two-input state, and its upper bound allows every adaptive two-call tester.

The theorem separates a finite ensemble decision task, not an arbitrary pair of fixed channels and not a complete memory-dimension hierarchy. The middle class has hard resources $N = Q = 2$; the last construction is one acquisition group of size two, with square cost four.

16.2. A swap-filter identity. For a Hermitian matrix X , write X_+ for its positive part.

Lemma 16.3 (Negative mass of a filtered swap). *Let $A, B \succeq 0$ be $d \times d$ matrices and $K = (\sqrt{A} \otimes \sqrt{B})S(\sqrt{A} \otimes \sqrt{B})$. Then*

$$(16.5) \quad \text{tr}(-K)_+ = \frac{1}{2} \left[(\text{tr} \sqrt{\sqrt{A}B\sqrt{A}})^2 - \text{tr}(AB) \right] \leq \frac{d-1}{2d} \text{tr} A \text{tr} B.$$

The same negative mass is obtained from $(U \otimes V)S(U \otimes V)^$ when $U^*U = A$ and $V^*V = B$, even if the receiving spaces are larger. If $\text{tr} A \text{tr} B > 0$, equality in the upper bound holds if and only if $A = (\text{tr} A)I/d$ and $B = (\text{tr} B)I/d$.*

Proof. First suppose A, B are positive definite. The matrix K is similar to $(A \otimes B)S$. Conjugation of the latter by $I \otimes B$ gives $(AB \otimes I)S$. The matrix AB is similar to $\sqrt{AB}\sqrt{A}$; let its positive eigenvalues be $\lambda_1, \dots, \lambda_d$. Conjugating by the tensor square of an eigenbasis matrix leaves S unchanged. The resulting matrix has eigenvalues λ_i on the one-dimensional spaces indexed by (i, i) , and $\pm\sqrt{\lambda_i\lambda_j}$ on the two-dimensional spaces indexed by $(i, j), (j, i)$, $i < j$. Thus its negative mass is $\sum_{i < j} \sqrt{\lambda_i\lambda_j}$, which gives the equality. By Cauchy–Schwarz, $\sum_i \lambda_i \geq d^{-1}(\sum_i \sqrt{\lambda_i})^2$. Moreover, the polar decomposition and Hilbert–Schmidt Cauchy–Schwarz give $\|\sqrt{A}\sqrt{B}\|_1 \leq \sqrt{\text{tr} A \text{tr} B}$. These inequalities give the bound. Approximation by positive definite matrices proves the singular case. The polar decompositions of U, V identify their filtered swap with K on its support, adding only zero eigenvalues in the orthogonal complement. For the equality assertion, put $f = \|\sqrt{A}\sqrt{B}\|_1$ and $a = \text{tr} A$, $b = \text{tr} B$. Equality in the stated bound, with $ab > 0$, forces equality in both $\text{tr}(AB) \geq f^2/d$ and $f^2 \leq ab$. The variational formula for the trace norm and equality in Hilbert–Schmidt Cauchy–Schwarz imply $B = (b/a)A$: indeed a maximizing unitary U has $\sqrt{B}U = c\sqrt{A}$, and multiplication by its adjoint gives this proportionality. Equality in the first inequality makes all eigenvalues of $\sqrt{AB}\sqrt{A}$ equal and positive. Consequently $A = (a/d)I$ and $B = (b/d)I$. These scalar matrices plainly attain the bound. This also excludes equality at nonzero singular factors. $\square$

The normalized equality statement and its singular case are expanded in the proof of Lemma 17.1: normalized fidelity one forces equality of the two density matrices, and eigenvalue Cauchy–Schwarz then forces all d eigenvalues to be $1/d$.

16.3. Payoff and complete normalization. We use the fully transposed, unnormalized Choi representative

$$(16.6) \quad J^\sharp(E) = [(\text{Id} \otimes \mathcal{M}_E)(|\Omega\rangle\langle\Omega|)]^\top = \sum_y E_y \otimes |y\rangle\langle y|, \quad |\Omega\rangle = \sum_i |ii\rangle.$$

Its output trace is I . A simultaneous full transpose of the usual tester representative makes its contraction with $J^\sharp$ the Born probability. Factor order is I_1, Y_1, I_2, Y_2 ; expressions grouping I_1, I_2 use the canonical permutation of these factors. Dephasing the output factors does not change any score. Write $T_{0,yz}$ for the input block of the tester decision C . The gain over always guessing the alternative is

$$(16.7) \quad P - p_E = \sum_{y,z} \text{tr}(T_{0,yz} W_{yz}), \quad W_{yz} = p_C C_y \otimes C_z - p_E \sum_b w_b E_y^b(t) \otimes E_z^b(t).$$

The averaged tensor in the second term is $(1 - t^2)I/d^2$ plus t^2 times (16.1). With $k = t^2/(dL)$, substitution gives

$$(16.8) \quad W_{yy} = -kS, \quad W_{yz} = -\frac{2k}{d-1}\Pi_- \quad (y \neq z).$$

These operators are real in the coordinate basis; a full transpose therefore does not alter the following congruences.

Proof of Theorem 16.2. Classical adaptation. Each block of a measure-and-reprepare tester is a sum of positive products $A \otimes B$. Since $\text{tr}[(A \otimes B)S] = \text{tr}(AB) \geq 0$, equal labels contribute no positive gain in (16.7); different labels cannot contribute positively either. Hence $P_{\text{ca}} \leq p_E$, with equality by the constant decision. This proof also covers limits of such testers.

All unit-reset protocols. Purify the first acquisition and write its state as $\sum_i |i\rangle C_0 |i\rangle$, with $\text{tr} C_0^* C_0 = 1$. After the first device label y , combine all intervening receiver operations into an instrument with recorded outcome h . For an upper bound, retain its Stinespring environment in the receiver. The first conditional reference operator then has the form $C_{yh} E_y^\top C_{yh}^*$, where

$$A_{yh} = C_{yh}^* C_{yh}, \quad \sum_h A_{yh} = C_0^* C_0 \quad \text{for each } y.$$

Retaining this environment only enlarges receiver processing and does not import it into the second acquisition. Conditional reset permits a purification of that fresh acquisition with a map D_{yh} satisfying $B_{yh} = D_{yh}^* D_{yh}$ and $\text{tr} B_{yh} = 1$. Its complete system is independent of the old receiver conditional on (y, h) . The two-reference payoff operator at record (y, h, z) is therefore

$$(C_{yh} \otimes D_{yh}) W_{yz}^\top (C_{yh} \otimes D_{yh})^*.$$

For different labels it is nonpositive. For equal labels, maximizing any final effect bounded by the identity gives at most its positive mass, which by Lemma 16.3 is $k(d-1) \text{tr} A_{yh}/(2d)$. Summing over y, h gives

$$P_{\text{reset}} - p_E \leq k(d-1)/2 = \frac{t^2(d-1)}{2dL}.$$

Public randomization can be included in the recorded history. A normal instrument with countably many outcomes is treated by trace-class sums, or by finite truncations retaining a remainder outcome and then taking the trace-norm limit. The bound is independent of receiver size. Thus no finite-dimensional restriction on receiver storage has entered the upper bound. For $t > 0$, the equality case of Lemma 16.3 also shows that a nonrandomized protocol attaining this bound in the displayed purified normal form must satisfy $A_{yh} = (\text{tr} A_{yh})I/d$ and $B_{yh} = I/d$ on every nonzero branch. All deficits in the summed upper bound are nonnegative, so equality forces their branchwise vanishing. Summing over h then gives $C_0^* C_0 = I/d$. Maximal mixing is thus a consequence of equality, not a restriction imposed on the competing protocols. This statement concerns the purified normal form, not uniqueness of its dilations or of the final readout.

For attainment, prepare $|\Omega\rangle/\sqrt{d}$ independently on each input and reference. Declare C precisely when the two device labels agree and the references lie in Π_- . The event probability under C is $a = (d-1)/(2d^2)$. Under any $E^b(t)$ it is $(1-t^2)a$: indeed $\text{tr} E_y^b(t) = 1$, $\text{tr} E_y^b(t)^2 = t^2 + (1-t^2)/d$, and $\text{tr}[\Pi_-(E \otimes E)] = [(\text{tr} E)^2 - \text{tr} E^2]/2$. The factor d^{-2} comes from the two normalized input-reference pairs. The resulting gain is $a[p_C - p_E(1-t^2)] = t^2(d-1)/(2dL)$.

All adaptive protocols. The deterministic normalization of any two-call tester is $T_0 + T_1 = \Xi \otimes I_{Y_2}$, where, after dephasing Y_1 , Ξ has input blocks $\Xi_y \succeq 0$ satisfying

$$\text{tr}_{I_2} \Xi_y = \rho \quad (1 \leq y \leq d), \quad \text{tr} \rho = 1.$$

Here is a direct realization of these constraints in the present classical-output interface. Purify the initial state as $|\psi\rangle \in I_1 \otimes R$, and let $\mathcal{A}_y$ be the trace-preserving intervening map from R to $I_2 \otimes R'$ after observing y . Put

$$X_y = (\text{Id}_{I_1} \otimes \mathcal{A}_y)(|\psi\rangle\langle\psi|), \quad \Xi_y = \text{tr}_{R'} X_y, \quad \rho = \text{tr}_R |\psi\rangle\langle\psi|.$$

Positivity and trace preservation give the displayed marginal equations. If F_{yz} is the final receiver effect for decision C , then

$$T_{0,yz} = \text{tr}_{R'} [(I_{I_1 I_2} \otimes \sqrt{F_{yz}}) X_y (I_{I_1 I_2} \otimes \sqrt{F_{yz}})].$$

The Choi contraction (16.6) gives its Born probability, and $0 \preceq F_{yz} \preceq I$ gives $0 \preceq T_{0,yz} \preceq \Xi_y$. This also proves positivity of the complement, without an input-dimension factor from a normalized Choi state. In particular $\text{tr} \Xi_y = 1$. Since $-kS \preceq kI$ and the other payoff blocks are nonpositive,

$$P_{\text{all}} - p_E \leq k \sum_y \text{tr} \Xi_y = kd = t^2/L.$$

Prepare any normalized antisymmetric pure state on the two inputs and declare C when their labels agree. The event probability is $1/d$ under C , and $(1 - t^2)/d$ under every $E^b(t)$, because $(P_y^b \otimes P_y^b)\Pi_- = 0$. The gain is t^2/L . This is a legitimate parallel two-call strategy and attains the adaptive upper. The case $t = 0$ gives equal values $1/2$ directly. $\square$

Corollary 16.4 (Seven qubit devices). *Take the scalar qubit measurement with prior $2/5$ and each of the six ordered Pauli measurements with prior $1/10$. With the device fixed for both calls and the binary decision specified above, the exact success probabilities are*

$$P_{\text{ca}} = 3/5, \quad P_{\text{reset}} = 13/20, \quad P_{\text{all}} = 4/5.$$

Proof. Use Lemma 16.1 and substitute $d = 2, t = 1$. $\square$

The antisymmetric readout and comparison mechanism have antecedents in measurement comparison [55]; reductions between measurement and state discrimination also have a substantial history [17]. Here the assertion is the joint exact optimization over the three specified classes, including unrestricted reset feedback, on one finite classical-output ensemble and its full-rank deformation.

16.4. Quantitative stability of the operational gap.

Corollary 16.5 (A stable score separation). *Suppose every device in the finite game is replaced by a classical-output measurement channel at diamond distance at most δ from its specified channel. Keep the priors and the two-call resource classes fixed. Each of their optimal success probabilities changes by at most δ . If the displayed reset implementation also has total hard pathwise preparation defect at most ε in the sense of Proposition 14.4, its score exceeds the optimal perturbed classical-adaptation score whenever*

$$(16.9) \quad 2\delta + \varepsilon/2 < \frac{t^2(d-1)}{2dL}.$$

Proof. A two-call hybrid changes each normalized output in unhalved trace norm by at most 2δ . A fixed decision event changes by at most δ , and averaging over the unchanged priors preserves this bound. Taking suprema over a fixed strategy class gives the same two-sided bound for its optimum. The preparation certificate changes each implemented output by at most ε , hence its decision probability by at most $\varepsilon/2$. Subtract the classical upper from the reset lower in (16.4). $\square$

These are conditional norm estimates, not physical calibration claims. In the preceding corollary the same δ bounds the scalar channel and each entire basis measurement channel individually. Priors and the once-drawn-index law stay fixed; index-dependent channel perturbations are permitted. Corollary 17.3 additionally treats prior and law perturbations, with the unhalved norm factors explicit. They make the relative scale needed to preserve a small score gap explicit.

16.5. Hard budget conventions beyond the normalized domain.

Proposition 16.6 (Normalization of arbitrary hard thresholds). *For positive integer reservations, $N_\pi \leq Q_\pi \leq N_\pi^2$. Given real hard bounds $n, q \geq 1$, put*

$$M = \min\{\lfloor n \rfloor, \lfloor q \rfloor\}, \quad R = \min\{\lfloor q \rfloor, M^2\}.$$

The class $\{\pi : N_\pi \leq n, Q_\pi \leq q\}$ equals $\{\pi : N_\pi \leq M, Q_\pi \leq R\}$, and $M \leq R \leq M^2$. Consequently Theorem 14.3 applies with (M, R) .

Proof. On each path $\sum n(h) \leq \sum n(h)^2 \leq (\sum n(h))^2$. Taking hard suprema proves the first assertion. Bounded integer costs have integer suprema, so both constraints can be floored. The first inequality reduces the call bound to M and the last reduces the quadratic bound to R . Conversely the reduced bounds imply the original ones. Finally $M \leq \lfloor q \rfloor$ and $M \leq M^2$. $\square$

The normalization $N \leq Q$ allows the N -singleton lower construction; it does not say that every admissible individual policy has $Q_\pi \geq N$. If a hard threshold is below one, only the zero-call class is available. Fractional-duration reservations and expected-cost constraints are different models. None is substituted for a hard integer reservation here.

17. AN EXACT HIERARCHY WITH EQUAL HYPOTHESIS PRIORS

The strict separation does not require the prior in (16.3). Equal hypothesis priors lead to a different optimal receiver readout and to another exact hierarchy. The finite weighted second-moment family is unchanged.

17.1. A trace-norm bound. Let S be the swap on $\mathbb{C}^d \otimes \mathbb{C}^d$ and $\Pi_\pm = (I \pm S)/2$.

Lemma 17.1 (The centered filtered swap). *For $d \geq 2$, $A, B \succeq 0$, and $L = \sqrt{A} \otimes \sqrt{B}$,*

$$(17.1) \quad \|L(I - dS)L\|_1 \leq \left(d - \frac{1}{d}\right) \operatorname{tr} A \operatorname{tr} B.$$

For nonzero factors equality holds precisely when A and B are positive scalar multiples of I . The same norm and bound hold for rectangular factors having Gram matrices A, B .

Proof. Put $K = LSL$ and $f = \|\sqrt{A}\sqrt{B}\|_1$. The spectrum calculation in Lemma 16.3 gives $\|K\|_1 = f^2$, including singular factors by continuity. Since $I - dS = 2\Pi_- - (d-1)S$, the triangle inequality gives

$$\begin{aligned} \|L(I - dS)L\|_1 &\leq 2\operatorname{tr}(L\Pi_-L) + (d-1)\|K\|_1 \\ &= \operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB) + (d-1)f^2 \\ &\leq \operatorname{tr} A \operatorname{tr} B + (d-1-d^{-1})f^2 \\ &\leq (d-d^{-1})\operatorname{tr} A \operatorname{tr} B. \end{aligned}$$

The penultimate step is eigenvalue Cauchy–Schwarz, $\operatorname{tr}(AB) \geq f^2/d$; the last is $f^2 \leq \operatorname{tr} A \operatorname{tr} B$.

Here is the equality argument, also supplying a detailed treatment of singular nonzero factors in Lemma 16.3. Normalize $a = A/\operatorname{tr} A$, $b = B/\operatorname{tr} B$. The variational formula for the trace norm selects a unitary U with $\operatorname{Re} \operatorname{tr}(\sqrt{a}\sqrt{b}U) = \|\sqrt{a}\sqrt{b}\|_1$. Both $\sqrt{a}$ and $\sqrt{b}U$ have Hilbert–Schmidt norm one. Equality in Cauchy–Schwarz, with the phase chosen as above, forces $\sqrt{b}U = \sqrt{a}$;

multiplication by the adjoint gives $b = a$. This implication uses no invertibility. Equality in $\text{tr}(AB) \geq f^2/d$ then gives $\text{tr}(a^2) = 1/d$. The d nonnegative eigenvalues of a , including zeros, sum to one, so equality in their Cauchy–Schwarz inequality forces all of them to equal $1/d$. Thus nonzero singular factors cannot attain equality. Since $d - 1 - d^{-1} > 0$, equality in (17.1) forces both of these equalities. Scalar factors attain the bound directly from the eigenvalues $1 - d$ on Π_+ and $1 + d$ on Π_- . Polar decompositions of rectangular factors preserve the nonzero spectrum and add only zero eigenvalues, proving the last assertion. $\square$

17.2. Equal-prior optimization.

Theorem 17.2 (Equal-prior operational hierarchy). *Let (w_b, P_y^b) be any weighted exact second-moment family satisfying (16.1). Choose the scalar measurement $C_y = I/d$ with probability $1/2$, or choose the alternative family with probability $1/2$ and then draw one basis index b with law w_b . In the latter case the same device $E_y^b(t) = (1 - t)I/d + tP_y^b$ is used at both calls. For $d \geq 2$ and $0 \leq t \leq 1$, the three classes of Definition 15.1 have exact optimal success probabilities*

$$(17.2) \quad \begin{aligned} P_{\text{ca}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(d-1)}{2d(d+1)}, \\ P_{\text{reset}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(d-1)}{2d^2}, \\ P_{\text{all}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2}{2d}. \end{aligned}$$

Both inequalities are strict for $t > 0$, including the full-rank interval $0 < t < 1$; all three values equal $1/2$ at $t = 0$. The reset optimum is attained with independent maximally entangled input–reference pairs and no receiver feedback. Its upper includes all unit-reset receiver instruments and arbitrary receiver dimension. The unrestricted optimum is attained by a normalized antisymmetric two-input state and bounds every adaptive strategy. If the alternative basis is instead redrawn independently at each call, its two-slot averaged channel is the scalar product channel and all three equal-prior values are $1/2$.

Proof. Put $a = t^2/[2d^2(d+1)]$. Subtracting the alternative-weighted second moment from the scalar-weighted product gives

$$(17.3) \quad W_{yy} = a(I - dS), \quad W_{yz} = -\frac{a}{d-1}(I - dS) \quad (y \neq z).$$

Indeed the diagonal moment is $(1 - t^2)I/d^2 + t^2(I + S)/[d(d+1)]$, and the off-diagonal moment is $(1 - t^2)I/d^2 + t^2(dI - S)/[d(d^2 - 1)]$. These substitutions prove (17.3). Both S and $\Pi_{\pm}$ are real in the chosen basis; their full transpose does not alter these payoff blocks. The gain over the constant alternative decision is $\sum_{yz} \text{tr}(T_{C,yz} W_{yz})$.

Classically adaptive upper. After a complete first readout, a branch has an input effect $A \succeq 0$, with $\sum_h A_{yh} = \rho_1$ for each y and $\text{tr} \rho_1 = 1$. The second single-call tester has normalization $B \succeq 0$, $\text{tr} B = 1$, and decision blocks $0 \preceq T_z \preceq B$. For $\text{tr} A > 0$ set $\hat{A} = A/\text{tr} A$. Contracting (17.3) with A in the first slot gives $a \text{tr} A(I - d\hat{A})$ for $z = y$ and its negative divided by $d - 1$ for $z \neq y$. For a Hermitian Q and $0 \preceq T \preceq B$, $\text{tr}(TQ) \leq \text{tr}(BQ_+)$. Therefore the branch gain is at most

$$a \text{tr} A \text{tr}[B|I - d\hat{A}] \leq a(d-1) \text{tr} A.$$

The final inequality follows from $0 \preceq \hat{A} \preceq I$ and $d \geq 2$. The zero branch contributes zero. Summing $\sum_{yh} \text{tr} A_{yh} = d$ proves the first upper bound. It covers arbitrary second-call references and decisions, not only unassisted classical probes. Norm closure preserves it.

To attain it, use the same fixed pure input $|u\rangle$ twice and declare C on unequal labels. Under C the equal-label probability is $1/d$; under the averaged alternative it is

$$\sum_{by} w_b \langle u, E_y^b(t)u \rangle^2 = \frac{1}{d} + \frac{t^2(d-1)}{d(d+1)}.$$

This follows directly from the second moments. The corresponding Bayes gain is half their difference, namely $ad(d-1)$.

Unit-reset upper. Use Lemma 15.3 without any restriction on its rectangular receiver spaces. On a branch (y, h) put

$$X = (C_{yh} \otimes D_{yh})(I - dS)(C_{yh} \otimes D_{yh})^*.$$

The elementary identity $\max_{0 \preceq F \preceq I} \text{tr}(FX) = \text{tr}(X_+)$ and (17.3) give total branch gain at most

$$a \text{tr}(X_+) + a \text{tr}((-X)_+) = a\|X\|_1 \leq a(d - d^{-1}) \text{tr} A_{yh},$$

since $\text{tr} b_{yh} = 1$. There are $d-1$ opposite-sign output blocks, which cancel their denominator. Summing over y, h gives $a(d^2 - 1) = t^2(d-1)/(2d^2)$. Countable outcomes and null histories use unnormalized positive tails.

For attainment, prepare two independent normalized maximally entangled pairs $|\Omega\rangle/\sqrt{d}$. On equal device labels declare C when the references lie in Π_- ; on unequal labels declare C when the references lie in Π_+ . This is one final joint receiver measurement, with complementary effect $I - F_C$ on each classical block. Its event probabilities are

$$(17.4) \quad q_C = \frac{(d-1)(d+2)}{2d^2}, \quad q_b = \frac{(d-1)(d+2-2t^2)}{2d^2} \quad \text{for every } b.$$

To verify them, the receiver block is $(E_y^b(t))^\top \otimes (E_z^b(t))^\top / d^2$ and $\text{tr}[\Pi_\pm(A \otimes B)] = [\text{tr} A \text{tr} B \pm \text{tr}(AB)]/2$. Here $\text{tr} E_y = 1$, $\text{tr} E_y^2 = t^2 + (1-t^2)/d$, and $\text{tr} E_y E_z = (1-t^2)/d$ for $y \neq z$. Counting the d equal-label and $d(d-1)$ unequal-label blocks gives (17.4). Half the difference is the reset upper.

Unrestricted upper. By Proposition 15.10, $T_{C,yz} \preceq \Xi_y$ and $\text{tr} \Xi_y = 1$. The sum of the positive payoff parts for a fixed first label is

$$(W_{yy})_+ + \sum_{z \neq y} (W_{yz})_+ = a(d+1)\Pi_- + a(d-1)\Pi_+ \preceq a(d+1)I.$$

Thus the gain is at most $ad(d+1) = t^2/(2d)$. Any normalized state supported on the nonzero antisymmetric input subspace attains it by declaring C on equal labels: their probability is $1/d$ under C and $(1-t^2)/d$ under every alternative.

The gaps are

$$(17.5) \quad P_{\text{reset}}^{\text{eq}} - P_{\text{ca}}^{\text{eq}} = \frac{t^2(d-1)}{2d^2(d+1)}, \quad P_{\text{all}}^{\text{eq}} - P_{\text{reset}}^{\text{eq}} = \frac{t^2}{2d^2}.$$

Finally $\sum_b w_b \mathcal{M}_{E^b(t)} = \mathcal{M}_C$ as a channel identity. Two independent draws give its tensor square, hence the same two-slot statistics even for adaptive testers. This proves the last assertion and the endpoint statements. $\square$

The equal priors concern the two hypotheses, not a uniform prior on all individual devices. The result removes prior tuning, but still uses a shared latent device. It does not assert a gap for every fixed pair. Together with Theorem 15.4, it gives a general optimization principle and an explicitly solved family, rather than identifying these two levels of generality.

17.3. Seven explicit qubit devices. For $d = 2$ write

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The devices are $C = (I/2, I/2)$ and, for $A \in \{X, Y, Z\}$ and $\epsilon \in \{-1, 1\}$,

$$E^{A,\epsilon}(t) = \left(\frac{I + t\epsilon A}{2}, \frac{I - t\epsilon A}{2} \right).$$

The six alternatives have conditional weight $1/6$. Both outcome orders are included: they cancel the first Pauli moments and supply the required ordered second moments. At $t = 1$ they are the six ordered Pauli eigenbasis measurements. For the prior of Theorem 16.2, C has mass $2/5$ and each alternative mass $1/10$; for Theorem 17.2, these masses are $1/2$ and $1/12$.

For reset acquisition prepare

$$|\phi^+\rangle_{I_1 R_1} \otimes |\phi^+\rangle_{I_2 R_2}, \quad |\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}.$$

The inputs are independent. The two retained references are measured jointly only after both calls. Let $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$, $\Pi_- = |\psi^-\rangle\langle\psi^-|$ and $\Pi_+ = I_4 - \Pi_-$ on $R_1 \otimes R_2$. The receiver block for any two devices is $E_y^\top \otimes F_z^\top/4$. Under $J^\sharp$, the decision tester block is $F_{C,yz}^\top/4$, where

$$F_{C,yz} = \mathbf{1}_{y=z} \Pi_- \quad \text{for the prior } 2/5, \quad F_{C,yz} = \begin{cases} \Pi_-, & y = z, \\ \Pi_+, & y \neq z \end{cases} \quad \text{for equal priors.}$$

The other block is $(I_4 - F_{C,yz})^\top/4 \succeq 0$. Their sum is $I_4/4$ for each (y, z) , with partial trace $I_2/2$ over I_2 and trace one. This verifies the causal normalization in the order I_1, Y_1, I_2, Y_2 , using the canonical permutation to group the input copies. The transpose is required for arbitrary complex receiver effects, even though the displayed antisymmetric projectors are real.

At $t = 1$, the first reset event has probabilities $1/8$ under C and zero under every alternative. Its Bayes score is $(2/5)(1/8) + (3/5) = 13/20$. The equal-prior reset event has probabilities $1/2$ and $1/4$, respectively, giving $\frac{1}{2}(\frac{1}{2}) + \frac{1}{2}(\frac{3}{4}) = 5/8$. For unrestricted acquisition put the normalized state $|\psi^-\rangle$ on $I_1 \otimes I_2$, not on the references, and declare C on equal labels. The probabilities are $1/2$ and zero, yielding $4/5$ for the first prior and $3/4$ for equal priors. For classical adaptation, the biased-prior optimum is the constant decision $3/5$. For equal priors, two inputs $|0\rangle$ and the unequal-label event have probabilities $1/2$ under C and $1/3$ averaged over alternatives, giving $7/12$. Consequently the two exact triples on the same devices are

$$\frac{3}{5} < \frac{13}{20} < \frac{4}{5}, \quad \frac{7}{12} < \frac{5}{8} < \frac{3}{4}.$$

The class upper bounds, not these evaluations alone, prove optimality.

17.4. An open neighborhood of the equal-prior experiment.

Corollary 17.3 (Channel, prior, and ensemble stability). *Fix the family and $t > 0$. Replace the scalar device and each entire basis measurement channel individually by channels within unhalved diamond distance δ . The perturbations may depend on the basis index and may change all its effects, but must form legal channels on the same interfaces. Keep the once-drawn-index rule. Replace the scalar prior $1/2$ by q and the conditional basis law w by ν , and put*

$$\eta = \frac{1}{2} \sum_b |\nu_b - w_b|, \quad \kappa = \delta + |q - \frac{1}{2}| + \eta.$$

For each of the three fixed operational classes, $|\tilde{P}_{\text{class}} - P_{\text{class}}^{\text{eq}}| \leq \kappa$. Thus both class gaps remain strict when $2\kappa < t^2(d-1)/[2d^2(d+1)]$. If the exhibited reset protocol has, in addition, a uniform

reference-complete diamond certificate of preparation defect whose hard pathwise sum is at most ε , its implemented score still exceeds the perturbed classical optimum provided

$$2\kappa + \varepsilon/2 < \frac{t^2(d-1)}{2d^2(d+1)}.$$

Proof. For a fixed protocol and device index, the two-call hybrid bounds output trace distance by 2δ . Normalization makes the change of any decision probability at most δ , with the factor $1/2$ for unhalved trace distance. Changing the binary prior changes success by at most $|q - 1/2|$; changing the alternative law changes an average of $[0, 1]$ -valued successes by at most η . The bound is uniform in the protocol, so taking suprema preserves it in both directions. Subtract and use (17.5). The preparation hybrid changes each implemented output by at most ε , hence its success by at most $\varepsilon/2$. This last statement concerns a certified implementation of the displayed reset protocol, not an enlargement of the ideal optimization class. $\square$

17.5. Quantitative rigidity of near-optimal reset strategies. The equality condition for the centered swap has a stable form. It controls the input Gram matrices in a chosen purified normal form, rather than specifying unique instruments, dilations, or receiver readouts. For $d \geq 2$ set

$$c_d = d - 1 - d^{-1} > 0, \quad K_d = d \left(1 + \frac{9}{2c_d} \right).$$

Lemma 17.4 (Stable centered-swap equality). *For density matrices a, b on $\mathbb{C}^d$, let*

$$\Delta(a, b) = d - d^{-1} - \|(\sqrt{a} \otimes \sqrt{b})(I - dS)(\sqrt{a} \otimes \sqrt{b})\|_1.$$

Then $\Delta(a, b) \geq 0$ and

$$(17.6) \quad \|a - I/d\|_1^2 \leq K_d \Delta(a, b), \quad \|b - I/d\|_1^2 \leq K_d \Delta(a, b).$$

No invertibility is required.

Proof. Put $f = \|\sqrt{a}\sqrt{b}\|_1$, $\eta = 1 - f^2$, and $e = \text{tr}(ab) - f^2/d$. The proof of Lemma 17.1 gives $\eta, e \geq 0$ and

$$(17.7) \quad \Delta(a, b) \geq c_d \eta + e.$$

Choose a unitary V extending the right polar factor so that $Z = \sqrt{a}\sqrt{b}V = \sqrt{\sqrt{a}b\sqrt{a}} \succeq 0$. This is possible also when the matrices are singular. The Hilbert–Schmidt norm, denoted by $\|\cdot\|_2$, satisfies

$$\|\sqrt{b}V - \sqrt{a}\|_2^2 = 2(1 - f), \quad \|Z - a\|_2 \leq \sqrt{2\eta}, \quad \|Z - fI/d\|_2^2 = e.$$

Here $\|\sqrt{a}\|_{\text{op}} \leq 1$ and $1 - f \leq \eta$. Since also $1 - f \leq \sqrt{\eta}$, the triangle inequality yields

$$\|a - I/d\|_2 \leq (\sqrt{2} + d^{-1/2})\sqrt{\eta} + \sqrt{e}.$$

Weighted Cauchy–Schwarz bounds its square by $[1 + (\sqrt{2} + d^{-1/2})^2/c_d](c_d\eta + e)$. For $d \geq 2$, $(\sqrt{2} + d^{-1/2})^2 \leq 9/2$; and $\|T\|_1^2 \leq d\|T\|_2^2$ for a $d \times d$ matrix T . These facts and (17.7) prove the first estimate. Interchanging a, b proves the second, since swap conjugation leaves the centered-swap norm unchanged. $\square$

Corollary 17.5 (Near-optimal acquisition marginals). *Fix the equal-prior experiment of Theorem 17.2 with $t > 0$. Suppose a unit-reset strategy achieves reward $v \geq P_{\text{reset}}^{\text{eq}} - \varepsilon$. Choose its purified normal form from Lemma 15.3. On nonzero branches put*

$$a_{yh} = A_{yh}/\text{tr } A_{yh}, \quad \mu_{yh} = \text{tr } A_{yh}/d.$$

Then $\sum_{yh} \mu_{yh} = 1$ and

$$(17.8) \quad \sum_{y,h} \mu_{yh} \left(\|a_{yh} - I/d\|_1^2 + \|b_{yh} - I/d\|_1^2 \right) \leq \frac{4K_d d(d+1)}{t^2} \varepsilon.$$

For a public mixture the same conclusion holds after adjoining the seed and multiplying these weights by its law. Finite and countable recorded instruments are both allowed.

Proof. Write $\alpha = t^2/[2d^2(d+1)]$ for the coefficient in (17.3). The complete reset upper, before applying the centered-swap bound, gives

$$v \leq \frac{1}{2} + \alpha \sum_{yh} \text{tr } A_{yh} [d - d^{-1} - \Delta(a_{yh}, b_{yh})].$$

Purification does not obstruct recovering the original decision rule, so this inequality bounds its achieved reward as well. The common Gram identity gives $\sum_{yh} \text{tr } A_{yh} = d$. Subtract from the attained optimum to obtain $\sum_{yh} \mu_{yh} \Delta(a_{yh}, b_{yh}) \leq \varepsilon/(\alpha d)$. Apply Lemma 17.4 to both factors and substitute α . Positive trace tails justify countable sums. For a public mixture, apply the same upper to each component and average before subtracting its reward from the common optimum. $\square$

In particular, the total μ -weight on which either normalized Gram matrix is at trace distance at least $u > 0$ from I/d is at most $4K_d d(d+1)\varepsilon/(t^2 u^2)$. These are Gram weights, not probabilities under a selected hypothesis. At zero loss the result recovers branchwise isotropy. The estimate is nontrivial at the relative scale ε/t^2 ; it does not turn fixed calibration error into vanishing-signal robustness. It asserts neither uniqueness of a dilation nor a device-independent certification of the reset cut.

18. RANK-RESOLVED RECEIVER RESOURCES

The distinction between classical and quantum receivers extends to an attained hierarchy at each pair of retained-register dimensions. This section keeps the reset cut fixed and resolves its two sides separately. It supplies a general variational characterization and an exactly solved spectrum, rather than identifying all notions of bounded quantum memory.

18.1. Two specified recording cuts. Use the finite experiment of Definition 15.1. Fix integers $1 \leq k \leq d_1$ and $1 \leq \ell \leq d_2$.

Definition 18.1 (Retained-register dimensions). The class $\mathfrak{R}_{k,\ell}$ consists of unit-reset strategies for which, conditional on every recorded first history, the entire retained old quantum system has dimension at most k and the entire fresh reference accompanying the second input has dimension at most ℓ . The old system and the complete fresh input–reference system are in a tensor product at that cut. All other quantum registers are discarded before the second call; classical records may be retained without a cardinality bound. The initial first-call reference and temporary workspaces before this cut have no dimension restriction. The final measurement on the two retained receivers is arbitrary. Mixed preparations, recorded normal instruments, public randomization, null histories, discarded terminal calls and finite-dimensional tester closure have the conventions of Definition 15.1. Denote the optimal reward by $P_{k,\ell}$.

In particular, this is not a bound on the largest workspace throughout an implementation. Enlarging the old output by keeping an instrument environment would violate the stated resource. The refinement in the next lemma records that environment’s Kraus index *classically* instead, without retaining it coherently.

Lemma 18.2 (Dimension-preserving refinement). *For the optimization over $\mathfrak{R}_{k,\ell}$, the normal form (15.1)–(15.2) is exact with*

$$(18.1) \quad \text{rank } A_{yh} \leq k, \quad \text{rank } b_{yh} \leq \ell.$$

Every implemented competitor can be refined into such branches while preserving its score if the added classical indices are ignored in its decisions. Conversely every finite Gram collection satisfying these ranks and the common-barycenter constraints is realizable in $\mathfrak{R}_{k,\ell}$.

Proof. Purify the first input, retaining a reference whose relevant Schmidt support has dimension at most d_1 . A completely positive branch of the subsequent map from that support to $\mathbb{C}^k$ has a finite Kraus representation $X \mapsto \sum_q V_{yhq} X V_{yhq}^*$, with $V_{yhq} : \mathbb{C}^{r_0} \rightarrow \mathbb{C}^k$. Refine h to (h, q) , record q classically, and use the original fresh preparation and final effects. Summing over q recovers the old channel and its decision statistics. Thus $C_{yhq} = V_{yhq} C_0$ has rank at most k , and completeness still gives $\sum_{h,q} C_{yhq}^* C_{yhq} = \rho$ for every y . An original mixed first preparation is recovered by discarding its purification before this old-output cut; this discarding is part of that same completely positive branch, not an additional retained quantum register.

A mixed fresh state on $I_2 \otimes \mathbb{C}^\ell$ has a finite spectral mixture of normalized pure states in that same space. Sample and record its mixture index, use the corresponding vector, and ignore that index in the original decision. Its coefficient map $D : I_2 \rightarrow \mathbb{C}^\ell$ has Gram rank at most ℓ . Multiplying the old coefficient map by the square root of this mixing probability preserves the common Gram sum. This is a classical refinement, not a purification into a larger fresh reference. It is independent of the old quantum output conditional on the refined record. Countable outcomes are handled by the trace tails of Lemma 15.2. Public mixtures and tester limits preserve the resulting value bound.

Conversely choose $C_0^* C_0 = \rho$ on its Schmidt support. Factor each A_{yh} as $C_{yh}^* C_{yh}$ with $C_{yh} : I_1 \rightarrow \mathbb{C}^k$, and each b_{yh} as $D_{yh}^* D_{yh}$ with $D_{yh} : I_2 \rightarrow \mathbb{C}^\ell$. The maps $V_{yh} = C_{yh} C_0^+$ satisfy $\sum_h V_{yh}^* V_{yh} = I$ on that support. They give a recorded trace-preserving instrument. The normalized fresh vector $|D_{yh}\rangle$ is prepared independently, and polar isometries transfer the prescribed leaf POVMs to $\mathbb{C}^k \otimes \mathbb{C}^\ell$. Zero Gram branches are simply zero maps. The original unnormalized formulas define all null histories. $\square$

18.2. An attained hierarchy and its dual. Write $\mathcal{D}_d^{\leq q}$ for the compact set of density matrices of rank at most q . In the notation of (15.4), put

$$(18.2) \quad g_y^\ell(a) = \max_{b \in \mathcal{D}_{d_2}^{\leq \ell}} \Phi_y(a, b),$$

$$(18.3) \quad c_y^{k,\ell}(\rho) = \max \left\{ \sum_h \lambda_h g_y^\ell(a_h) : a_h \in \mathcal{D}_{d_1}^{\leq k}, \quad \lambda_h \geq 0, \quad \sum_h \lambda_h a_h = \rho \right\}.$$

Weights sum to one by taking traces. All density matrices have such a representation, since rank-one atoms are allowed.

Theorem 18.3 (Rank-resolved variational principle). *For every finite two-call experiment,*

$$(18.4) \quad \begin{aligned} P_{k,\ell} &= \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y^{k,\ell}(\rho) \\ &= \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y^\ell(a) \quad (a \in \mathcal{D}_{d_1}^{\leq k}), \end{aligned}$$

where the H_y are Hermitian. Both extrema are attained. At an optimal barycenter ρ , at most $(\text{rank } \rho)^2$ nonzero receiver outcomes after each first label suffice; a final quantum receiver of

dimension kl suffices. The values increase monotonically in either dimension, with

$$P_{1,d_2} = P_{\text{ca}}, \quad P_{d_1,d_2} = P_{\text{reset}}.$$

The contact, spectral and quantitative defect criteria of Theorem 15.5 and Corollary 15.6 hold with g_y^ℓ and the rank-restricted atoms.

Proof. The function Φ_y is continuous and homogeneous in its first positive argument. The rank-bounded sets are compact, so g_y^ℓ is continuous and its graph on $\mathcal{D}_{d_1}^{\leq k}$ has compact convex hull. The largest height over each ρ is an attained, upper semicontinuous concave envelope. Lemma 18.2 gives both operational directions. The atom-elimination proof of Theorem 15.4 uses only linear dependence of atoms on the support of ρ , and does not change any atom. It therefore preserves the rank restriction and gives the same branch bound. Compactness supplies all fresh states and final POVMs.

For clarity, dual attainment does not require the rank-bounded atom set to be convex. The homogeneous hypograph of its concave envelope is a closed convex cone, finite on every positive matrix and nonnegative there. At the common matrix I/d_1 with every height -1 , all these cones are strictly feasible. The conic argument of Theorem 15.5 applies without alteration: its dual is precisely majorization of the atom graph in (18.4). Bounded reward gives a finite value. Slater duality yields no gap and an attained Hermitian majorant. Subtracting the primal value from the dual upper gives the same three nonnegative defects, proving the contact and quantitative assertions. Inclusion of the rank-bounded atom sets proves monotonicity. The two endpoint identifications are Theorems 15.8 and 15.4. $\square$

The dual inequalities are universal on their rank-bounded domains; sampling is not a certificate. In particular, strict improvement between any two nested dimension pairs is equivalent to a feasible finite strategy for the larger pair exceeding an attained dual upper for the smaller pair. This criterion covers arbitrary fixed pairs, without asserting that every such pair exhibits a gap.

18.3. Rank-sensitive swap bounds. The rank in the next lemma is an upper bound on each of the two Gram ranks; the active overlap may have a smaller rank.

Lemma 18.4 (Swap bounds at finite rank). *Let $1 \leq r \leq d$, $d \geq 2$, and let $A, B \succeq 0$ have ranks at most r . With $L = \sqrt{A} \otimes \sqrt{B}$,*

$$(18.5) \quad \text{tr}[-LSL]_+ \leq \frac{r-1}{2r} \text{tr } A \text{tr } B,$$

$$(18.6) \quad \|L(I - dS)L\|_1 \leq \left(d - \frac{1}{r}\right) \text{tr } A \text{tr } B.$$

The bounds also hold if only the smaller of the two ranks is at most r , and for rectangular factors with the same Grams. Both bounds are sharp: take $A = aP/r$, $B = bP/r$, where P is any rank- r projection and $a, b \geq 0$. For nonzero factors, equality in the first bound with $r > 1$ requires these forms with a common P ; equality in the second requires them when $(d, r) \neq (2, 1)$.

Proof. The nonzero eigenvalues of $\sqrt{AB}\sqrt{A}$ number at most r . Consequently, for $f = \|\sqrt{A}\sqrt{B}\|_1$,

$$\text{tr}(AB) \geq f^2/r, \quad f^2 \leq \text{tr } A \text{tr } B.$$

Insert the first inequality into Lemma 16.3 to get (18.5). The triangle step of Lemma 17.1 gives

$$\|L(I - dS)L\|_1 \leq \text{tr } A \text{tr } B - \text{tr}(AB) + (d-1)f^2 \leq \text{tr } A \text{tr } B + (d-1-r^{-1})f^2.$$

The coefficient is nonnegative, proving (18.6). These arguments only use the smaller rank, not a common support assumption. The spectrum identities hold for singular factors by the proved continuity argument, and polar isometries handle rectangular maps.

For equality, when the coefficient of f^2 is positive, normalized fidelity one gives $A/\text{tr } A = B/\text{tr } B$ without an invertibility assumption, as in Lemma 17.1. Equality in the rank- r Cauchy–Schwarz inequality then forces exactly r equal nonzero eigenvalues, proving the common-projection forms. For such forms the centered spectrum is $(1-d)ab/r^2$ with multiplicity $r(r+1)/2$ and $(1+d)ab/r^2$ with multiplicity $r(r-1)/2$; the negative swap has mass $ab(r-1)/(2r)$. This proves sharpness. The excluded centered case has zero coefficient and different equality cases: for $d=2, r=1$, orthogonal pure factors also attain the bound. For the first inequality at $r=1$, the negative mass is always zero. $\square$

18.4. The complete two-cut spectrum of the basis experiment. The following formula resolves every pair of retained-register thresholds on the same once-selected finite basis family. No additional design is introduced at a new threshold.

Theorem 18.5 (Exact dimension spectra). *Fix $d \geq 2$, $0 \leq t \leq 1$, and a weighted exact second-moment family satisfying (16.1). Under the once-drawn device rule, let $P_{k,\ell}^b$ use the priors of Theorem 16.2, and let $P_{k,\ell}^{\text{eq}}$ use the equal hypothesis priors of Theorem 17.2. For $1 \leq k, \ell \leq d$, set $r = \min\{k, \ell\}$ and $L_0 = 2(d+1) - t^2$. Then*

$$(18.7) \quad P_{k,\ell}^b = \frac{d+1}{L_0} + \frac{t^2(r-1)}{2rL_0},$$

$$(18.8) \quad P_{k,\ell}^{\text{eq}} = \frac{1}{2} + \frac{t^2(dr-1)}{2dr(d+1)}.$$

These are optima over the complete classes $\mathfrak{R}_{k,\ell}$. For $t > 0$ they strictly increase at every increase of the smaller threshold, and attain the unrestricted reset values precisely at $k = \ell = d$. The two-cut resource dependence is therefore exactly through $\min\{k, \ell\}$ for these experiments.

Proof. Use the refined normal form of Lemma 18.2; there is no retained dilation environment outside the counted receivers. Every branch has $\text{tr } b_{yh} = 1$ and smaller Gram rank at most r , while $\sum_{yh} \text{tr } A_{yh} = d$. For the biased priors the equal-label payoff is $-t^2 S/(dL_0)$ and all unequal-label payoffs are nonpositive. Equation (18.5) bounds the gain by $t^2(r-1)/(2rL_0)$. For equal priors write $\alpha = t^2/[2d^2(d+1)]$. The $d-1$ opposite-sign payoff blocks cancel their denominator, so the branch gain is at most $\alpha \|(C_{yh} \otimes D_{yh})(I - dS)(C_{yh} \otimes D_{yh})^*\|_1$. Equation (18.6) and the Gram sum give $\alpha d(d-1/r)$, which is (18.8) minus $1/2$. Classical mixtures and trace-norm closure preserve both upper bounds.

Here is a single finite construction attaining them. Index the input basis cyclically modulo d , and let $J_h : \mathbb{C}^r \rightarrow \mathbb{C}^d$ include the r consecutive coordinates starting at h , $0 \leq h < d$. Put $P_h = J_h J_h^*$; then $\sum_h P_h = rI$. Choose the first coefficient map $C_0 = I/\sqrt{d}$ and, after every first device label, the receiver instrument and fresh coefficient maps

$$(18.9) \quad V_h = J_h^*/\sqrt{r}, \quad C_{yh} = J_h^*/\sqrt{dr}, \quad D_{yh} = J_h^*/\sqrt{r}.$$

Thus $\sum_h V_h^* V_h = I$, $A_{yh} = P_h/(dr)$, $b_{yh} = P_h/r$ and $\sum_h A_{yh} = I/d$. The fresh vector has norm one and is independent of the old receiver. Both retained quantum registers have dimension r ; h is classical. The initial reference has dimension d , as explicitly allowed before the recording cut in Definition 18.1.

On these receivers the filtered swap is $S_r/(dr^2)$ and the centered one is $(I_{r^2} - dS_r)/(dr^2)$. For the biased prior, declare C only on equal labels and the antisymmetric receiver projection. For equal priors, declare C on the antisymmetric projection on equal labels and on the symmetric projection on unequal labels. Complementary effects complete a POVM. Their spectral signs attain every branch bound above. At $r=1$ the antisymmetric projection is zero; the second readout reduces to the unequal-label rule. At $r=d$ the instrument's redundant classical randomization

may be omitted. All constructions remain normalized at $t = 0$ and on projective null histories at $t = 1$. $\square$

Corollary 18.6 (Adjacent gains and dimension certificates). *For $1 \leq r < d$, abbreviate the values in Theorem 18.5 by P_r^b and P_r^{eq} . Their adjacent gains are*

$$P_{r+1}^b - P_r^b = \frac{t^2}{2L_0 r(r+1)}, \quad P_{r+1}^{\text{eq}} - P_r^{\text{eq}} = \frac{t^2}{2d(d+1)r(r+1)}.$$

For the equal-prior perturbations of Corollary 17.3, with its score error κ , every perturbed class value moves by at most κ . In particular an achieved ideal-reset score strictly above $P_r^{\text{eq}} + \kappa$ requires both counted receiver dimensions to exceed r . The adjacent gap survives if 2κ is smaller than its displayed value. A certified additional preparation defect ε changes the implemented score by at most $\varepsilon/2$ and may be added to these error budgets.

Proof. Subtract the consecutive exact formulas. The hybrid and probability law bounds are uniform in a fixed protocol, independently of its receiver dimensions, so they hold for every rank-restricted supremum. If either dimension were at most r , its nominal value would be at most P_r^{eq} by Theorem 18.5. This proves the asserted certificate and the perturbation statements. $\square$

These dimension certificates assume the specified reset architecture and calibration bounds. They are not device-independent conclusions from an uncalibrated score. The entire two-cut hierarchy still lies inside the reset class; the unrestricted coherent-acquisition optimum of Theorem 17.2 is a different resource.

18.5. Relation to constrained-memory discrimination. Memory-constrained tester optimization and its relation to constrained separability are established in [51], including the classical adaptive boundary. Theorem 18.3 is a rank-envelope characterization for the specified classical-output, fresh-preparation cut, and Theorem 18.5 solves its two-coordinate values on one fixed finite family. It does not replace the general channel-memory formulations of that work. The autonomous multi-time hierarchy of [52] carries coherent information between successive acquisitions and compiles arbitrary finite-time testers with a counter. Here increasing the two receiver dimensions never removes the reset cut. Consequently saturation means the full reset optimum, not the unrestricted strategy norm. These different saturation statements are compatible and cannot be interchanged.

19. INITIAL SPECTRA AND RIGIDITY OF RANK-OPTIMAL RECEIVERS

The dimension spectrum determines an optimal value. We now determine which initial spectra can attain it, and how a small loss in value restricts the acquisition marginals. Throughout this section the basis index is drawn once and retained for both calls, the hypothesis priors are equal, and $0 < t \leq 1$. The family satisfies (16.1). Put

$$r = \min\{k, \ell\}, \quad p_r = \frac{1}{2} + \frac{t^2(dr-1)}{2dr(d+1)}, \quad \alpha = \frac{t^2}{2d^2(d+1)}.$$

All ranks and norms refer to the fixed input copies. Trace norms are unhalved. An initial Gram matrix ρ means a pure first acquisition $|C_0\rangle$ with $C_0^* C_0 = \rho$, in the convention of Lemma 15.3; its input marginal is $\rho^\top$. The receiver cut is exactly Definition 18.1.

19.1. The complete set of optimal initial spectra.

Theorem 19.1 (Optimal initial Gram matrices). *A fixed initial Gram matrix $\rho \in \mathcal{D}_d$ admits a strategy in $\mathfrak{R}_{k,\ell}$ attaining p_r if and only if*

$$(19.1) \quad \rho \preceq I/r.$$

Every such ρ has a decomposition

$$(19.2) \quad \rho = \sum_{h=1}^s w_h P_h / r, \quad s \leq d, \quad w_h > 0, \quad \sum_h w_h = 1,$$

where the P_h are rank- r projections commuting with ρ . This decomposition gives an attaining strategy with the same receiver instrument after every first device label and with both retained quantum registers of dimension r .

Except when $(d, r) = (2, 1)$, every attaining strategy in the dimension-preserving refined normal form has, on each nonzero branch,

$$(19.3) \quad A_{yh} = w_{yh} P_{yh} / r, \quad b_{yh} = P_{yh} / r, \quad \text{rank } P_{yh} = r.$$

Together with the common Gram sums and maximizing leaf decisions, these conditions are also sufficient. They specify Gram data, not unique dilations or unique final measurements.

Proof. For a normalized branch write $a = A / \text{tr } A$ and $b = b_{yh}$. The equal-prior upper in Theorem 18.5, before applying its final inequality, is

$$(19.4) \quad v \leq \frac{1}{2} + \alpha \sum_{y,h} w_{yh} \|(\sqrt{a_{yh}} \otimes \sqrt{b_{yh}})(I - dS)(\sqrt{a_{yh}} \otimes \sqrt{b_{yh}})\|_1.$$

The smaller Gram rank is at most r , and $\sum_{y,h} w_{yh} = d$. All branch deficits from the bound $d - 1/r$ are nonnegative. Attainment therefore forces equality on every branch of positive weight. When $(d, r) \neq (2, 1)$, Lemma 18.4 gives (19.3). For each fixed y , summing yields $\rho = \sum_h w_{yh} P_{yh} / r \preceq I / r$, since $\sum_h w_{yh} = 1$. In the exceptional case $r = 1$, the claimed spectral condition holds for every density matrix anyway. The fixed- ρ compact-envelope argument of Theorem 18.3 gives attainment of its constrained value, so the same conclusion applies when an optimum was initially defined through tester closure.

We give a finite construction for the converse. Diagonalize ρ with eigenvalues λ_i and set $x_i = r\lambda_i$ and $s_i = \sum_{j \leq i} x_j$, $s_0 = 0$. Then $0 \leq x_i \leq 1$ and $s_d = r$. For almost every $u \in [0, 1)$, mark the r points $u, u+1, \dots, u+r-1$ in $[0, r)$. Select coordinate i if its half-open interval $[s_{i-1}, s_i)$ contains a marked point. Each interval has length at most one, so exactly r distinct coordinates are selected. The probability of selecting coordinate i is x_i . The selected subset is constant between successive fractional parts of the s_i . There are at most d such intervals of positive length, because s_0 and s_d have the same fractional part. Their lengths are the weights w_h , and their selected coordinate projections are P_h . Taking diagonal entries proves (19.2). Endpoints have zero measure and may be assigned either adjacent choice. Rational eigenvalues give rational weights by this construction; no rational eigenbasis is presumed.

Choose isometries $J_h : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $J_h J_h^* = P_h$ and put

$$C_{yh} = \sqrt{w_h/r} J_h^*, \quad D_{yh} = J_h^* / \sqrt{r}, \quad V_{yh} = C_{yh} C_0^+.$$

On the initial Schmidt support, the common Gram sum gives $\sum_h V_{yh}^* V_{yh} = I$ and $V_{yh} C_0 = C_{yh}$. Thus V_{yh} is a recorded instrument, and $|D_{yh}\rangle$ is an independently prepared normalized fresh vector. The same maps may be used for every y . The receiver centered swap is $w_h(I_{r^2} - dS_r)/r^2$. Declare the scalar hypothesis on the antisymmetric receiver projection for equal labels and on the symmetric projection for unequal labels. Complementary effects complete the measurement. These decisions attain each positive-part bound, proving sufficiency, including $r = 1$. Applying the same construction without a prescribed commuting eigenbasis proves sufficiency of (19.3) with the common Gram sums. Countable normal forms are covered by the positive trace tails of Lemma 15.2. $\square$

In particular the initial state need not be maximally mixed when $r < d$. Its largest Schmidt weight must be at most $1/r$. When $r = d$, the condition reduces to $\rho = I/d$. The finite interval construction concerns a known initial spectrum, not construction of the second-moment family itself.

19.2. Counting the first acquisition as well. Let $\mathfrak{R}_{q;k,\ell}$ impose the additional condition that the entire reference accompanying the first input has dimension at most q , conditional on the public preparation record. Mixed states on $I_1 \otimes \mathbb{C}^q$ are allowed. The old receiver after its recorded instrument has dimension at most k and the independent second reference at most ℓ , as before. Workspaces discarded before a specified cut do not enlarge a counted register. Classical records and the final joint receiver measurement have the same conventions as Definition 18.1.

Corollary 19.2 (Three specified dimension cuts). *For $1 \leq q, k, \ell \leq d$, put $s = \min\{q, k, \ell\}$. On the same finite family the complete classes $\mathfrak{R}_{q;k,\ell}$ have exact optimal values*

$$(19.5) \quad \begin{aligned} P_{q;k,\ell}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(ds - 1)}{2ds(d + 1)}, \\ P_{q;k,\ell}^{\text{b}} &= \frac{d + 1}{2(d + 1) - t^2} + \frac{t^2(s - 1)}{2s[2(d + 1) - t^2]}. \end{aligned}$$

The second line uses exactly the prior of Theorem 16.2. Both optima are attained by two independently prepared maximally entangled vectors on any fixed rank- s input subspace, with no intervening receiver instrument and no feedback. No first-call reference of dimension d is required unless $s = d$.

Proof. Resolve a mixed first preparation into a finite pure-state mixture in $I_1 \otimes \mathbb{C}^q$ and record its mixture index classically. This recovers its old score if the index is ignored, without adding a quantum purification. For each component rank $C_0 \leq q$. The Kraus refinement of Lemma 18.2 then gives rank $A_{yh} \leq \min\{q, k\}$, while rank $b_{yh} \leq \ell$. The two rank-sensitive swap bounds apply with s . The trace sums, constant-decision baselines and positive-part optimizations in Theorem 18.5 give both upper bounds. Averaging over initial mixtures and public seeds, and then taking norm limits, preserves these bounds.

For attainment fix an isometry $J : \mathbb{C}^s \rightarrow \mathbb{C}^d$ and use $C_0 = D = J^*/\sqrt{s}$. Keep the first receiver unchanged and prepare the second vector freshly. For every first label the old Gram and the fresh Gram are both JJ^*/s . The two final spectral readouts used in Theorem 18.5 attain the bounds. All three counted registers have dimension s . This proves the assertion also at $s = 1$, where the antisymmetric receiver projection is zero. $\square$

The three bounds apply at specified cuts. They are not a cap on the total workspace of an arbitrary implementation or a removal of the reset architecture. In particular, saturation remains the reset value, not the unrestricted coherent-acquisition value.

19.3. A stable common-projection theorem. For density matrices a, b with smaller rank at most r , set

$$\Delta_{d,r}(a, b) = d - r^{-1} - \|(\sqrt{a} \otimes \sqrt{b})(I - dS)(\sqrt{a} \otimes \sqrt{b})\|_1.$$

When $(d, r) \neq (2, 1)$ define

$$(19.6) \quad c_{d,r} = d - 1 - r^{-1} > 0, \quad K_{d,r} = r + \frac{(3 + \sqrt{2})^2}{c_{d,r}}.$$

Lemma 19.3 (Quantitative common support). *For $d \geq 2$, $1 \leq r \leq d$ and $(d, r) \neq (2, 1)$, there exists a rank- r projection P such that*

$$(19.7) \quad \max\{\|a - P/r\|_1^2, \|b - P/r\|_1^2\} \leq K_{d,r} \Delta_{d,r}(a, b).$$

Only the smaller Gram rank need be bounded by r ; neither invertibility nor equality of the two supports is assumed.

Proof. Put $f = \|\sqrt{a}\sqrt{b}\|_1$, $\eta = 1 - f^2$ and $e = \text{tr}(ab) - f^2/r$. The overlap rank bound gives $e \geq 0$, and $f \leq 1$ gives $\eta \geq 0$. The proof of Lemma 18.4 gives

$$(19.8) \quad \Delta_{d,r}(a, b) \geq c_{d,r}\eta + e.$$

Extend a right polar factor to a unitary U so that $Z = \sqrt{a}\sqrt{b}U = \sqrt{\sqrt{ab}\sqrt{a}} \succeq 0$. The Hilbert–Schmidt identity and the Schatten product inequality give

$$\|\sqrt{b}U - \sqrt{a}\|_2^2 = 2(1 - f), \quad \|Z - a\|_1 \leq \sqrt{2(1 - f)} \leq \sqrt{2\eta}.$$

Choose any rank- r projection P containing the range of Z . This is possible since $\text{rank } Z \leq r$. Since $\text{tr } Z = f$, one has

$$\|Z - fP/r\|_2^2 = e, \quad \|Z - fP/r\|_1 \leq \sqrt{re}.$$

These formulas remain valid at $f = 0$, without normalizing Z . As $1 - f \leq \sqrt{\eta}$, the triangle inequality yields

$$\|a - P/r\|_1 \leq (1 + \sqrt{2})\sqrt{\eta} + \sqrt{re}.$$

The normalized coefficient vectors for $\sqrt{a}$ and $\sqrt{b}U$ have inner product f and receiver marginals a, b . Their pure-state trace distance is $2\sqrt{1 - f^2}$, by its rank-two spectral calculation. Contractivity under partial trace therefore gives $\|a - b\|_1 \leq 2\sqrt{\eta}$, the trace-distance/root-fidelity estimate of Fuchs–van de Graaf [58, Theorem 1]. Consequently

$$\|b - P/r\|_1 \leq (3 + \sqrt{2})\sqrt{\eta} + \sqrt{re}.$$

Weighted Cauchy–Schwarz bounds the square of either right side by $[r + (3 + \sqrt{2})^2/c_{d,r}](c_{d,r}\eta + e)$. Now use (19.8). Every argument used matrices on finite supports and a unitary extension of a partial isometry; no inverse of a singular matrix or limit of an equality statement occurs. $\square$

The constants in this lemma are explicit but not asserted to be optimal. The projection may vary between branches; the theorem does not force all receivers to occupy one common subspace throughout the protocol.

Corollary 19.4 (Near-optimal projection decompositions). *Assume $(d, r) \neq (2, 1)$ and a strategy in $\mathfrak{R}_{k,\ell}$ has reward $v \geq p_r - \varepsilon$. Choose a dimension-preserving refined normal form; on its nonzero branches put*

$$w_{yh} = \text{tr } A_{yh}, \quad a_{yh} = A_{yh}/w_{yh}, \quad \mu_{yh} = w_{yh}/d, \quad B_\varepsilon = \frac{2K_{d,r}d(d+1)}{t^2}\varepsilon.$$

There are rank- r projections P_{yh} such that

$$(19.9) \quad \sum_{y,h} \mu_{yh} (\|a_{yh} - P_{yh}/r\|_1^2 + \|b_{yh} - P_{yh}/r\|_1^2) \leq 2B_\varepsilon,$$

$$(19.10) \quad \frac{1}{d} \sum_y \left\| \rho - \sum_h w_{yh} P_{yh}/r \right\|_1^2 \leq B_\varepsilon.$$

Finite and countable recorded instruments are allowed. The weights μ_{yh} sum to one and are Gram weights, not probabilities under a chosen hypothesis. For public mixtures the same statements hold with the public seed adjoined to the refined record, using the common purified initial Gram matrix.

Proof. Subtract (19.4) from the attained class value. Since $\sum_{y,h} w_{yh} = d$, this gives

$$\sum_{y,h} \mu_{yh} \Delta_{d,r}(a_{yh}, b_{yh}) \leq \frac{\varepsilon}{\alpha d} = \frac{2d(d+1)}{t^2} \varepsilon.$$

Apply Lemma 19.3 to each branch and sum. For the second estimate, each fixed y has $\sum_h w_{yh} = 1$ and $\sum_h w_{yh} a_{yh} = \rho$. The triangle inequality followed by Jensen's inequality bounds its squared residual by $\sum_h w_{yh} \|a_{yh} - P_{yh}/r\|_1^2$. Average over y . The proof uses nonnegative sums with finite total weight, so countable traces pass to the limit. The seed construction in Lemma 15.3, followed by the dimension-preserving refinement, preserves the common Gram sum and the achieved reward. $\square$

The total μ -weight on branches for which either deviation is at least $u > 0$ is at most $2B_\varepsilon/u^2$. This is a quantitative statement at the relative loss scale ε/t^2 , not a claim of vanishing error at fixed nonzero calibration defect.

19.4. A spectral loss certificate. Let

$$\mathcal{C}_{d,r} = \{\sigma \in \mathcal{D}_d : \sigma \preceq I/r\}, \quad e_r(\rho) = \text{tr}(\rho - I/r)_+.$$

Proposition 19.5 (Distance to the optimal initial spectra). *For every density matrix ρ ,*

$$(19.11) \quad \inf_{\sigma \in \mathcal{C}_{d,r}} \|\rho - \sigma\|_1 = 2e_r(\rho).$$

For $(d, r) \neq (2, 1)$, any strategy in $\mathfrak{R}_{k,\ell}$ with initial Gram ρ and achieved equal-prior reward v satisfies

$$(19.12) \quad p_r - v \geq \frac{2t^2}{K_{d,r}d(d+1)} e_r(\rho)^2.$$

Proof. Let Q be the spectral projection of ρ above $1/r$. For $\sigma \preceq I/r$, one has $\text{tr} Q(\rho - \sigma) \geq e_r(\rho)$. As $\text{tr}(\rho - \sigma) = 0$, positive-part optimization gives $\|\rho - \sigma\|_1 \geq 2e_r(\rho)$. Conversely, in an eigenbasis of ρ , cap every eigenvalue exceeding $1/r$ and redistribute the removed mass among eigenvalues below $1/r$. There is enough capacity since $d/r \geq 1$. This produces a density matrix in $\mathcal{C}_{d,r}$ at trace distance exactly $2e_r(\rho)$.

Every matrix $\sum_h w_{yh} P_{yh}/r$ in (19.10) belongs to $\mathcal{C}_{d,r}$. Set $\varepsilon = p_r - v \geq 0$ there and use (19.11); rearranging $4e_r(\rho)^2 \leq 2K_{d,r}d(d+1)(p_r - v)/t^2$ proves the bound. $\square$

The estimate concerns the actual first-input marginal, up to its fixed transpose, and a specified ideal reset architecture. Under the calibrated perturbations of Corollary 17.3, the same strategy's nominal reward differs from its observed reward by at most the stated channel, prior and latent-law error; a preparation defect adds its stated half-trace contribution. That total error must be added to $p_r - v$ before applying the certificate. There is no uncalibrated inference of the reset cut from a score.

19.5. The exceptional qubit face.

Proposition 19.6 (Pinching at the exceptional endpoint). *Let $d = 2$, $r = 1$, and suppose a is pure. Put $\mathcal{P}_a(b) = aba + (I - a)b(I - a)$ and $\Delta = \Delta_{2,1}(a, b)$. Then*

$$(19.13) \quad \|b - \mathcal{P}_a(b)\|_1^2 = 2\Delta - \Delta^2 \leq 2\Delta.$$

In particular $\Delta = 0$ if and only if $[a, b] = 0$. If instead b is the pure factor, interchange a, b .

Proof. By simultaneous unitary conjugation take $a = |0\rangle\langle 0|$ and $b = \begin{pmatrix} x & z \\ z & 1-x \end{pmatrix}$. The nonzero block of the centered filtered swap is $\sqrt{b}(I - 2a)\sqrt{b}$. Its trace is $1 - 2x$ and its determinant is $-\det b$. Its trace norm, also when b is singular, is therefore

$$\sqrt{(1 - 2x)^2 + 4 \det b} = \sqrt{1 - 4|z|^2} = 1 - \Delta.$$

The pinching difference has eigenvalues $\pm|z|$. Squaring gives (19.13). Vanishing is exactly $z = 0$, or $[a, b] = 0$. $\square$

Thus the exceptional equality set includes a pure factor with any commuting mixed qubit state, not just identical or orthogonal pure factors. A common rank-one projection cannot replace pinching here. The initial-spectrum theorem remains valid because $\mathcal{C}_{2,1} = \mathcal{D}_2$. This exception is separate from the zero-signal case $t = 0$, in which every strategy has reward $1/2$.

20. EXACT VALUES AT A PRESCRIBED INITIAL SPECTRUM

The equality set of a resource bound does not determine the value away from that set. We now solve this remaining optimization for the biased basis experiment of Theorem 16.2. Besides giving the exact spectral loss, the solution distinguishes protocols that merely retain a receiver from protocols that communicate a receiver outcome to the next preparation.

Fix $d \geq 2$, a weighted exact second-moment family satisfying (16.1), and $0 \leq t \leq 1$. The alternative basis is selected once and used at both calls. In this section only, the priors are those of (16.3); write

$$L_0 = 2(d+1) - t^2, \quad p_E = (d+1)/L_0.$$

Fix the *pure* first acquisition through its Gram matrix $\rho = C_0^* C_0 \in \mathcal{D}_d$. Its input marginal is $\rho^\top$. The initial acquisition is not replaced by a classically resolved mixture. Subsequent instruments, mixtures of fresh states, public seeds and tester limits retain the conventions of Definition 18.1.

For a nonnegative vector u , put $v_i = \sqrt{u_i}$ and define

$$(20.1) \quad \chi(u) = \lambda_{\max}(vv^* - \text{diag } u).$$

Appending zeros does not change this nonnegative number. If at least two entries of u are positive, it is the unique positive solution of

$$(20.2) \quad \sum_i \frac{u_i}{u_i + \chi} = 1.$$

If at most one entry is positive, set $\chi(u) = 0$.

Let $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ be the eigenvalues of ρ . For $1 \leq r \leq d$, choose the smallest $m \in \{0, \dots, r-1\}$ for which

$$\lambda_{m+1} \leq \tau, \quad \tau = \frac{\sum_{i>m} \lambda_i}{r-m},$$

and define the rank- r majorant

$$(20.3) \quad q^{(r)}(\lambda) = (\lambda_1, \dots, \lambda_m, \underbrace{\tau, \dots, \tau}_{r-m}, 0, \dots, 0).$$

The choice always exists at $m = r-1$. The preceding failed test, if $m > 0$, implies $\lambda_m > \tau$, so the displayed vector is decreasing. Zero tails are allowed; “rank- r ” here means rank at most r . Write $\mathcal{H}_r(\rho) = \chi(q^{(r)}(\lambda(\rho)))$.

Theorem 20.1 (Prescribed-spectrum value). *For the once-selected finite family and the fixed pure first acquisition above, the optimum over the complete class $\mathfrak{R}_{k,\ell}$, $1 \leq k, \ell \leq d$, is*

$$(20.4) \quad P_{k,\ell}^b(\rho) = p_E + \frac{t^2}{2L_0} \mathcal{H}_r(\rho), \quad r = \min\{k, \ell\}.$$

The optimum is attained by an instrument with at most d recorded outcomes, the same instrument after every first device label, and fresh references of dimension at most r . Both retained quantum registers may have dimension r . The second preparation may depend on the recorded receiver outcome, but need not depend on the first device label. This construction acts on the prescribed initial acquisition; it does not optimize over its Gram matrix.

For $r = d$ no instrument is needed and the answer is given directly by the secular equation for $\lambda(\rho)$. The theorem gives an exact value for the specified priors. The equal-prior spectral rigidity of Section 19 remains a separate result.

20.1. The scalar function and a rank-constrained leaf.

Lemma 20.2 (Secular concavity). *The function χ is continuous, symmetric, homogeneous of degree one, coordinatewise nondecreasing and concave on each nonnegative orthant. It is strictly Schur-concave on each fixed positive trace simplex: if $u \prec w$ and u, w are not permutations, then $\chi(u) > \chi(w)$. Here vectors are padded to a common length.*

Proof. On a support containing at least two coordinates, the matrix in (20.1) has strictly positive off-diagonal entries. A maximizing Rayleigh vector can be chosen nonnegative, and its coordinates on this support must be positive. Its eigenvalue equation is $(u_i + \chi)x_i = \sqrt{u_i}\langle v, x \rangle$. The Rayleigh value is positive, and summing gives (20.2). Its left side strictly decreases from the support size to zero, proving uniqueness. The support-one case follows directly from the matrix formula. That formula also gives continuity, symmetry and homogeneity, including at the boundary.

For $a > 0$, the root characterization says

$$\chi(u) \geq a \iff \sum_i \frac{u_i}{u_i + a} \geq 1.$$

Each summand is increasing and concave in u_i , so every such superlevel set is convex and coordinatewise increasing. If $c = \chi(u) > 0$ and $e = \chi(w) > 0$, convexity of the unit superlevel set applied with weights $c/(c+e), e/(c+e)$ gives $\chi(u+w) \geq c+e$. If one value vanishes the same inequality follows from coordinate monotonicity. Homogeneity now proves concavity.

Symmetry and concavity imply Schur concavity by convex combinations of permutations. For strictness, if $\chi(w) > 0$, strict concavity of $x/(x + \chi(w))$ shows that $u \prec w$, without permutation equality, implies $\sum_i u_i/(u_i + \chi(w)) > 1$. The root for u is therefore larger. If $\chi(w) = 0$, w has only one positive entry and any nonpermutation vector majorized by it has at least two, again giving strictness. $\square$

For $A, B \succeq 0$ let

$$\mathcal{N}(A, B) = \text{tr}[-(\sqrt{A} \otimes \sqrt{B})S(\sqrt{A} \otimes \sqrt{B})]_+.$$

Lemma 20.3 (Optimal fresh Gram at a fixed old Gram). *If $a_1 \geq \dots \geq a_d \geq 0$ are the eigenvalues of A , then*

$$(20.5) \quad \max_{B \in \mathcal{D}_d, \text{rank } B \leq \ell} \mathcal{N}(A, B) = \frac{1}{2} \chi(a_1, \dots, a_\ell).$$

A maximizing B commutes with A and is supported on its leading ℓ eigenspaces. If $c = \chi(a_1, \dots, a_\ell) > 0$, an explicit choice on that support is

$$(20.6) \quad b_i = \frac{a_i/(a_i + c)^2}{\sum_{j \leq \ell} a_j/(a_j + c)^2}.$$

No invertibility of A is required.

Proof. Lemma 16.3 expresses $2\mathcal{N}(A, B)$ as $(\text{tr } \sqrt{Z})^2 - \text{tr } Z$, where $Z = \sqrt{A}B\sqrt{A}$. First work on the support of A . Compressing B to that support leaves Z unchanged and does not increase its rank. If the compressed trace is positive, normalizing it can only increase the nonnegative, homogeneous objective. If the trace vanishes the objective is zero. We may therefore take A invertible on its support and $\text{tr}(A^{-1}Z) = 1$.

For fixed eigenvalues $z_1 \geq \dots \geq z_s > 0$ of Z , $s \leq \ell$, rearrangement gives

$$\sum_{i=1}^s z_i/a_i \leq \text{tr}(A^{-1}Z) = 1.$$

For completeness, in eigenbases the trace is a sum of products weighted by squared entries of a unitary, a doubly stochastic matrix. Its minimum over that larger convex set is achieved at a permutation; exchanging two inverted pairs puts the largest z_i on the largest a_i . This proves the inequality. Put Z' diagonal with these z_i on the leading eigenspaces of A and $B' = A^{-1/2}Z'A^{-1/2}$. Then $0 < \text{tr } B' \leq 1$; normalizing B' proves that a commuting maximizer with the asserted support suffices.

For such a B , write $x_i = \sqrt{b_i}$, so $\|x\|_2 = 1$. The objective is

$$2\mathcal{N}(A, B) = \left(\sum_{i \leq \ell} \sqrt{a_i} x_i \right)^2 - \sum_{i \leq \ell} a_i x_i^2.$$

The Rayleigh maximum is (20.1), with a nonnegative maximizer. Its eigenvector equation yields (20.6). With at most one positive a_i the maximum is zero, achieved by any rank-one Gram on the support, or any state if $A = 0$. This handles every singular case explicitly. $\square$

20.2. A spectral evaluation of the concave envelope. The next lemma records the majorization step separately. It applies to every continuous symmetric concave function, not only to the leaf objective just derived. The underlying majorization and ensemble methods are classical; their operational use in entanglement transformation and assistance is discussed in [59, 60, 61].

Lemma 20.4 (Rank-capped spectral envelopes). *Let f be a continuous symmetric concave function on the probability simplex of length r . For a density matrix of rank at most r , pad its eigenvalue vector to length r before applying f . Then*

$$(20.7) \quad \max_{\substack{\rho = \sum_h w_h a_h, \ w_h \geq 0 \\ a_h \in \mathcal{D}_d^{\leq r}}} \sum_h w_h f(\lambda(a_h)) = f(q_1^{(r)}(\lambda(\rho)), \dots, q_r^{(r)}(\lambda(\rho))).$$

The maximum admits at most d atoms commuting with ρ , all with the same spectrum $q^{(r)}$. In particular $\mathcal{H}_r$ is concave on $\mathcal{D}_d$; its homogeneous extension to positive matrices is concave and superadditive.

Proof. Write $q = q^{(r)}(\lambda)$. Its construction gives $\lambda \prec q$. Indeed its initial m partial sums agree with those of λ ; each subsequent entry of λ is at most τ , and its total sum is one. Moreover q is majorized by every decreasing probability vector z of rank at most r which majorizes λ . Up to m this follows from the majorization inequalities. Between m and r the decreasing entries of z give

$$\sum_{i \leq j} z_i \geq \frac{r-j}{r-m} \sum_{i \leq m} z_i + \frac{j-m}{r-m} \geq \sum_{i \leq m} \lambda_i + (j-m)\tau = \sum_{i \leq j} q_i.$$

After r both totals are one. Thus q is the least such majorant.

For any finite decomposition, the variational formula for the sum of the j largest eigenvalues gives

$$\lambda(\rho) \prec z := \sum_h w_h \lambda(a_h),$$

where the eigenvalue vectors are decreasing and padded to length d . The vector z has rank at most r . Concavity and Schur concavity give $\sum_h w_h f(\lambda(a_h)) \leq f(z) \leq f(q)$.

For the converse, $\lambda \prec q$ implies that λ lies in the convex hull of the permutations of q . One direct verification is as follows. For every real vector b , rearrangement followed by summation by parts in the partial-sum inequalities gives $b \cdot \lambda \leq b^\downarrow \cdot \lambda \leq b^\downarrow \cdot q = \max_\pi b \cdot \pi q$. A separating hyperplane is therefore impossible. This convex hull lies in a hyperplane of dimension at most $d - 1$, so Carathéodory reduces the combination to at most d permutations. In an eigenbasis of ρ they give commuting density matrices a_h of spectrum q . Their f values coincide, proving attainment.

Concavity follows also by combining any two attaining decompositions. For a countable decomposition, the same upper follows from the Ky Fan inequalities and Jensen's inequality by convergence of its bounded trace weights. At zero trace define the homogeneous extension to be zero. Trace-weighted convex combinations prove its concavity and superadditivity. Apply this with $f = \chi$ for the final assertion. $\square$

20.3. Proof of the prescribed-spectrum theorem.

Proof of Theorem 20.1. Use the dimension-preserving normal form of Lemma 18.2. The fixed initial Gram gives $\sum_h A_{yh} = \rho$ for every y . Put $w_{yh} = \text{tr } A_{yh}$ and $a_{yh} = A_{yh}/w_{yh}$ on nonzero branches. These are Gram weights; they are not substituted for the hypothesis-dependent history probabilities. Mixed fresh states and receiver channels are refined classically, without increasing either retained quantum dimension.

The biased payoff blocks are $-\kappa S$ for equal labels and $-2\kappa\Pi_-/(d-1) \preceq 0$ for unequal labels, where $\kappa = t^2/(dL_0)$. Thus only equal labels can improve on the constant alternative decision. Lemma 20.3 gives

$$P - p_E \leq \frac{\kappa}{2} \sum_{y,h} \chi(\lambda_1(A_{yh}), \dots, \lambda_\ell(A_{yh})).$$

Each summand is at most $\mathcal{H}_r(A_{yh})$. If $k \leq \ell$, the rank of A_{yh} is at most $r = k$ and the two expressions agree. If $\ell < k$, the first $r = \ell$ coordinates of its rank- r majorant are coordinatewise at least its leading r eigenvalues; use the monotonicity of χ . Superadditivity then gives

$$P - p_E \leq \frac{\kappa}{2} \sum_y \mathcal{H}_r(\rho) = \frac{t^2}{2L_0} \mathcal{H}_r(\rho).$$

The same bound survives public mixing, countable normal instruments and tester closure, by the fixed-Gram normal form and trace tails. At $t = 0$ all devices coincide and both sides reduce to $1/2$.

For attainment take the commuting decomposition $\rho = \sum_h w_h a_h$ supplied by Lemma 20.4, all a_h with spectrum $q^{(r)}$. On each support choose rectangular $C_h, D_h : \mathbb{C}^d \rightarrow \mathbb{C}^r$ with

$$C_h^* C_h = w_h a_h, \quad D_h^* D_h = b_h,$$

where b_h is the permuted optimizing fresh Gram in (20.6). If $\chi(q^{(r)}) = 0$, use any pure fresh Gram and the constant decision. Otherwise b_h is a density matrix of rank at most r , supported with a_h . The maps $V_h = C_h C_0^+$ satisfy $V_h C_0 = C_h$ and $\sum_h V_h^* V_h = I$ on the initial Schmidt support: positivity of the Gram sum implies that every C_h vanishes on $\ker \rho$. They therefore form an instrument on the given initial receiver; complete it arbitrarily on an unused orthogonal complement.

Use this same instrument after every y and prepare $|D_h\rangle$ independently conditional on its recorded outcome. For equal labels, the positive spectral projection of the filtered negative swap attains $\kappa w_h \chi(q^{(r)})/2$. For unequal labels choose the zero scalar-hypothesis effect. Complements complete each binary readout. Summing the d first labels gives equality. These unnormalized

formulas also define the endpoint $t = 1$ and null histories. The construction proves attainment without an extra retained environment. $\square$

20.4. When a receiver message is strictly useful. The following comparison fixes the initial acquisition and changes only the permission to send a receiver outcome to the fresh source. Let $P_{d,r}^{\text{nm}}(\rho)$ be the optimum with old receiver dimension d and fresh reference dimension r , when the fresh state may depend on the first device label and an independent public seed, but on no outcome obtained by measuring the old receiver. Arbitrary receiver processing and the final joint readout remain allowed. Any old processing without such communication can be postponed to the final readout. This is a subclass of $\mathfrak{R}_{d,r}$, not the complete-measure-and-reprepare class.

Corollary 20.5 (Exact value of a receiver message). *For the fixed experiment and initial Gram above,*

$$(20.8) \quad P_{d,r}^{\text{nm}}(\rho) = p_E + \frac{t^2}{2L_0} \chi(\lambda_1, \dots, \lambda_r).$$

If $t > 0$, then

$$(20.9) \quad P_{d,r}^{\text{b}}(\rho) > P_{d,r}^{\text{nm}}(\rho) \iff 2 \leq r < \text{rank } \rho.$$

The strict gain is realized with at most d receiver outcomes and without first-device-label dependence of the fresh state.

Proof. Without a receiver message each first-label branch retains Gram ρ . Lemma 20.3 gives the displayed upper, and the commuting optimizer with the positive-part final decisions attains it. Public mixtures cannot improve an already maximized linear score. If $r = 1$, both secular values vanish. If $r \geq \text{rank } \rho$, no tail is moved and $q^{(r)} = \lambda$. In the remaining case the leading r eigenvalues are all positive, and $q^{(r)}$ increases at least one of them without decreasing any. For a vector with at least two positive entries the root strictly increases when any positive coordinate is increased, as follows immediately from (20.2). This proves both directions. $\square$

For an explicit example take $d = 3$, $r = 2$ and $\rho = \text{diag}(3/4, 1/8, 1/8)$. Then

$$q^{(2)} = (3/4, 1/4, 0), \quad \chi(q^{(2)}) = \sqrt{3}/4, \quad \chi(\lambda_1, \lambda_2) = \sqrt{6}/8.$$

With $L_0 = 8 - t^2$, the exact advantage of the message is

$$(20.10) \quad \frac{t^2(2\sqrt{3} - \sqrt{6})}{16L_0} > 0 \quad (t > 0).$$

Use two Gram atoms $a_1 = \text{diag}(3/4, 1/4, 0)$ and $a_2 = \text{diag}(3/4, 0, 1/4)$ with weights $1/2$. With $C_0 = \sqrt{\rho}$, explicit old-receiver maps are

$$V_1 = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

They satisfy $\sum_h V_h^* V_h = I_3$ and $C_h = V_h C_0$, $C_h^* C_h = a_h/2$. The fresh Grams are respectively $\text{diag}(1/2, 1/2, 0)$ and $\text{diag}(1/2, 0, 1/2)$; normalized fresh vectors have the corresponding two equal Schmidt weights. The final positive-part readouts used in the proof attain the claimed scores. Thus the improvement is a realizable recorded instrument, not a change of initial state or postselection on a favorable history.

Corollary 20.6 (Spectral loss and exact saturation). *For $r > 1$, the difference from the optimized rank- r value of Theorem 18.5 is exactly*

$$(20.11) \quad P_{k,\ell}^{\text{b}} - P_{k,\ell}^{\text{b}}(\rho) = \frac{t^2}{2L_0} \left(\frac{r-1}{r} - \mathcal{H}_r(\rho) \right).$$

At $t > 0$ it vanishes exactly when $\rho \preceq I/r$. For $t > 0$ and a fixed ρ of rank s , its constrained optimum strictly increases as the smaller receiver dimension increases from 1 to s , and is constant thereafter. The smallest dimension attaining its full-reset value is therefore s , including the trivial case $s = 1$. At $t = 0$ every dimension gives the same value $1/2$.

Proof. The uniform vector of length r is majorized by every probability vector of that length. Lemma 20.2 gives $\chi(q) \leq (r-1)/r$, with equality only at that uniform vector. The construction of $q^{(r)}$ makes this equivalent to $\lambda_1 \leq 1/r$. Subtracting (20.4) from (18.7) proves the exact loss identity.

The least-majorant property gives $q^{(r+1)} \prec q^{(r)}$. If $r < s$, these have respectively $r+1$ and r positive entries and are not permutations, so strict Schur concavity gives a strict increase. At $r \geq s$ both vectors equal $\lambda(\rho)$. This proves the saturation statement. $\square$

The secular root and the majorant are functions of a supplied spectrum. Bisection of (20.2) evaluates the value without an optimization over physical protocols; a finite convex decomposition then realizes it. Real input spectra and real weights are permitted. These statements do not assume a rational eigenbasis or give an efficient construction of the unknown-device design. In particular, the general gain from a receiver message proved here is compatible with the no-feedback globally attaining preparations in Corollary 19.2: those preparations optimize the initial Gram, whereas (20.9) holds it fixed.

21. RELATION TO CHANNEL METROLOGY AND MEASUREMENT INFORMATION

21.1. Support spans, channel estimation and finite pairs. For the measurement Kraus operators $L_{j,a} = |j\rangle\langle a|\sqrt{E_j}$, the real Hermitian Kraus-product span is exactly $\mathcal{W}_E$. For the orbit $L_{j,a}(t) = L_{j,a}e^{-itG}$ the effective generator is G . Zhou–Jiang [40, Theorems 1–3 and Section V] establish the Kraus-span criterion for the linear or quadratic asymptotic channel Fisher-information rate and give error-correction attainability. We do not claim a new metrological dichotomy or a new correction principle. Theorems 4.1 and 5.1 identify its complete support-kernel form and give finite-angle trace-distance inequalities with an explicit $4mt^2\|G\|_{\text{op}}^2$ remainder. Theorem 6.2 adds a normalized first-order realization for general two-sided C^2 effect curves; its $\sqrt{N}|t| + O(Nt^2)$ bound handles rank changes without a smooth exact Kraus hypothesis. Theorem 7.1 separately budgets certified control errors. Neither constitutes a new claim to the underlying metrological exponents.

Sieniawski–Demkowicz-Dobrzański [26, Section IV, equations (16)–(23)] integrate channel Fisher-information bounds to bound adaptive discrimination and compare metrological scaling with perfect finite discrimination. The same horizontal method is a principal antecedent of (1.4). Our support-kernel identity is stated in measurement coordinates and our lower (5.9) comes from an actual corrected finite-angle channel. It does not claim the exact optimal tester or perfect discrimination.

21.2. Strategy classes and finite-use comparisons. Our independent inputs are products of *complete* probe–reference systems. Public mixtures preserve the bound; a correlated reference shared between groups does not satisfy this definition. Block- b preparations interpolate between those inputs and unrestricted entangled parallel inputs. Classical feedback permits earlier labels to change later preparations and is not a parallel strategy; fully adaptive control additionally retains quantum memory. The independent-block theorem is not a classification of every feedback model. Theorem 12.4 treats the separately defined classical-feedback reset class with arbitrary receiver memory; it still excludes coherent import into a fresh block.

Harrow–Hassidim–Leung–Watrous [46, Sections 3–4] construct channels perfectly distinguishable by two adaptive calls but not by any finite entangled parallel strategy. Those are perfect-discrimination statements about a fixed pair of general channels. Corollary 11.5 concerns comparison orders in shrinking fixed-tangent neighborhoods of classical-output measurements; it asserts neither equality of distances nor perfect discrimination. There is no contradiction with that separation.

Salek–Hayashi–Winter [47, Theorem 9 and Remark 10] compare asymptotic Chernoff and Hoeffding rates for classical feed-forward with no quantum memory at the input against nonadaptive product strategies. Their restrictions and fixed-pair error exponents differ from the present hard call cap and finite trace separation. In particular their product result is not an upper bound on our entangled parallel inputs.

Li–Hirche–Tomamichel [48, Theorems 3.1–3.3] characterize sequential error-exponent regions under expected and high-probability stopping constraints in their specified adaptive, block and nonadaptive models, under finite max-relative-entropy hypotheses. Our budget instead bounds the number of calls on every record, and our parameter s varies jointly with that budget. Their general instrument discussion is not invoked as a theorem of unrestricted quantum-memory optimality here.

Bounds in terms of sums of squared entangled group sizes, and GHZ attainability of metrological enhancements, are standard [33, 34]. Theorem 11.4 does not claim these mechanisms as new. It proves the finite-pair trace law for a normalized measurement tube, controls arbitrary legal quadratic remainders, and uses a readout independent of that unknown remainder. The factor $\sqrt{Nb}$ comes with both a block-state Bures upper and an integer-block Bernoulli lower, not just a Fisher-information scaling.

21.3. Memory-constrained testers and retained receiver systems. Ohst–Zhang–Nguyen–Plávala–Quintino [51] formulate discrimination with prescribed auxiliary dimensions as optimization over constrained separable operators. Their Definition 6 specifies single-call memory-limited testers, while Theorem 9 characterizes convex combinations of memoryless testers by a convergent symmetric-extension hierarchy; Sections 5.2–5.4 treat multi-call quantum and classical memory and affine constraints on the factors of the lifted decomposition. These are dimension and causal-transfer restrictions, not a bound on the squared number of calls between fresh preparations.

Their Definition 23, equation (64), describes two-call classically adaptive testers as $T_i = \sum_z R_z \otimes S_{i|z}$, with single-call testers R and $S_{\cdot|z}$. The first complete output is measured and only the classical result chooses the second tester. Their Theorem 24 shows that this class and the parallel class are not nested; both lie in the adaptive class. Our inclusion and explicit negative-partial-transpose witness in Proposition 14.5 answer the relevant comparison without identifying these classical testers with all unit-reset protocols. Reset protocols may keep an unmeasured old receiver register for a final joint measurement. It is independent of the complete next acquisition, but need not be separable from all other receiver systems after receiver processing.

The constrained-separability formulation in their Section 6.2, equations (65)–(68), optimizes over the stated classically adaptive set. It is not a characterization of our larger retained-receiver set: its cross-use separability requirement excludes the tester (14.5). Conditional tensor independence at preparation is not the same condition as separability of every final tester effect across uses. Consequently an upper bound for that smaller optimization cannot be used as an upper for the reset class merely by dropping a dimension cap. Their general adaptive optimization, when applicable to the same finite channel ensemble, can of course bound a reset subclass; that inclusion does not supply a sharp Q -dependent shrinking-tube law. We do not claim that no other lifted constrained-separability representation of reset preparation could be developed.

Thus arbitrary receiver memory with zero coherent import has an explicit operational relation to their complete-measurement class, but it is not identified with any fixed auxiliary-dimension

slice of their hierarchy. The contribution here is not a new memory hierarchy or a new separability method. It is the support-coordinate local profile and hard-budget theorem, including the unknown quadratic remainder and a matching lower in those resources. The Bell-reference witness alone establishes a model boundary; Theorem 16.2 now supplies a strict exact score hierarchy for the specified finite ensemble, whereas the general memory-constrained discrimination problem and independent priority assessment remain distinct.

21.4. Channel Bures bounds and block testing. Huang–Meyer–Nuradha–Wilde [49, Theorem 17, Remark 18] restate Yuan–Fung’s parallel bound [25, equation (23)]. For aligned canonical isometries put $M_W = V_0^*(W \otimes I)V_1$, where $\|W\| \leq 1$, $a_W = \|I - \text{Re } M_W\|$ and $b_W = \|I - M_W\|$. Their Theorems 17 and 19 give, respectively,

$$d_B^2(\mathcal{N}_0^{\otimes n}, \mathcal{N}_1^{\otimes n}) \leq n \inf_W \{2a_W + (n-1)b_W^2\},$$

$$\sup_{\text{adaptive controls}} d_B^2(\mathcal{P}_0^{(n)}, \mathcal{P}_1^{(n)}) \leq n \inf_W \{2a_W + (n-1)\sqrt{2a_W} b_W\}.$$

Channel Bures distance takes the supremum over complete probe–reference inputs. For our chosen surrogate dilations, $2a_W = \|W_s - W_0\|^2 \leq \alpha^2 s^2$ and $b_W \leq \alpha s$; either expression yields $d_B \leq \alpha n s$. Restoring the actual tube member costs nrs^2 in diamond norm and hence $s\sqrt{rn}$ in Bures distance. Thus the block upper specializes established isometric-extension bounds; it is not a new fidelity principle. The finite-tube contribution is the uniform remainder, complete support cases and matching width-dependent lower. The reset theorem additionally handles history-dependent allocation and receiver instruments by conditional fidelity, rather than unconditional tensorization.

Ghosal–Halder–Patra–Sen [50, Section III, Theorem 1] use block-i.i.d. probes, with arbitrary within-block correlations and independent identical repetition. For a closed convex null channel set $\mathbf{R}$ and a simple alternative $\mathcal{N}$, their memory-assisted fixed-block Stein exponent is

$$\sup_{\sigma_{RA \otimes \ell}} \inf_{\Phi \in \mathbf{R}} \frac{1}{\ell} D\left((\text{id}_R \otimes \Phi^{\otimes \ell})(\sigma) \parallel (\text{id}_R \otimes \mathcal{N}^{\otimes \ell})(\sigma)\right).$$

Their memory-free expression restricts σ to the probes. The block hierarchy is a direct antecedent of our resource taxonomy. Our blocks may have unequal sizes and nonidentical states; the reset class also has classical feedback and adaptive within-block control. We consider finite unhalved trace separation in a shrinking fixed tube, not a composite Stein exponent at fixed block size. Neither theorem is substituted for the other, and no first introduction of block strategies is claimed.

21.5. Semidefinite tangent geometry and nonemptiness. The positive-semidefinite tangent condition $QHQ \succeq 0$ and second-order corrections are classical variational geometry; Bonnans–Cominetti–Shapiro [45] study parabolic tangent sets and semidefinite second-order regularity. We use no new primitive cone theorem. Proposition 10.1 provides a common normalization, explicit coefficient and interval, and a simultaneous operator-norm remainder bound for the measurement tuple. The sufficient bound can be large and is not a minimal feasibility threshold. Its exact certificate does not determine optimal discrimination constants or synthesize quantum controls.

21.6. Fisher frames and covariance embeddings. Saini–Kiukas–Burgarth–Gilchrist [42, Theorem 1] compare classical and quantum Fisher information for a varying full-rank *state* measured by one fixed informationally complete POVM, using a state-dependent frame operator. Here the varying object is the measurement, the tangent has coupled normalization, and the loss permits N adaptive uses with a reference. In the rank-one case our support span equals the effect span, so informational completeness excludes the linear orbit regime by Corollary 5.4. That intersection does not identify the two quadratic forms or their resource models.

Mayer–Yun [43, Theorem 3.6, Definition 3.7 and Remark 4.4] use a kernel embedding of operator-valued measures, a kernel-dependent discrepancy, and a tomographic Gram operator

whose nullspace records unobserved state directions. Our support span contains entire matrix blocks $P_j \mathcal{H}_d P_j$, rather than one observable per effect; this distinction disappears for rank-one effects but not in general. Theorem 4.1 concerns a coupled measurement tangent, and (1.4) concerns adaptive finite-use loss. Neither kernel embedding nor injectivity alone supplies those statements.

21.7. Known symmetry and tomography primitives. The full primary text of Yoshida–Okigami–Posta–Grinko [44] distinguishes known-symmetry estimation from known-pair discrimination. Theorem 7 gives invariant-state copy complexity $\Theta((D_{\text{st}} - 1 + \log(1/\eta))/\epsilon^2)$, where $D_{\text{st}} = \sum_{\lambda} m_{\lambda}^2$; $D_{\text{st}} = 1$ is a known singleton. Corollary 9 gives parallel covariant-channel bounds

$$O((D'_G + K_G \log(1/\eta))/\epsilon_{\diamond}^2), \quad O((C'_G + \log(1/\eta))/\epsilon_{\text{tr}}^2),$$

in one-use halved diamond distance and normalized-Choi trace distance, respectively, for $0 < \epsilon \leq 1$, $0 < \eta \leq 1/2$. Here $D_G = \sum_{\lambda} m_{\lambda} M_{\lambda}$ counts Choi commutant parameters before trace preservation, $K_G = \sum_{\lambda} m_{\lambda}$, $D'_G = D_G + K_G \log(4L)$, and $C'_G = d_{\mathcal{I}}^{-1}(\sum_{\lambda} m_{\lambda} \sqrt{d_{\lambda} M_{\lambda}})^2$; L counts occupied blocks. Theorem 13 does not give a general matching inverse-square diamond lower: its bound is $\Omega_{\xi}(D_G/\epsilon_{\diamond}^{\xi/2})$ under its affine-dimension hypothesis.

For permutation covariance on ℓ systems of fixed local dimension q , Theorems 12 and 15 match the Choi order $\ell^{q^4 - q^2} \epsilon_{\text{tr}}^{-2}$ at constant confidence. Theorem 18 approximates the Hayashi measurement in $O((s+q) \text{polylog}(s, q, \tau^{-1}, \zeta^{-1}))$ gates, with coupled output error at most τ except with probability ζ . This is not a diamond-error certificate for our recovery. Known-symmetry estimation, its parameter dimensions, and that specialized gate construction do not themselves give our support kernel or finite-pair curve theorem. Nor do our theorems claim faster learning on those symmetry classes.

For ordinary tomography the main-text substitutions remain explicit. Mele–Bittel [5, Theorem III.3] give the sufficient constant-confidence order $O(dkr\epsilon^{-2})$ in halved diamond loss, with $r \leq dk$ for measurement channels. Zambrano–Ramos-Calderer–Kueng [29, Theorem 2] give the sufficient single-copy worst-input outcome-TV bound

$$m \geq \frac{8[d^2 + \epsilon(d^2 + 1)/6]}{\epsilon^2} \log \frac{2^{k+1} 9^{2d}}{\eta}.$$

Either loss bounds the component norm error by ϵ after output dephasing where needed. A legal estimator need not be balanced: apply Theorem S24.1 before using the interior metric. The resulting future loss is at most $8k\sqrt{N}\epsilon$. With $\epsilon = \delta/(8k\sqrt{N})$ the respective sufficient orders are

$$O(d^2 k^4 N \delta^{-2}), \quad O(k^2 N \delta^{-2} d^2 [d + k + \log(1/\eta)]).$$

These are upper-bound substitutions, not optimality claims about the cited estimators. Our component procedure credits the existing binary operator-norm primitive and uses a different confidence allocation. The full calculations and confidence restrictions are retained in the supplement, Section S3.

21.8. What remains beyond the theorems. The orbit theorem and Theorem 6.2 provide matching scales on each fixed two-sided smooth curve with nonzero first derivative, including second-order rank openings. Their constants need not be uniform across curves. Arbitrary-pair midpoint equivalence and full-body multi-outcome entropy are not consequences of these local statements. Growing- k minimax sharpness, polynomial entropy-optimal dictionaries, and general readout synthesis remain separate mathematical questions. Independent expert priority assessment is not inferred from the absence of an identical statement in this targeted comparison.

21.9. Tangent neighborhoods and independent acquisition. Theorem 8.5 fixes E, H and an explicit second-order remainder allowance, rather than assuming a particular smooth family of Kraus operators. It treats the full one-sided positive tangent cone. A nonzero missing-support compression has a classical impossible-event witness; only within the cone’s lineality space does the support-generator test select the coherent branch. The latter still uses the credited correction mechanism. Theorem 9.1 separates these mechanisms by an explicit product probe–reference restriction. Its upper uses standard root-fidelity inequalities and tensor multiplicativity [1, Chapter 3], not a new fidelity principle or a bound on all nonadaptive entangled strategies. The intervening parallel class is addressed instead by the new proof in Section 11. Its conclusion matches the adaptive local order and does not promote the product upper to that class.

The higher-order statement requires the finite remainder (9.1). Proposition 9.3 instead proves a two-scale law when an intermediate support-opening term is present. These results address one-sided openings and selected higher-order approaches without identifying a first jet with an arbitrary-pair global geometry. No general boundary entropy or unrestricted growing-outcome learning law is inferred.

21.10. Autonomous coherent-memory hierarchies. Zonnios–Binder [52, Proposition 2] prove monotonicity of their memory-parametrized process distinguishability and its upper bound by the strategy norm. Their Theorem 1 compiles any finite-time tester into repeated use of one instrument with an internal counter and sufficient coherent memory; Corollary 1 gives finite-time completeness. Their unknown object is a multi-time process, and their resource is the coherent ancilla dimension of an autonomous repeated-instrument machine. Our reset class instead allows public time-dependent history rules and unrestricted receiver storage, but forbids coherent import across a specified fresh-preparation cut. Its hard square cost Q is not an ancilla dimension. Neither their completeness theorem nor our fixed-tube profile law supplies the other’s conclusion. The exact finite-ensemble hierarchy of Theorem 16.2 likewise compares three acquisition/receiver classes, not every dimension slice of their hierarchy. Their work is a direct antecedent for systematic coherent-memory interpolation; no first such framework is claimed here.

21.11. Classical outputs and accessible post-measurement states. Eid–Quintino [53] consider the quantum state returned by the associated Lüders instrument in addition to the classical label. Our unknown channel returns only that label. The references used in Theorem 16.2 were supplied by the experimenter; they are not residual quantum outputs of the device. Thus an instrument discrimination advantage cannot be imported into the present interface.

21.12. What the exact hierarchy adds. The strict tester inclusion in Proposition 14.5 alone is not an operational score theorem. Theorem 16.2 provides one finite classical-output ensemble, exact upper bounds for all three classes, and matching realizations in every dimension and along a full-rank deformation. Ohst et al. [51, Theorem 24] already separate other strategy classes by channel-discrimination examples; the present assertion does not claim a first memory advantage. Its middle optimization permits arbitrary unmeasured receiver storage and all unit-reset feedback. Lemma 16.3 is the step that prevents a separable-tester upper from being incorrectly assigned to this larger class. These exact ensemble values and the local allocation orders answer different questions; neither is substituted for the other.

21.13. Finite reset optimization. The constrained-separability formulations of Ohst et al. [51] and the finite-time compilation theorem of Zonnios–Binder [52] are direct antecedents for optimization with memory restrictions. Theorem 15.4 instead fixes a complete fresh-preparation cut and retains an arbitrary old receiver; it realizes a common-barycenter concave-envelope formula with a finite branch bound for every finite two-call measurement decision experiment. It does not classify all auxiliary-dimension slices of those models. Concave envelopes, Carathéodory reduction

and conic duality are established tools [56, 57], not separate novelty claims. The equal-prior values in Theorem 17.2 are an exact application, with all class uppers proved rather than inferred from exhibited strategies. Antisymmetric comparison and finite design mechanisms remain credited to [55, 54]. Independent specialist assessment of the resulting optimization statements remains appropriate.

Rank-resolved recording cuts. Theorems 18.3 and 18.5 separate an arbitrary-experiment value characterization from an explicitly solved finite family. The resource counts the old receiver after its recorded instrument and the fresh reference during the second call; it does not count the unrestricted initial reference. In contrast, [51, Sections 5–6] formulates constrained channel-memory optimization through separability with affine constraints, including complete classically adaptive testers in Definition 23. The present rank envelope gives the exact common-barycenter form for a reset measurement interface, with explicit two-coordinate values and adjacent gains on the same ensemble. Neither a new general memory concept nor priority over constrained-memory optimization is asserted. The autonomous process hierarchy of [52, Theorem 1 and Corollary 1] saturates the unrestricted strategy norm at sufficient coherent transfer memory. Our hierarchy preserves fresh reset at every threshold and saturates only the reset optimum. Its conditional dimension witnesses require that architecture and calibrated channels.

21.14. Spectral equality and its quantitative form. The initial-state constraint $\rho \preceq I/r$ is the convex hull of normalized rank- r projections. We use this elementary spectral geometry as a tool; neither its convex-hull identity nor the standard trace-distance/fidelity inequality is a priority claim. Theorem 19.1 identifies this entire set with the attainable initial Grams of the specified reset discrimination optimum. Lemma 19.3 and Proposition 19.5 convert its score deficit into quantitative common-support and initial-spectrum constraints. The proof includes the familiar purification estimate, credited to Fuchs–van de Graaf [58], but derives its use with singular, unequal-rank factors directly. The exceptional qubit equality face is treated separately by Proposition 19.6. The dimension cap on the initial reference is independent of the old-output cut. The three-cut formula does not replace the stationary multi-time hierarchy of Zonnios–Binder or the constrained tester framework of Ohst et al.; its endpoint is still the reset value.

21.15. Spectral transformations and receiver messages. Majorization under local pure-state transformations is established by Nielsen [59]; Jonathan–Plenio [60] characterize transformations into ensembles. Maximizing an average function over decompositions also has a long history in entanglement of assistance [61]. We use the spectral and convex mechanisms with this attribution. Lemma 20.4 includes the elementary majorization proof, rather than treating that mechanism as new. The contribution of Theorem 20.1 is the exact operational evaluation for a prescribed initial acquisition under the complete reset class, including the rank-constrained fresh-Gram optimizer and the necessary and sufficient strict-message criterion. It is not an identification with a general entanglement monotone or a new proof of the classical majorization theorems. The once-selected device, specified cuts and biased priors remain part of the statement.

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